AMENDMENT OF SOLICITATION/MODIFICATION OF CONTRACT

1. CONTRACT ID CODE: J	PAGE OF PAGES: 1 of 3		
2. AMENDMENT/MODIFICATION NO.: U0002	3. EFFECTIVE DATE: 05-Aug-2025	4. REQUISITION/PURCHASE REQ. NO.: W22W9K50135988	5. PROJECT NO. (If applicable)
6. ISSUED BY CODE: W912QR U. S. ARMY ENGINEER DISTRICT, LOUISVILLE 600 DR. MARTIN LUTHER KING, JR. PLACE ROOM 821 LOUISVILLE KY 40202-2239		7. ADMINISTERED BY (If other than item 6) CODE **See Item 6**	
8. NAME AND ADDRESS OF CONTRACTOR (No., Street, County, State and Zip Code)		X 9A. AMENDMENT OF SOLICITATION NO.: W912QR25R0027	
		X 9B. DATED (SEE ITEM 11): 23-Jun-2025	
		10A. MOD. OF CONTRACT/ORDER NO.	
CODE	FACILITY CODE	10B. DATED (SEE ITEM 13)	

11. THIS ITEM ONLY APPLIES TO AMENDMENTS OF SOLICITATIONS

[X] The above numbered solicitation is amended as set forth in Item 14. The hour and date specified for receipt of Offer [ ] is extended, [X] is not extended.

Offer must acknowledge receipt of this amendment prior to the hour and date specified in the solicitation or as amended by one of the following methods: (a) By completing Items 8 and 15, and returning 1 copies of the amendment; (b) By acknowledging receipt of this amendment on each copy of the offer submitted; or (c) By separate letter or telegram which includes a reference to the solicitation and amendment numbers. FAILURE OF YOUR ACKNOWLEDGMENT TO BE RECEIVED AT THE PLACE DESIGNATED FOR THE RECEIPT OF OFFERS PRIOR TO THE HOUR AND DATE SPECIFIED MAY RESULT IN REJECTION OF YOUR OFFER. If by virtue of this amendment you desire to change an offer already submitted, such change may be made by telegram or letter, provided each telegram or letter makes reference to the solicitation and this amendment, and is received prior to the opening hour and date specified.

12. ACCOUNTING AND APPROPRIATION DATA (If required)

13. THIS ITEM APPLIES ONLY TO MODIFICATIONS OF CONTRACTS/ORDERS.
IT MODIFIES THE CONTRACT/ORDER NO. AS DESCRIBED IN ITEM 14.

[ ] A. THIS CHANGE ORDER IS ISSUED PURSUANT TO: (Specify authority) THE CHANGES SET FORTH IN ITEM 14 ARE MADE IN THE CONTRACT ORDER NO. IN ITEM 10A.

[ ] B. THE ABOVE NUMBERED CONTRACT/ORDER IS MODIFIED TO REFLECT THE ADMINISTRATIVE CHANGES (such as changes in paying office, appropriation date, etc.) SET FORTH IN ITEM 14, PURSUANT TO THE AUTHORITY OF FAR 43.103(B).

[ ] C. THIS SUPPLEMENTAL AGREEMENT IS ENTERED INTO PURSUANT TO AUTHORITY OF:

[ ] D. OTHER (Specify type of modification and authority)

E. IMPORTANT: Contractor [ ] is not, [ ] is required to sign this document and return ______ copies to the issuing office.

14. DESCRIPTION OF AMENDMENT/MODIFICATION (Organized by UCF section headings, including solicitation/contract subject matter where feasible.)

See Summary of Changes for Amendment 002

Except as provided herein, all terms and conditions of the document referenced in Item 9A or 10A, as heretofore changed, remains unchanged and in full force and effect.

15A. NAME AND TITLE OF SIGNER (Type or print)		16A. NAME AND TITLE OF CONTRACTING OFFICER (Type or print) TEL: EMAIL:	
15B. CONTRACTOR/OFFEROR (Signature of person authorized to sign)	15C. DATE SIGNED	16B. UNITED STATES OF AMERICA BY (Signature of Contracting Officer)	16C. DATE SIGNED

SECTION SF 30 BLOCK 14 CONTINUATION PAGE

SUMMARY OF CHANGES

SECTION SF 30 - BLOCK 14 CONTINUATION PAGE (SF 30)

The following have been added by full text:

AMENDMENT 002

Summary of Changes

The following have been changed for Amendment #0002

WPAFB – RENOVATE RESEARCH FACILITY FOR MS&A, BUILDING 20145

Specification Revisions:

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02 41 00 DEMOLITION AND DECONSTRUCTION
05 51 00 METAL STAIRS
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09 90 00 PAINTS AND COATINGS
31 63 33 INJECTION GROUTED MICROPILES

.

CID Revisions:

FURNITURE FIXTURES AND EQUIPMENT:
STRUCTURAL INTERIOR DESIGN - FLOORING:

Drawing Revisions:

G-004: SECURITY PLAN / GENERAL NOTES:
GS105: CODE COMPLIANCE PLAN:
S-001: GENERAL NOTES:
S-103: GENERAL NOTES:
S-501: SHEAR CONNECTIONS DESIGN CRITERIA:
S-506: LID GUARDRAIL DETAILS:
S-601: GENERAL NOTES:
A-005: PARTITON TYPES:
A-110: FIRST FLOOR REFLECTED CEILING PLAN:
A-115: SECURITY AND ACOUSTIC WALLS FIRST FLOOR PLAN:
A-118: ACCESS FLOORING FIRST FLOOR PLAN:
A-301: WEST TO EAST CROSS SECTION:

A-302: WALL SECTION:
A-404: ENLARGED CASEWORK PLAN / TOILET ACCESSORY SCHEDULE / TYPICAL MOUNTING HEIGHTS:
A-501: WALL DETAIL:
A-503: WALL DETAIL:
A-601: DOOR SCHEDULE:
I-001: STAIR NOTES:
I-501: TYPICAL SIGN INSTALLATION:
I-601: LEGEND OF COLORS AND FINISHES:
F-001: GENERAL FIRE ALARM NOTES:

P-104: THIRD FLOOR PIPING PLAN:
M-001: DUCTWORK CONSTRUCTION AND LEAKAGE TESTING:
M-101: BASEMENT DUCTWORK PLAN:
M-102: FIRST FLOOR DUCTWORK PLAN:
M-103: SECOND FLOOR DUCTWORK PLAN:
M-104: THIRD FLOOR DUCTWORK PLAN:
M-106: FIRST FLOOR PIPING PLAN:
M-401: PIPE CONNECTION SECTION / MECHANICAL ROOM SECTION:
M-502: TYPICAL Z DUCT AND ACOUSTICAL PENETRATION DETAIL:
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M-801: FIRST FLOOR PIPING DIAGRAM:
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E-601: RISER DIGRAM:
E-605: MCC-D (REVISED):
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T-601: COMMUNICATION ONE-LINE DIAGRAM:
T-604: CER 274 ELEVATION / SHEET KEY NOTES:
T-605: CER 374 ELEVATION / SHEET KEY NOTES:

(End of Summary of Changes)

AMENDMENT 0002

PROJECT TABLE OF CONTENTS

ATTACHMENTS TO SPECIFICATIONS AS FOLLOWS:

APPENDICES TO SPECIFICATIONS AS FOLLOWS:

ADDITIONAL APPENDICES AT END OF SPECIFICATIONS:

- End of Project Table of Contents --

SECTION 02 41 00

DEMOLITION AND DECONSTRUCTION
05/10, CHG 2: 02/19
AMENDMENT 0002

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AIR-CONDITIONING, HEATING AND REFRIGERATION INSTITUTE (AHRI)

AHRI Guideline K	(2009) Guideline for Containers for Recovered Non-Flammable Fluorocarbon Refrigerants

AMERICAN SOCIETY OF SAFETY PROFESSIONALS (ASSP)

ASSP A10.6	(2006) Safety & Health Program Requirements for Demolition Operations - American National Standard for Construction and Demolition Operations

CARPET AND RUG INSTITUTE (CRI)

CRI 104	(2015) Carpet Installation Standard for Comnmercial Carpet
CRI 105	(2015) Carpet Installation Standard for Residential Carpet

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1	(2024) Safety and Health Requirements Manual

U.S. DEFENSE LOGISTICS AGENCY (DLA)

DLA 4145.25	(Jun 2000; Reaffirmed Oct 2010) Storage and Handling of Liquefied and Gaseous Compressed Gases and Their Full and Empty Cylinders http://www.aviation.dla.mil/UserWeb/aviationengineerin

U.S. DEPARTMENT OF DEFENSE (DOD)

DOD 4000.25-1-M	(2006) MILSTRIP - Military Standard Requisitioning and Issue Procedures
MIL-STD-129	(2014; Rev R; Change 1 2018; Change 2 2019) Military Marking for Shipment and Storage

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

40 CFR 61	National Emission Standards for Hazardous Air Pollutants
40 CFR 82	Protection of Stratospheric Ozone
49 CFR 173.301	Shipment of Compressed Gases in Cylinders and Spherical Pressure Vessels

1.2 PROJECT DESCRIPTION

1.2.1 Definitions

1.2.1.1 Demolition

Demolition is the process of wrecking or taking out any load-supporting structural member of a facility together with any related handling and disposal operations.

1.2.1.2 Deconstruction

Deconstruction is the process of taking apart a facility with the primary goal of preserving the value of all useful building materials.

1.2.1.3 Demolition Plan

Demolition Plan is the planned steps and processes for managing demolition activities and identifying the required sequencing activities and disposal mechanisms.

1.2.1.4 Deconstruction Plan

Deconstruction Plan is the planned steps and processes for dismantling all or portions of a structure or assembly, to include managing sequencing activities, storage, re-installation activities, salvage and disposal mechanisms.

1.2.2 Demolition/Deconstruction Plan

Prepare a Demolition Plan and Deconstruction Plan and submit proposed salvage, demolition, deconstruction, and removal procedures for approval before work is started. Include in the plan procedures for careful removal and disposition of materials specified to be salvaged, coordination with other work in progress, a disconnection schedule of utility services, a detailed description of methods and equipment to be used for each operation and of the sequence of operations. Coordinate with Waste Management Plan in accordance with Section 01 35 43 GENERAL ENVIRONMENTAL PROTECTION REQUIREMENTS (WPAFB). Provide procedures for safe conduct of the work in accordance with EM 385-1-1. Plan shall be approved by Contracting Officer prior to work beginning.

1.2.3 General Requirements

Do not begin demolition or deconstruction until authorization is received from the Contracting Officer. Remove rubbish and debris from the project site; do not allow accumulations inside or outside the buildings. The work includes demolition, deconstruction, salvage of identified items and materials, and removal of resulting rubbish and debris. Remove rubbish and debris from Government property daily, unless otherwise directed. Store materials that cannot be removed daily in areas specified by the Contracting Officer. In the interest of occupational safety and health, perform the work in accordance with EM 385-1-1, Chapter 23, Demolition, and other applicable Sections.

1.3 ITEMS TO REMAIN IN PLACE

Take necessary precautions to avoid damage to existing items to remain in place, to be reused, or to remain the property of the Government. Repair or replace damaged items as approved by the Contracting Officer. Coordinate the work of this section with all other work indicated. Construct and maintain shoring, bracing, and supports as required. Ensure that structural elements are not overloaded. Increase structural supports or add new supports as may be required as a result of any cutting, removal, deconstruction, or demolition work performed under this contract. Do not overload structural elements. Provide new supports and reinforcement for existing construction weakened by demolition, deconstruction, or removal work. Repairs, reinforcement, or structural replacement require approval by the Contracting Officer prior to performing such work.

1.3.1 Existing Construction Limits and Protection

Do not disturb existing construction beyond the extent indicated or necessary for installation of new construction. Provide temporary shoring and bracing for support of building components to prevent settlement or other movement. Provide protective measures to control accumulation and migration of dust and dirt in all work areas. Remove snow, dust, dirt, and debris from work areas daily.

1.3.2 Weather Protection

For portions of the building to remain, protect building interior and materials and equipment from the weather at all times. Where removal of existing roofing is necessary to accomplish work, have materials and workmen ready to provide adequate and temporary covering of exposed areas.

1.3.3 Trees

Protect trees within the project site which might be damaged during demolition or deconstruction, and which are indicated to be left in place, by a 6 foot high fence. Erect and secure fence a minimum of 5 feet from the trunk of individual trees or follow the outer perimeter of branches or clumps of trees. Replace any tree designated to remain that is damaged during the work under this contract with like-kind or as approved by the Contracting Officer.

1.3.4 Utility Service

Maintain existing utilities indicated to stay in service and protect against damage during demolition and deconstruction operations.

1.3.5 Facilities

Protect electrical and mechanical services and utilities. Where removal of existing utilities and pavement is specified or indicated, provide approved barricades, temporary covering of exposed areas, and temporary services or connections for electrical and mechanical utilities. Floors, roofs, walls, columns, pilasters, and other structural components that are designed and constructed to stand without lateral support or shoring, and are determined to be in stable condition, must remain standing without additional bracing, shoring, or lateral support until demolished or deconstructed, unless directed otherwise by the Contracting Officer. Ensure that no elements determined to be unstable are left unsupported and place and secure bracing, shoring, or lateral supports as may be required as a result of any cutting, removal, deconstruction, or demolition work performed under this contract.

1.4 BURNING

The use of burning at the project site for the disposal of refuse and debris will not be permitted.

1.5 AVAILABILITY OF WORK AREAS

See 01 11 00 SUMMARY OF WORK, paragraph 1.5, for phasing requirements.

1.6 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Demolition Plan; G
Deconstruction Plan; G
Existing Conditions

SD-07 Certificates

Notification

SD-11 Closeout Submittals

Receipts

1.7 QUALITY ASSURANCE

Submit timely notification of demolition and deconstruction and renovation projects to Federal, State, regional, and local authorities in accordance with 40 CFR 61, Subpart M. Notify the State's environmental protection agency and the Contracting Officer in writing 10 working days prior to the commencement of work in accordance with 40 CFR 61, Subpart M. Comply with federal, state, and local hauling and disposal regulations. In addition to the requirements of the "Contract Clauses," conform to the safety requirements contained in ASSP A10.6. Comply with the Environmental Protection Agency requirements specified. Use of explosives will be permitted.

1.7.1 Dust and Debris Control

Prevent the spread of dust and debris to occupied portions of the building and avoid the creation of a nuisance or hazard in the surrounding area. Do not use water if it results in hazardous or objectionable conditions such as, but not limited to, ice, flooding, or pollution. Vacuum and dust the work area daily.

1.8 PROTECTION

1.8.1 Protection of Personnel

Before, during and after the demolition and deconstruction work continuously evaluate the condition of the structure being demolished and deconstructed and take immediate action to protect all personnel working in and around the project site. No area, section, or component of floors, roofs, walls, columns, pilasters, or other structural element will be allowed to be left standing without sufficient bracing, shoring, or lateral support to prevent collapse or failure while workmen remove debris or perform other work in the immediate area.

1.9 FOREIGN OBJECT DAMAGE (FOD)

Aircraft and aircraft engines are subject to FOD from debris and waste material lying on airfield pavements. Remove all such materials that may appear on operational aircraft pavements due to the Contractor's operations. If necessary, the Contracting Officer may require the Contractor to install a temporary barricade at the Contractor's expense to control the spread of FOD potential debris. The barricade shall include a fence covered with a fabric designed to stop the spread of debris. Anchor the fence and fabric to prevent displacement by winds. Remove barricade when no longer required.

1.10 RELOCATIONS

Perform the removal and reinstallation of relocated items as indicated with workmen skilled in the trades involved. Repair or replace items to be relocated which are damaged by the Contractor with new undamaged items as approved by the Contracting Officer.

1.11 EXISTING CONDITIONS

Before beginning any demolition or deconstruction work, survey the site and examine the drawings and specifications to determine the extent of the work. Record existing conditions in the presence of the Contracting Officer showing the condition of structures and other facilities adjacent to areas of alteration or removal. Photographs sized 4 inch will be acceptable as a record of existing conditions. Include in the record the elevation of the top of foundation walls, finish floor elevations, possible conflicting electrical conduits, plumbing lines, alarms systems, the location and extent of existing cracks and other damage and description of surface conditions that exist prior to before starting work. It is the Contractor's responsibility to verify and document all required outages which will be required during the course of work, and to note these outages on the record document. Submit survey results.

PART 2 PRODUCTS

2.1 FILL MATERIAL

a. Comply with excavating, backfilling, and compacting procedures for soils used as backfill material to fill basements, voids, depressions or excavations resulting from demolition or deconstruction of structures.

PART 3 EXECUTION

3.1 EXISTING FACILITIES TO BE REMOVED

Inspect and evaluate existing structures onsite for reuse. Existing construction scheduled to be removed for reuse shall be disassembled. Dismantled and removed materials are to be separated, set aside, and prepared as specified, and stored or delivered to a collection point for reuse, remanufacture, recycling, or other disposal, as specified. Materials shall be designated for reuse onsite whenever possible.

3.1.1 Structures

a. Remove existing structures indicated to be removed to three (3) feet below lowest floor. Break up basement slabs to permit drainage.

b. Deconstruct structures in a systematic manner from the top of the structure to the ground. Complete demolition work above each tier or floor before the supporting members on the lower level are disturbed. Demolish concrete and masonry walls in small sections. Remove structural framing members and lower to ground by means of derricks, platforms hoists, or other suitable methods as approved by the Contracting Officer.

c. Locate demolition and deconstruction equipment throughout the structure and remove materials so as to not impose excessive loads to supporting walls, floors, or framing.

3.1.2 Utilities and Related Equipment

3.1.2.1 General Requirements

Do not interrupt existing utilities serving occupied or used facilities, except when authorized in writing by the Contracting Officer. Do not interrupt existing utilities serving facilities occupied and used by the Government except when approved in writing and then only after temporary utility services have been approved and provided. Do not begin demolition or deconstruction work until all utility disconnections have been made. Shut off and cap utilities for future use, as indicated.

3.1.2.2 Disconnecting Existing Utilities

Remove existing utilities uncovered by work and terminate in a manner conforming to the nationally recognized code covering the specific utility and approved by the Contracting Officer. When utility lines are encountered but are not indicated on the drawings, notify the Contracting Officer prior to further work in that area. Remove meters and related equipment and deliver to a location in accordance with instructions of the Contracting Officer.

3.1.3 Paving and Slabs

Remove sawcut concrete slabs to a depth of 36 inches below existing adjacent floor level. Pavement and slabs not to be used in this project shall be removed from the Installation at Contractor's expense.

3.1.4 Masonry

Sawcut and remove masonry so as to prevent damage to surfaces to remain, to removed materials being salvaged and to facilitate the installation of new work. Where new masonry adjoins existing, the new work shall abut or tie into the existing construction. Provide square, straight edges and corners where existing masonry adjoins new work and other locations.

3.1.5 Concrete

Saw concrete along straight lines to a depth of a minimum 2 inch. Make each cut in walls perpendicular to the face and in alignment with the cut in the opposite face. Break out the remainder of the concrete provided that the broken area is concealed in the finished work, and the remaining concrete is sound. At locations where the broken face cannot be concealed, grind smooth or saw cut entirely through the concrete.

3.1.6 Structural Steel

Dismantle structural steel at field connections and in a manner that will prevent bending or damage. Salvage for recycle structural steel, steel joists, girders, angles, plates, columns and shapes. Flame-cutting torches are permitted when other methods of dismantling are not practical. Transport steel joists and girders as whole units and not dismantled. Transport structural steel shapes to a designated recycling facility, stacked according to size, type of member and length, and stored off the ground, protected from the weather.

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3.1.7 Miscellaneous Metal

Salvage shop-fabricated items such as access doors and frames, steel gratings, metal ladders, wire mesh partitions, metal railings, metal windows and similar items as whole units. Salvage light-gage and cold-formed metal framing, such as steel studs, steel trusses, metal gutters, roofing and siding, metal toilet partitions, toilet accessories and similar items. Scrap metal and wiring shall be recycled at the WPAFB Recycling Center. Recycle scrap metal as part of demolition and deconstruction operations. Provide separate containers to collect scrap metal and transport to a scrap metal collection or recycling facility, in accordance with the Waste Management Plan.

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3.1.8 Carpet

Remove existing carpet for reclamation in accordance with manufacturer recommendations and as follows. Remove used carpet in large pieces, roll tightly, and pack neatly in a container. Remove adhesive according to recommendations of the Carpet and Rug Institute (CRI). Adhesive removal solvents shall comply with CRI 104/CRI 105. Recycle removed carpet cushion.

3.1.9 Acoustic Ceiling Tile

Remove, neatly stack, and recycle acoustic ceiling tiles. Recycling may be available with manufacturer. Otherwise, priority shall be given to a local recycling organization. Recycling is not required if the tiles contain or may have been exposed to asbestos material.

3.1.10 Patching

Where removals leave holes and damaged surfaces exposed in the finished work, patch and repair these holes and damaged surfaces to match adjacent finished surfaces, using on-site materials when available. Where new work is to be applied to existing surfaces, perform removals and patching in a manner to produce surfaces suitable for receiving new work. Finished surfaces of patched area shall be flush with the adjacent existing surface and shall match the existing adjacent surface as closely as possible as to texture and finish. Patching shall be as specified and indicated, and shall include:

a. Concrete and Masonry: Completely fill holes and depressions, left as a result of removals in existing masonry walls to remain, with an approved masonry patching material, applied in accordance with the manufacturer's printed instructions.

b. Where existing partitions have been removed leaving damaged or missing resilient tile flooring, patch to match the existing floor tile.

3.1.11 Air Conditioning Equipment

Remove air conditioning, refrigeration, and other equipment containing refrigerants without releasing chlorofluorocarbon refrigerants to the atmosphere in accordance with the Clean Air Act Amendment of 1990. Recover all refrigerants prior to removing air conditioning, refrigeration, and other equipment containing refrigerants and dispose of in accordance with the paragraph entitled "Disposal of Ozone Depleting Substance (ODS)."

3.1.12 Cylinders and Canisters

Remove all fire suppression system cylinders and canisters and dispose of in accordance with the paragraph entitled "Disposal of Ozone Depleting Substance (ODS)."

3.1.13 Locksets on Swinging Doors

Remove all locksets from all swinging doors indicated to be removed and disposed of. Deliver the locksets and related items to a designated location for receipt by the Contracting Officer after removal.

3.1.14 Mechanical Equipment and Fixtures

Disconnect mechanical hardware at the nearest connection to existing services to remain, unless otherwise noted. Disconnect mechanical equipment and fixtures at fittings. Remove service valves attached to the unit. Salvage each item of equipment and fixtures as a whole unit; listed, indexed, tagged, and stored. Salvage each unit with its normal operating auxiliary equipment. Transport salvaged equipment and fixtures, including motors and machines, to a designated storage area as directed by the Contracting Officer. Do not remove equipment until approved. Do not offer low-efficiency equipment for reuse.

3.1.14.1 Preparation for Storage

Remove water, dirt, dust, and foreign matter from units; tanks, piping and fixtures shall be drained; interiors, if previously used to store flammable, explosive, or other dangerous liquids, shall be steam cleaned. Seal openings with caps, plates, or plugs. Secure motors attached by flexible connections to the unit. Change lubricating systems with the proper oil or grease.

3.1.14.2 Piping

Disconnect piping at unions, flanges and valves, and fittings as required to reduce the pipe into straight lengths for practical storage. Store salvaged piping according to size and type. If the piping that remains can become pressurized due to upstream valve failure, end caps, blind flanges, or other types of plugs or fittings with a pressure gage and bleed valve shall be attached to the open end of the pipe to ensure positive leak control. Carefully dismantle piping that previously contained gas, gasoline, oil, or other dangerous fluids, with precautions taken to prevent injury to persons and property. Store piping outdoors until all fumes and residues are removed. Box prefabricated supports, hangers, plates, valves, and specialty items according to size and type. Wrap sprinkler heads individually in plastic bags before boxing. Classify piping not designated for salvage, or not reusable, as scrap metal.

3.1.14.3 Ducts

Classify removed duct work as scrap metal.

3.1.14.4 Fixtures, Motors and Machines

Remove and salvage fixtures, motors and machines associated with plumbing, heating, air conditioning, refrigeration, and other mechanical system installations. Salvage, box and store auxiliary units and accessories with the main motor and machines. Tag salvaged items for identification, storage, and protection from damage. Classify broken, damaged, or otherwise unserviceable units and not caused to be broken, damaged, or otherwise unserviceable as debris to be disposed of by the Contractor.

3.1.15 Elevators and Hoists

Remove elevators, hoists, and similar conveying equipment and salvage as whole units, to the most practical extent. Remove and prepare items for salvage without damage to any of the various parts. Salvage and store rails for structural steel with the equipment as an integral part of the unit.

3.2 DISPOSITION OF MATERIAL

3.2.1 Reuse of Equipment

Remove and store equipment listed in the Demolition Deconstruction Plan to be reused or relocated to prevent damage, and reinstall as the work progresses. Coordinate the re-use of materials and equipment with the re-use requirements in accordance with Section 01 35 43 GENERAL ENVIRONMENTAL PROTECTION REQUIREMENTS (WPAFB). Capture re-use of materials in the diversion calculations for the project.

3.2.2 Salvaged Equipment

Remove equipment that are to remain the property of the Government, and deliver to a storage site at the facility.

a. Salvage items to the maximum extent possible.

b. Remove salvaged items to remain the property of the Government in a manner to prevent damage, and packed or crated to protect the items from damage while in storage or during shipment. Items damaged during removal or storage must be repaired or replaced to match existing items. Properly identify the contents of containers.

c. Government to dismantle, store, and reinstall existing video wall and all associated equipment. Coordinate with phasing plan under 01 11 00 SUMMARY OF WORK, paragraph 1.5.

3.2.3 Disposal of Ozone Depleting Substance (ODS)

Class I and Class II ODS are defined in Section, 602(a) and (b), of The Clean Air Act. Prevent discharge of Class I and Class II ODS to the atmosphere. Place recovered ODS in cylinders meeting AHRI Guideline K suitable for the type ODS (filled to no more than 80 percent capacity) and provide appropriate labeling. Recovered ODS shall be removed from Government property and disposed of in accordance with 40 CFR 82. Products, equipment and appliances containing ODS in a sealed, self-contained system (e.g. residential refrigerators and window air conditioners) shall be disposed of in accordance with 40 CFR 82. Submit Receipts or bills of lading, as specified. Submit a shipping receipt or bill of lading for all containers of ozone depleting substance (ODS) shipped to the Defense Depot, Richmond, Virginia.

3.2.3.1 Special Instructions

No more than one type of ODS is permitted in each container. A warning/hazardous label shall be applied to the containers in accordance with Department of Transportation regulations. All cylinders including but not limited to fire extinguishers, spheres, or canisters containing an ODS shall have a tag with the following information:

a. Activity name and unit identification code

b. Activity point of contact and phone number

c. Type of ODS and pounds of ODS contained

d. Date of shipment

e. National stock number (for information, call (804) 279-4525).

3.2.3.2 Fire Suppression Containers

Deactivate fire suppression system cylinders and canisters with electrical charges or initiators prior to shipment. Also, safety caps must be used to cover exposed actuation mechanisms and discharge ports on these special cylinders.

3.2.4 Transportation Guidance

Ship all ODS containers in accordance with MIL-STD-129, DLA 4145.25 (also referenced one of the following: Army Regulation 700-68, Naval Supply Instruction 4440.128C, Marine Corps Order 10330.2C, and Air Force Regulation 67-12), 49 CFR 173.301, and DOD 4000.25-1-M.

3.3 CLEANUP

Remove debris and rubbish from basement and similar excavations. Remove and transport the debris in a manner that prevents spillage on streets or adjacent areas. Apply local regulations regarding hauling and disposal.

3.4 DISPOSAL OF REMOVED MATERIALS

3.4.1 Regulation of Removed Materials

Dispose of debris, rubbish, scrap, and other nonsalvageable materials resulting from removal operations with all applicable federal, state and local regulations as contractually specified in the Waste Management Plan. Storage of removed materials on the project site is prohibited.

3.4.2 Removal from Government Property

Transport waste materials removed from demolished and deconstructed structures, except waste soil, from Government property for legal disposal. Dispose of waste soil as directed.

3.5 REUSE OF SALVAGED ITEMS

Recondition salvaged materials and equipment designated for reuse before installation. Replace items damaged during removal and salvage operations or restore them as necessary to usable condition.

-- End of Section --

SECTION 05 51 00

METAL STAIRS
02/17, CHG 1: 05/17
AMENDMENT 0002

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN INSTITUTE OF STEEL CONSTRUCTION (AISC)

AISC 360	(2016) Specification for Structural Steel Buildings

AMERICAN IRON AND STEEL INSTITUTE (AISI)

AISC/AISI 121	(2007) Standard Definitions for Use in the Design of Steel Structures

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B18.2.1	(2012; Errata 2013) Square and Hex Bolts and Screws (Inch Series)
ASME B18.6.1	(2016) Wood Screws (Inch Series)
ASME B18.6.3	(2013; R 2017) Machine Screws, Tapping Screws, and Machine Drive Screws (Inch Series)
ASME B18.21.1	(2009; R 2016) Washers: Helical Spring-Lock, Tooth Lock, and Plain Washers (Inch Series)

AMERICAN WELDING SOCIETY (AWS)

AWS D1.1/D1.1M	(2020; Errata 1 2021) Structural Welding Code - Steel

ASTM INTERNATIONAL (ASTM)

ASTM A6/A6M	(2021) Standard Specification for General Requirements for Rolled Structural Steel Bars, Plates, Shapes, and Sheet Piling
ASTM A29/A29M	(2020) Standard Specification for General Requirements for Steel Bars, Carbon and Alloy, Hot-Wrought
ASTM A36/A36M	(2019) Standard Specification for Carbon Structural Steel
ASTM A53/A53M	(2022) Standard Specification for Pipe,

	Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless
ASTM A108	(2013) Standard Specification for Steel Bar, Carbon and Alloy, Cold-Finished
ASTM A123/A123M	(2017) Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products
ASTM A153/A153M	(2016a) Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware
ASTM A283/A283M	(2013) Standard Specification for Low and Intermediate Tensile Strength Carbon Steel Plates
ASTM A307	(2021) Standard Specification for Carbon Steel Bolts, Studs, and Threaded Rod 60 000 PSI Tensile Strength
ASTM A512	(2006; R 2012) Standard Specification for Cold-Drawn Buttweld Carbon Steel Mechanical Tubing
ASTM A568/A568M	(2019a) Standard Specification for Steel, Sheet, Carbon, Structural, and High-Strength, Low-Alloy, Hot-Rolled and Cold-Rolled, General Requirements for
ASTM A575	(2020) Standard Specification for Steel Bars, Carbon, Merchant Quality, M-Grades
ASTM A653/A653M	(2022) Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process
ASTM A924/A924M	(2020) Standard Specification for General Requirements for Steel Sheet, Metallic-Coated by the Hot-Dip Process
ASTM A1008/A1008M	(2021a) Standard Specification for Steel, Sheet, Cold-Rolled, Carbon, Structural, High-Strength Low-Alloy, High-Strength Low-Alloy with Improved Formability, Solution Hardened, and Bake Hardenable
ASTM A1011/A1011M	(2018a) Standard Specification for Steel Sheet and Strip, Hot-Rolled, Carbon, Structural, High-Strength Low-Alloy, High-Strength Low-Alloy with Improved Formability, and Ultra-High Strength

NATIONAL ASSOCIATION OF ARCHITECTURAL METAL MANUFACTURERS (NAAMM)

NAAMM MBG 531	(2017) Metal Bar Grating Manual

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 101	(2021) Life Safety Code

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Iron and Steel Hardware

Steel Shapes, Plates, Bars, and Strips

Metal Stair System

SD-03 Product Data

Structural Steel Plates, Shapes, and Bars

Structural Steel Tubing

Hot-Rolled Carbon Steel Sheets and Strips

Cold-Finished Steel Bars

Hot-Rolled Carbon Steel Bars

Cold-Rolled Carbon Steel Sheets

Galvanized Carbon Steel Sheets

Cold-Drawn Steel Tubing

Protective Coating

Steel Pan Stairs

Steel Stairs

SD-08 Manufacturer's Instructions

Structural Steel Plates, Shapes, and Bars

Structural Steel Tubing

Hot-Rolled Carbon Steel Sheets and Strips

Cold-Finished Steel Bars

Hot-Rolled Carbon Steel Bars

Cold-Rolled Carbon Steel Sheets

Galvanized Carbon Steel Sheets

Cold-Drawn Steel Tubing

Protective Coating

1.3 QUALITY CONTROL

1.3.1 Qualifications for Welding Work

Certify welder qualification by tests in accordance with AWS D1.1/D1.1M, or under an equivalent approved qualification test. In addition, perform tests on test pieces in positions and with clearances equivalent to those actually encountered. If a test weld fails to meet requirements, ensure that two test welds are retested immediately and that each test weld is made and passes. Failure in the immediate retest requires that the welder be retested after further practice or training and a complete set of test welds be made.

PART 2 PRODUCTS

2.1 SYSTEM DESCRIPTION

Submit complete and detailed fabrication drawings for all iron and steel hardware, and for all steel shapes, plates, bars, and strips used in accordance with the design specifications referenced in this section. Submit accompanying calculations stamped by a licensed professional engineer. The calculations shall include reactions to base structure to allow for designer review.

2.2 FABRICATION

Preassemble items in the shop to the greatest extent possible. Disassemble units only to the extent necessary for shipping and handling. Clearly mark units for reassembly and coordinated installation.

For the fabrication of work exposed to view, use only materials that are smooth and free of surface blemishes, including pitting, seam marks, roller marks, rolled trade names, and roughness. Remove blemishes by grinding, or by welding and grinding, before cleaning and treating surfaces and applying surface finishes, including zinc coatings.

2.2.1 General Fabrication

Prepare and submit metal stair system shop drawings with detailed plans and elevations at scales not less than 1 inch to 1 foot and with details of sections and connections at scales not less than 3 inches to 1 foot. Also detail the placement drawings, diagrams, and templates for installation of anchorages, including concrete inserts, anchor bolts, and miscellaneous metal items having integral anchorage devices.

Use materials of size and thicknesses indicated or, if not indicated, of the size and thickness necessary to produce a finished product that is strong enough and durable enough for its intended use. Work the materials to the dimensions indicated on approved detail drawings, using proven methods of fabrication and support. Use the type of materials indicated or specified for the various components of work.

Form exposed work true to line and level, with accurate angles and surfaces and with straight sharp edges. Ease exposed edges to a radius of approximately 1/32 inch, and bend metal corners to the smallest radius possible without causing grain separation or otherwise impairing the work.

Continuously weld corners and seams in accordance with the recommendations of AWS D1.1/D1.1M. Grind exposed welds smooth and flush to match and blend with adjoining surfaces.

Form exposed connections with hairline joints that are flush and smooth, using concealed fasteners wherever possible. Use exposed fasteners of the type indicated or, if not indicated, use Phillips flat-head (countersunk) screws or bolts.

Provide and coordinate anchorage of the type indicated for the supporting structure. Fabricate anchoring devices, and space them as indicated and as necessary to provide adequate support for the intended use of the work.

Use hot-rolled steel bars for work fabricated from bar stock unless work is indicated or specified as fabricated from cold-finished or cold-rolled stock.

2.2.2 Steel Pan Stairs

2.2.2.1 General

Joining pieces by welding. Fabricate units so that bolts and other fastenings do not appear on finished surfaces. Make joints true and tight, and connections between parts lighttight. Grind continuous welds smooth where exposed.

Construct metal stair units to sizes and arrangements indicated to support a minimum live load of 100 pounds per square foot. Provide framing, hangers, columns, struts, clips, brackets, bearing plates, and other components as required for the support of stairs and platforms.

2.2.2.2 Stair Framing

Fabricate stringers of structural steel channels, or plates, or a combination thereof as indicated. Provide closures for exposed ends of strings.

Construct platforms of structural steel channel headers and miscellaneous framing members as indicated. Bolt headers to stringers and newels, and bolt framing members to stringers and headers.

2.2.2.3 Riser, Subtread, and Subplatform Metal Pans

Form metal pans of 0.1084-inch (12-gage) structural steel sheets, conforming to ASTM A1011/A1011M, Grade 36. Shape the pans to the configuration indicated.

Construct risers and subtread metal pans with steel angle supporting brackets, of the size indicated, welded to stringers. Secure metal pans to brackets with rivets or welds. Secure subplatform metal pans to platform frames with welds.

AMENDMENT 0002
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2.2.2.4 Not Used
AMENDMENT 0002
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2.2.2.5 Steel Floor Plate Treads and Platforms

Provide raised-pattern steel floor plate fabricated from steel complying with ASTM A36/A36M. Provide the pattern indicated or, if not indicated, as selected from the manufacturer's standard patterns.

Form treads of 1/4-inch thick steel floor plate with integral nosing and back-edge stiffener. Weld steel supporting brackets to strings, and weld treads to brackets.

2.2.2.6 Steel Framing for Concrete Stairs

When necessary, modify fabricated units to fit actual dimensions of the supporting structure. Join steel components by welding. Provide 14-gage steel risers unless otherwise indicated. Arrange components to receive finish materials as indicated.

2.2.3 Floor Grating Treads and Platforms

Provide floor grating treads and platforms conforming to ASTM A6/A6M, ASTM A29/A29M and NAAMM MBG 531, "Metal Bar Grating Manual." Provide the pattern, spacing, and bar sizes as indicated:

a. Galvanized finish, conforming to ASTM A123/A123M.

b. Manufacturer's baked-on primer for painted finishes.

Fabricate grating treads with steel plate nosings on one edge and with steel angle or steel plate carriers at each end for string connections. Secure treads to strings with bolts.

Match the nosings of grating platforms with the nosing of grating treads at landings. Provide toeplates where the open-sided edges of floor grating meet platform framing members.

2.2.4 Protective Coating

Shop-prime steelwork as indicated in accordance with AISC/AISI 121 , except surfaces of steel encased in concrete; welded surfaces; high-strength, bolt-connected surfaces; and surfaces of crane rails.

2.3 COMPONENTS

2.3.1 Steel Stairs

Provide steel stairs complete with stringers, steel-plate treads and risers, landings, columns, handrails, and necessary bolts and other fastenings. Shop-paint steel stairs and accessories.

2.3.1.1 Design Loads

Design stairs to sustain a live load of not less than 100 pounds per square foot, or a concentrated load of 300 lbs applied where it is most critical. Except for a commercial product, design and fabricate steel stairs to conform to AISC 360.

Refer to drawing S-202 for the design basis concept for supporting the stairs from the base structure.

Design stairs with slotted connections or sliding bearing supports to accommodate seismic movement per ASCE 7-16 Section 13.5.10. Accommodate 1 inch of movement in each direction for stairs from Basement to First Floor and from Third Floor to Lid. Accommodate 2.5 inches of movement in each direction for stairs from First Floor to Second Floor and from Second Floor to Third Floor. Design fire stairs to conform to NFPA 101.

2.3.1.2 Materials

Provide steel stairs of welded construction except that bolts may be used where welding is not practicable. Do not use screw or screw-type connections.

a. Structural Steel: ASTM A36/A36M.

b. Gratings for Treads and Landings: NAAMM MBG 531

c. Support steel floor plate on angle cleats welded to stringers or treads with integral cleats, welded or bolted to the stringer. Close exposed ends.

e. Before fabrication, obtain necessary field measurements and verify drawing dimensions.

f. Clean metal surfaces free of mill scale, flake rust, and rust pitting before shop finishing. Weld permanent connections. Finish welds flush and smooth on surfaces that will be exposed after installation.

2.3.2 Soffit Clips

Provide clips with holes for attaching metal furring for plastered soffits. Space the clips not more than 12 inches on center, and weld them to stair treads and platforms as required.

2.3.3 Fasteners

Select galvanized zinc-coated fasteners conforming to ASTM A153/A153M for exterior applications or where the fasteners are built into exterior walls or floor systems. Select the fasteners for the type, grade, and class required for the installation of steel stair items:

a. Standard/regular hexagon-head bolts and nuts, conforming to ASTM A307, Grade A.

b. Square-head lag bolts conforming to ASME B18.2.1.

c. Cadmium-plated steel machine screws, conforming to ASME B18.6.3.

d. Flat-head carbon steel wood screws, conforming to ASME B18.6.1.

e. Plain, round, general-assembly-grade, carbon steel washers, conforming to ASME B18.21.1.

f. Helical-spring, carbon steel lockwashers, conforming to ASME B18.2.1.

2.4 MATERIALS

2.4.1 Structural Steel Plates, Shapes and Bars

Structural size shapes and plates, conforming to ASTM A36/A36M, unless otherwise noted, except bent or cold-formed plates.

Steel plates - bent or cold-formed, conforming to ASTM A283/A283M, Grade C.

Steel bars and bar-size shapes, conforming to ASTM A36/A36M, unless otherwise noted for steel bars and bar-size shapes.

2.4.2 Structural Steel Tubing

Provide the following:

Structural steel tubing, hot-formed, welded or seamless, conforming to A1085 .

2.4.3 Hot-Rolled Carbon Steel Bars

Provide the following:

a. Hot-rolled carbon steel bars and bar-size shapes, conforming to ASTM A575, grade as selected by the fabricator.

2.4.4 Cold-Finished Steel Bars

Provide the following:

a. Cold-finished steel bars conforming to ASTM A108, grade as selected by the fabricator.

2.4.5 Hot-Rolled Carbon Steel Sheets and Strips

Provide the following:

a. Hot-rolled carbon sheets and strips conforming to ASTM A568/A568M and ASTM A1011/A1011M, pickled and oiled.

2.4.6 Cold-Rolled Carbon Steel Sheets

Provide the following:

a. Cold-rolled carbon steel sheets conforming to ASTM A1008/A1008M.

2.4.7 Galvanized Carbon Steel Sheets

Provide the following:

a. Galvanized carbon steel sheets conforming to ASTM A653/A653M, with galvanizing conforming to ASTM A653/A653M and ASTM A924/A924M.

2.4.8 Cold-Drawn Steel Tubing

Provide the following:

a. Cold-drawn steel tubing conforming to ASTM A512, sunk drawn, butt-welded, cold-finished, and stress-relieved.

2.4.9 Steel Pipe

Provide the following:

a. Steel pipe conforming to ASTM A53/A53M, type as selected, Grade B; primed finish, unless galvanizing is required; standard weight (Schedule 40).

PART 3 EXECUTION

3.1 PREPARATION

Clean surfaces thoroughly before installation. Prepare surfaces using the methods recommended by the manufacturer for achieving the best result for the substrate under the project conditions. Examine materials upon arrival at site. Notify the carrier and manufacturer of any damage.

Protect installed products until completion of project. Touch up, repair or replace, damaged products before substantial completion

3.2 INSTALLATION

Install in accordance with the manufacturer's instructions and approved submittals. Install in proper relationship with adjacent construction.

Install items at locations indicated, according to the manufacturer's instructions. Verify all measurements and take all field measurements necessary before fabrication. Ensure that exposed fastenings are compatible with generally match the color and finish of, and harmonize with the material to which they are applied. Include materials and parts necessary to complete each item, even though such work is not definitely shown or specified. Poor matching of holes for fasteners is cause for rejection. Conceal fastenings where practicable. Select thickness of metal and details of assembly and supports that adequately strengthen and stiffen the construction. Form joints exposed to the weather to exclude water.

3.2.1 Field Preparation

Remove rust-preventive coating just before field erection, using a remover approved by the coating manufacturer. Provide surfaces, when assembled, free of rust, grease, dirt and other foreign matter.

3.2.2 Field Welding

Comply with AWS D1.1/D1.1M in executing manual shielded-metal arc welding, (for appearance and quality of new welds) and in correcting existing welding.

3.2.3 Safety Nosings

Completely embed nosing in concrete before the initial set of the concrete occurs and finish flush with the top of the concrete surface.

3.2.4 Touchup Painting

Immediately after installation, clean all field welds, bolted connections, and abraded areas of the shop-painted material, and repaint exposed areas with the same paint used for shop painting. Apply paint by brush or spray to provide a minimum dry-film thickness of 2 mils.

-- End of Section --

SECTION 09 90 00

PAINTS AND COATINGS
02/21
AMENDMENT 0002

PART 1 GENERAL

1.1 RELATED REQUIREMENTS

1.1.1 Painting Included

Where a space or surface is indicated to be painted, include the following unless indicated otherwise.

a. Surfaces behind portable objects and surface mounted articles readily detachable by removal of fasteners, such as screws and bolts.

b. New factory finished surfaces that require identification or color coding and factory finished surfaces that are damaged during performance of the work.

c. Existing coated surfaces that are damaged during performance of the work.

1.1.1.1 Interior Painting

Includes new surfaces, existing uncoated surfaces, and existing coated surfaces of the building and appurtenances as indicated and existing coated surfaces made bare by cleaning operations. Where a space or surface is indicated to be painted, include the following items, unless indicated otherwise.

a. Exposed columns, girders, beams, joists, and metal deck; and

b. Other contiguous surfaces.

1.1.2 Painting Excluded

Do not paint the following unless indicated otherwise.

a. Surfaces concealed and made inaccessible by panelboards, fixed ductwork, machinery, and equipment fixed in place.

b. Surfaces in concealed spaces. Concealed spaces are defined as enclosed spaces above suspended ceilings, furred spaces, attic spaces, crawl spaces, elevator shafts and chases.

c. Steel to be embedded in concrete.

d. Copper, stainless steel, aluminum, anodized aluminum, brass, and lead except existing coated surfaces.

e. Hardware, fittings, and other factory finished items.

1.1.3 Mechanical and Electrical Painting

Includes field coating of interior and exterior new and existing surfaces.

a. Where a space or surface is indicated to be painted, include the following items unless indicated otherwise.

(1) Exposed piping, conduit, and ductwork;

(2) Supports, hangers, air grilles, and registers;

(3) Miscellaneous metalwork and insulation coverings.

b. Do not paint the following, unless indicated otherwise:

(1) New zinc-coated, aluminum, and copper surfaces under insulation

(2) New aluminum jacket on piping

(3) New interior ferrous piping under insulation.

1.1.3.1 Fire Extinguishing Sprinkler Systems

Clean, pretreat, prime, and paint new fire extinguishing sprinkler systems including valves, piping, conduit, hangers, supports, miscellaneous metalwork, and accessories. Apply coatings to clean, dry surfaces, using clean brushes.

1.1.4 Miscellaneous Painting

1.1.4.1 Lettering

Provide lettering as indicated on the drawings and in other specifications. Samples must be approved before application.

1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN CONFERENCE OF GOVERNMENTAL INDUSTRIAL HYGIENISTS (ACGIH)

ACGIH 0100 — (2017; Suppl 2020) Documentation of the Threshold Limit Values and Biological Exposure Indices

ASTM INTERNATIONAL (ASTM)

ASTM D235 — (2002; R 2012) Mineral Spirits (Petroleum Spirits) (Hydrocarbon Dry Cleaning Solvent)

ASTM D523 — (2014; R 2018) Standard Test Method for Specular Gloss

ASTM D4214 — (2007; R 2015) Standard Test Method for Evaluating the Degree of Chalking of Exterior Paint Films

ASTM D4263 — (1983; R 2018) Standard Test Method for Indicating Moisture in Concrete by the Plastic Sheet Method

ASTM D4444	(2013; R 2018) Standard Test Method for Laboratory Standardization and Calibration of Hand-Held Moisture Meters
ASTM D6386	(2016a) Standard Practice for Preparation of Zinc (Hot-Dip Galvanized) Coated Iron and Steel Product and Hardware Surfaces for Painting
ASTM F1869	(2016a) Standard Test Method for Measuring Moisture Vapor Emission Rate of Concrete Subfloor Using Anhydrous Calcium Chloride

CENTERS FOR DISEASE CONTROL AND PREVENTION (CDC)

Intelligence Bulletin 65	(2013) Occupational Exposure to Carbon Nanotubes and Nanofibers

MASTER PAINTERS INSTITUTE (MPI)

MPI 1	(2012) Aluminum Paint
MPI 2	(2012) Aluminum Heat Resistant Enamel (up to 427 C and 800 F
MPI 3	(2016) Primer, Alkali Resistant, Water Based
MPI 4	(2016) Interior/Exterior Latex Block Filler
MPI 9	(2016) Alkyd, Exterior Gloss (MPI Gloss Level 6)
MPI 17	(2016) Primer, Bonding, Water Based
MPI 19	(2012) Primer, Zinc Rich, Inorganic
MPI 21	(2012) Heat Resistant Coating, (Up to 205°C/402°F), MPI Gloss Level 6
MPI 22	(2012) Aluminum Paint, High Heat (up to 590° C/1100° F)
MPI 23	(2015) Primer, Metal, Surface Tolerant
MPI 27	(2016) Floor Enamel, Alkyd, Gloss (MPI Gloss Level 6)
MPI 31	(2012) Varnish, Polyurethane, Moisture Cured, Gloss (MPI Gloss Level 6)
MPI 39	(2018) Primer, Latex, for Interior Wood
MPI 42	(2012) Textured Coating, Latex, Flat
MPI 44	(2016) Latex, Interior, (MPI Gloss Level 2)
MPI 45	(2016) Primer Sealer, Interior Alkyd

MPI 46	(2016) Undercoat, Enamel, Interior
MPI 47	(2016) Alkyd, Interior, Semi-Gloss (MPI Gloss Level 5)
MPI 48	(2016) Alkyd, Interior, Gloss (MPI Gloss Level 6-7)
MPI 49	(2015) Alkyd, Interior, Flat (MPI Gloss Level 1)
MPI 50	(2015) Primer Sealer, Latex, Interior
MPI 51	(2016) Alkyd, Interior, (MPI Gloss Level 3)2
MPI 52	(2016) Latex, Interior, (MPI Gloss Level 3)
MPI 54	(2016) Latex, Interior, Semi-Gloss (MPI Gloss Level 5)
MPI 56	(2012) Varnish, Interior, Polyurethane, Oil Modified, Gloss
MPI 57	(2012) Varnish, Interior, Polyurethane, Oil Modified, Satin
MPI 59	(2016) Floor Paint, Alkyd, Low Gloss
MPI 60	(2016) Floor Paint, Latex, Low Gloss
MPI 68	(2016) Floor Paint, Latex, Gloss
MPI 71	(2012) Varnish, Polyurethane, Moisture Cured, Flat (MPI Gloss Level 1)
MPI 72	(2016) Polyurethane, Two-Component, Pigmented, Gloss (MPI Gloss Level 6-7)
MPI 76	(2016) Primer, Alkyd, Quick Dry, for Metal
MPI 77	(2015) Epoxy, Gloss
MPI 90	(2012) Stain, Semi-Transparent, for Interior Wood
MPI 95	(2015) Primer, Quick Dry, for Aluminum
MPI 101	(2016) Primer, Epoxy, Anti-Corrosive, for Metal
MPI 107	(2016) Primer, Rust-Inhibitive, Water Based
MPI 116	(2012) Block Filler, Epoxy
MPI 138	(2016) Latex, Interior, High Performance Architectural, (MPI Gloss Level 2)

MPI 139	(2016) Latex, Interior, High Performance Architectural, (MPI Gloss Level 3)
MPI 140	(2016) Latex, Interior, High Performance Architectural, (MPI Gloss Level 4)
MPI 141	(2016) Latex, Interior, High Performance Architectural, Semi-Gloss (MPI Gloss Level 5)
MPI 144	(2016) Latex, Interior, Institutional Low Odor/VOC, (MPI Gloss Level 2)
MPI 145	(2016) Latex, Interior, Institutional Low Odor/VOC, (MPI Gloss Level 3)
MPI 146	(2016) Latex, Interior, Institutional Low Odor/VOC, (MPI Gloss Level 4)
MPI 147	(May 2016) Latex, Interior, Institutional Low Odor/VOC, Semi-Gloss (MPI Gloss Level 5)
MPI 149	(2016) Primer Sealer, Interior, Institutional Low Odor/VOC
MPI 151	(2016) Light Industrial Coating, Interior, Water Based (MPI Gloss Level 3)
MPI 153	(2016) Light Industrial Coating, Interior, Water Based, Semi-Gloss (MPI Gloss Level 5)
MPI 154	(2016) Light Industrial Coating, Interior, Water Based, Gloss (MPI Gloss Level 6)
MPI ASM	(2019) Architectural Painting Specification Manual
MPI GPS-1-14	(2014) Green Performance Standard GPS-1-14
MPI GPS-2-14	(2014) Green Performance Standard GPS-2-14
MPI MRM	(2015) Maintenance Repainting Manual

SOCIETY FOR PROTECTIVE COATINGS (SSPC)

SSPC 7/NACE No.4	(2007) Brush-Off Blast Cleaning
SSPC Glossary	(2011) SSPC Protective Coatings Glossary
SSPC Guide 6	(2015) Guide for Containing Surface Preparation Debris Generated During Paint Removal Operations
SSPC Guide 7	(2015) Guide to the Disposal of Lead-Contaminated Surface Preparation Debris
SSPC PA 1	(2016) Shop, Field, and Maintenance

	Coating of Metals
SSPC SP 1	(2015) Solvent Cleaning
SSPC SP 2	(2018) Hand Tool Cleaning
SSPC SP 3	(2018) Power Tool Cleaning
SSPC SP 6/NACE No.3	(2007) Commercial Blast Cleaning
SSPC SP 10/NACE No. 2	(2007) Near-White Blast Cleaning
SSPC VIS 1	(2002; E 2004) Guide and Reference Photographs for Steel Surfaces Prepared by Dry Abrasive Blast Cleaning
SSPC VIS 3	(2004) Guide and Reference Photographs for Steel Surfaces Prepared by Hand and Power Tool Cleaning
SSPC VIS 4/NACE VIS 7	(1998; E 2000; E 2004) Guide and Reference Photographs for Steel Surfaces Prepared by Waterjetting
SSPC-SP WJ-1/NACE WJ-1	(2012) Clean to Bare Substrate, Waterjet Cleaning of Metals
SSPC-SP WJ-2/NACE WJ-2	(2012) Very Thorough Cleaning, Waterjet Cleaning of Metals
SSPC-SP WJ-3/NACE WJ-3	(2012) Thorough Cleaning, Waterjet Cleaning of Metals
SSPC-SP WJ-4/NACE WJ-4	(2012) Light Cleaning, Waterjet Cleaning of Metals

U.S. DEPARTMENT OF DEFENSE (DOD)

MIL-PRF-680	(2010; Rev C; Notice 1 2015) Degreasing Solvent
MIL-STD-101	(2014; Rev C) Color Code for Pipelines and for Compressed Gas Cylinders

U.S. ENVIRONMENTAL PROTECTION AGENCY (EPA)

EPA Method 24	(2000) Determination of Volatile Matter Content, Water Content, Density, Volume Solids, and Weight Solids of Surface Coatings

U.S. GENERAL SERVICES ADMINISTRATION (GSA)

FED-STD-313	(2018) Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910.1000	Air Contaminants
29 CFR 1910.1001	Asbestos
29 CFR 1910.1025	Lead
29 CFR 1926.62	Lead

1.3 DEFINITIONS

1.3.1 Qualification Testing

Qualification testing is the performance of all test requirements listed in the product specification. This testing is accomplished by MPI to qualify each product for the MPI Approved Product List, and may also be accomplished by Contractor's third-party testing lab if an alternative to Batch Quality Conformance Testing by MPI is desired.

1.3.2 Batch Quality Conformance Testing

Batch quality conformance testing determines that the product provided is the same as the product qualified to the appropriate product specification. This testing must be accomplished by an MPI testing lab.

1.3.3 Coating

SSPC Glossary; (1) A liquid, liquefiable, or mastic composition that is converted to a solid protective, decorative, or functional adherent film after application as a thin layer; (2) Generic term for paint, lacquer, enamel.

1.3.4 DFT or dft

Dry film thickness, the film thickness of the fully cured, dry paint or coating.

1.3.5 DSD

Degree of Surface Degradation, the MPI system of defining degree of surface degradation. Five levels are generically defined under the Assessment sections in the MPI MRM, MPI Maintenance Repainting Manual.

1.3.6 INT

MPI short term designation for an interior coating system.

1.3.7 Loose Paint

Paint or coating that can be removed with a dull putty knife.

1.3.8 mil / mils

The English measurement for 0.001 in or one one-thousandth of an inch.

1.3.9 MPI Gloss Levels

MPI system of defining gloss. Seven gloss levels (G1 to G7) are generically defined under the Evaluation sections of the MPI Manuals. Traditionally, Flat refers to G1/G2, Eggshell refers to G3, Semigloss refers to G5, and Gloss refers to G6.

Gloss levels are defined by MPI as follows:

Gloss Level	Description	Units at 60 degree angle	Units at 80 degree angle
G1	Matte or Flat	0 to 5	10 max
G2	Velvet	0 to 10	10 to 35
G3	Eggshell	10 to 25	10 to 35
G4	Satin	20 to 35	35 min
G5	Semi-Gloss	35 to 70	
G6	Gloss	70 to 85	
G7	High Gloss		

Gloss is tested in accordance with ASTM D523. Historically, the Government has used Flat (G1 / G2), Eggshell (G3), Semi-Gloss (G5), and Gloss (G6).

1.3.10 MPI System Number

The MPI coating system number in each MPI Division found in either the MPI Architectural Painting Specification Manual or the Maintenance Repainting Manual and defined as an exterior (EXT/REX) or interior system (INT/RIN).

1.3.11 Paint

SSPC Glossary; (1) Any pigmented liquid, liquefiable, or mastic composition designed for application to a substrate in a thin layer that is converted to an opaque solid film after application. Used for protection, decoration, identification, or to serve some other functional purposes; (2) Application of a coating material.

1.3.12 RIN

MPI short term designation for an interior coating system used in repainting projects or over existing coating systems.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

Samples of specified materials may be taken and tested for compliance with specification requirements.

SD-02 Shop Drawings

Piping Identification

SD-03 Product Data

Coating

Product Data Sheets

SD-04 Samples

Color

Textured Wall Coating System

SD-07 Certificates

Qualification Testing laboratory for coatings

Indoor Air Quality for Paints and Primers

Indoor Air Quality for Consolidated Latex Paints

SD-08 Manufacturer's Instructions

Application Instructions

Mixing

Manufacturer's Safety Data Sheets

SD-10 Operation and Maintenance Data

Coatings, Data Package 1

1.5 QUALITY ASSURANCE

1.5.1 Regulatory Requirements

1.5.1.1 Environmental Protection

In addition to requirements specified elsewhere for environmental protection, provide coating materials that conform to the restrictions of the local Air Pollution Control District and regional jurisdiction. Notify Contracting Officer of any paint specified herein which fails to conform.

1.5.1.2 Lead Content

Do not use coatings having a lead content over 0.06 percent by weight of nonvolatile content.

1.5.1.3 Chromate Content

Do not use coatings containing zinc-chromate or strontium-chromate.

1.5.1.4 Asbestos Content

Provide asbestos-free materials.

1.5.1.5   Mercury Content

Provide materials free of mercury or mercury compounds.

1.5.1.6   Silica

Provide abrasive blast media containing no free crystalline silica.

1.5.1.7   Human Carcinogens

Provide materials that do not contain ACGIH 0100 confirmed human carcinogens (A1) or suspected human carcinogens (A2).

1.5.1.8   Carbon Based Fibers / Tubes

Materials must not contain carbon based fibers such as carbon nanotubes or carbon nanofibers. Intelligence Bulletin 65 ranks toxicity of carbon nanotubes on a par with asbestos.

1.5.2   Coating Contractor's Qualification

Submit the name, address, telephone number, and e-mail address of the Contractor that will be performing all surface preparation and coating application. Submit evidence that key personnel have successfully performed surface preparation and application of coatings on on a minimum of three similar projects within the past three years. List information by individual and include the following:

a. Name of individual and proposed position for this work.

b. Information about each previous assignment including:

Position or responsibility

Employer (if other than the Contractor)

Name of facility owner

Mailing address and telephone number of facility owner

Name of individual in facility owner's organization who can be contacted as a reference

Location, size and description of structure

Dates work was carried out

Description of work carried out on structure

AMENDMENT 0002
****
1.5.3   Not Used

AMENDMENT 0002
****

1.5.4 Approved Products List

The current MPI, "Approved Product List" which lists paint by brand, label, product name and product code as of the date of Contract award, will be used to determine compliance with the submittal requirements of this specification. The Contractor may choose to use a subsequent MPI "Approved Product List", however, only one list may be used for the entire Contract and each coating system is to be from a single manufacturer. Provide all coats on a particular substrate from a single manufacturer. No variation from the MPI Approved Products List is acceptable.

1.5.5 Paints and Coatings Indoor Air Quality Certifications

Provide paint and coating products certified to meet indoor air quality requirements by MPI GPS-1-14, MPI GPS-2-14 or provide certification by other third-party programs. Provide current product certification documentation from certification body.

Provide certification of Indoor Air Quality for Paints and Primers. Provide certification of Indoor Air Quality for Consolidated Latex Paints. Submit required indoor air quality certifications in one submittal package.

1.5.6 Field Samples and Tests

The Contracting Officer may choose up to two coatings that have been delivered to the site to be tested at no cost to the Government. Take samples of each chosen product as specified in the paragraph SAMPLING PROCEDURE. Test each chosen product as specified in the paragraph TESTING PROCEDURE. Remove products from the job site which do not conform, and replace with new products that conform to the referenced specification. Test replacement products that failed initial testing as specified in the paragraph TESTING PROCEDURE at no cost to the Government.

1.5.6.1 Sampling Procedure

Select paint at random from the products that have been delivered to the job site for sample testing. The Contractor must provide one quart samples of the selected paint materials. Take samples in the presence of the Contracting Officer, and label, and identify each sample. Provide labels in accordance with the paragraph PACKAGING, LABELING, AND STORAGE.

1.5.6.2 Testing Procedure

Provide Batch Quality Conformance Testing for specified products, as defined by and performed by MPI. As an alternative to Batch Quality Conformance Testing, the Contractor may provide Qualification Testing for specified products above to the appropriate MPI product specification, using the third-party laboratory approved under the paragraph QUALIFICATION TESTING laboratory for coatings. Include the backup data and summary of the test results within the qualification testing lab report. Provide a summary listing of all the reference specification requirements and the result of each test. Clearly indicate in the summary whether the tested paint meets each test requirement. Note that Qualification Testing may take 4 to 6 weeks to perform, due to the extent of testing required.

Submit name, address, telephone number, FAX number, and e-mail address of the independent third party laboratory selected to perform testing of coating samples for compliance with specification requirements. Submit documentation that laboratory is regularly engaged in testing of paint samples for conformance with specifications, and that employees performing testing are qualified. If MPI is chosen to perform the Batch Quality Conformance testing, the above submittal information is not required, only a letter is required from the Contractor stating that MPI will perform the testing.

1.5.7 Textured Wall Coating System

Three complete samples of each indicated type, pattern, and color of textured wall coating system applied to a panel of the same material as that on which the coating system will be applied in the work. Provide samples of wall coating systems minimum 5 by 7 inches and of sufficient size to show pattern repeat and texture.

1.6 PACKAGING, LABELING, AND STORAGE

Provide paints in sealed containers that legibly show the Contract specification number, designation name, formula or specification number, batch number, color, quantity, date of manufacture, manufacturer's formulation number, manufacturer's directions including any warnings and special precautions, and name and address of manufacturer. Furnish pigmented paints in containers not larger than 5 gallons. Store paints and thinners in accordance with the manufacturer's written directions, and as a minimum, stored off the ground, under cover, with sufficient ventilation to prevent the buildup of flammable vapors, and at temperatures between 40 to 95 degrees F.

1.7 SAFETY AND HEALTH

Comply with applicable Federal, State, and local laws and regulations, and with the ACCIDENT PREVENTION PLAN, including the Activity Hazard Analysis as specified in Section 01 35 26.00 06 GOVERNMENTAL SAFETY REQUIREMENTS and in Appendix A of EM 385-1-1. Include in the Activity Hazard Analysis the potential impact of painting operations on painting personnel and on others involved in and adjacent to the work zone.

1.7.1 Toxic Materials

To protect personnel from overexposure to toxic materials, conform to the most stringent guidance of:

a. The applicable manufacturer's Safety Data Sheets (SDS) or local regulation.

b. 29 CFR 1910.1000.

c. ACGIH 0100, threshold limit values.

d. The appropriate OSHA standard in 29 CFR 1910.1025 and 29 CFR 1926.62 for surface preparation on painted surfaces containing lead. Additional guidance is given in SSPC Guide 6 and SSPC Guide 7. Refer to drawings for list of hazardous materials located on this project. Coordinate paint preparation activities with this specification section.

e. The appropriate OSHA standards in 29 CFR 1910.1001 for surface preparation of painted surfaces containing asbestos. Refer to drawings for list of hazardous materials located on this project.

Coordinate paint preparation activities with this specification section.

Submit manufacturer's Safety Data Sheets for coatings, solvents, and other potentially hazardous materials, as defined in FED-STD-313.

1.8 ENVIRONMENTAL REQUIREMENTS

Comply, at minimum, with manufacturer recommendations for space ventilation during and after installation.

1.8.1 Coatings

Do not apply coating when air or substrate conditions are:

a. Less than 5 degrees F above dew point;

b. Below 50 degrees F or over 95 degrees F, unless specifically pre-approved by the Contracting Officer and the product manufacturer. Do not, under any circumstances, violate the manufacturer's application recommendations.

PART 2 PRODUCTS

2.1 MATERIALS

Conform to the coating specifications and standards referenced in PART 3. Submit Product Data Sheets for specified coatings and solvents. Provide preprinted cleaning and maintenance instructions for all coating systems. Submit Manufacturer's Instructions on Mixing: Detailed mixing instructions, minimum and maximum application temperature and humidity, pot life, and curing and drying times between coats.

2.2 COLOR SELECTION OF FINISH COATS

Provide colors of finish coats as indicated or specified. Allow Contracting Officer to select colors not indicated or specified. Manufacturers' names and color identification are used for the purpose of color identification only. Named products are acceptable for use only if they conform to specified requirements. Products of other manufacturers are acceptable if the colors are approximately the colors indicated and the product conforms to specified requirements.

Provide color, texture, and pattern of wall coating systems as indicated . Submit manufacturer's samples of paint colors. Cross reference color samples to color scheme as indicated. Submit color stencil codes. Tint each coat progressively darker to enable confirmation of the number of coats.

PART 3 EXECUTION

3.1 PROTECTION OF AREAS AND SPACES NOT TO BE PAINTED

Prior to surface preparation and coating applications, remove, mask, or otherwise protect hardware, hardware accessories, machined surfaces, radiator covers, plates, lighting fixtures, public and private property, and other such items not to be coated that are in contact with surfaces to be coated. Following completion of painting, reinstall removed items by workmen skilled in the trades. Restore surfaces contaminated by coating materials, to original condition and repair damaged items.

3.2 SURFACE PREPARATION

Remove dirt, splinters, loose particles, grease, oil, and other foreign matter and substances deleterious to coating performance as specified for each substrate before application of paint or surface treatments. Remove oil and grease prior to mechanical cleaning. Schedule cleaning so that dust and other contaminants will not fall on wet, newly painted surfaces. Spot-prime exposed ferrous metals such as nail heads on or in contact with surfaces to be painted with water-thinned paints, with a suitable corrosion-inhibitive primer capable of preventing flash rusting and compatible with the coating specified for the adjacent areas. Refer to MPI ASM and MPI MRM for additional more specific substrate preparation requirements.

3.2.1 Additional Requirements for Preparation of Surfaces With Existing Coatings

Before application of coatings, perform the following on surfaces covered by soundly-adhered coatings, defined as those which cannot be removed with a putty knife:

a. Test existing finishes for lead before sanding, scraping, or removing. If lead is present, refer to paragraph Toxic Materials.

b. Wipe previously painted surfaces to receive solvent-based coatings, except stucco and similarly rough surfaces clean with a clean, dry cloth saturated with mineral spirits, ASTM D235 or as specified in MPI MRM. Wipe the surfaces dry with a clean, dry, lint free cloth. Wipe immediately preceding the application of the first coat of any coating, unless specified otherwise.

c. Sand existing glossy surfaces to be painted to reduce gloss. Brush, and wipe clean with a damp cloth to remove dust.

d. The requirements specified are minimum. Comply also with the application instructions of the paint manufacturer and specific surface preparation requirements as outlined in MPI MRM Exterior Surface Preparation and Interior Surface Preparation.

e. Thoroughly clean previously painted surfaces specified to be repainted damaged during construction of all grease, dirt, dust or other foreign matter.

f. Remove blistering, cracking, flaking and peeling or otherwise deteriorated coatings.

g. Remove chalk so that when tested in accordance with ASTM D4214, the chalk resistance rating is no less than 8.

h. Roughen slick surfaces. Repair damaged areas such as, but not limited to, nail holes, cracks, chips, and spalls with suitable material to match adjacent undamaged areas.

i. Feather and sand smooth edges of chipped paint.

j. Clean rusty metal surfaces in accordance with SSPC requirements. Use solvent, mechanical, or chemical cleaning methods to provide surfaces suitable for painting.

k. Provide new, proposed coatings that are compatible with existing coatings.

3.2.2 Substrate Repair

a. Repair substrate surface damaged during coating removal;

b. Sand edges of adjacent soundly-adhered existing coatings so they are tapered as smooth as practical to areas involved with coating removal; and

c. Clean and prime the substrate as specified.

3.3 PREPARATION OF METAL SURFACES

3.3.1 Existing and New Ferrous Surfaces

a. Ferrous Surfaces including Shop-coated Surfaces and Small Areas That Contain Rust, Mill Scale and Other Foreign Substances: Solvent clean or detergent wash in accordance with SSPC SP 1 to remove oil and grease. Where shop coat is missing or damaged, clean according to or .; Protect shop-coated ferrous surfaces from corrosion by treating and touching up corroded areas immediately upon detection.

b. Surfaces With More Than 20 Percent Rust, Mill Scale, and Other Foreign Substances: Clean entire surface in accordance with .

3.3.2 Final Ferrous Surface Condition:

3.3.2.1 Tool Cleaned Surfaces

Comply with SSPC SP 2 and SSPC SP 3. Use as a visual reference, photographs in SSPC VIS 3 for the appearance of cleaned surfaces.

3.3.2.2 Abrasive Blast Cleaned Surfaces

Comply with SSPC 7/NACE No.4, SSPC SP 6/NACE No.3, and SSPC SP 10/NACE No. 2. Use as a visual reference, photographs in SSPC VIS 1 for the appearance of cleaned surfaces.

3.3.2.3 Waterjet Cleaned Surfaces

Comply with SSPC-SP WJ-1/NACE WJ-1, SSPC-SP WJ-2/NACE WJ-2, SSPC-SP WJ-3/NACE WJ-3 or SSPC-SP WJ-4/NACE WJ-4. Use as a visual reference, photographs in SSPC VIS 4/NACE VIS 7 for the appearance of cleaned surfaces.

3.3.3 Galvanized Surfaces

a. New or Existing Galvanized Surfaces With Only Dirt and Zinc Oxidation Products: Clean with solvent, in accordance with SSPC SP 1. Completely remove coating by brush-off abrasive blast if the galvanized metal has been passivated or stabilized. Do not "passivate" or "stabilize" new galvanized steel to be coated. If the absence of hexavalent stain inhibitors is not documented, test as described in ASTM D6386, Appendix X2, and remove by one of the methods described therein.

b. Galvanized with Slight Coating Deterioration or with Little or No Rusting: Water jetting to SSPC-SP WJ-3/NACE WJ-3 to remove loose coating from surfaces with less than 20 percent coating deterioration and no blistering, peeling, or cracking. Use inhibitor as recommended by the coating manufacturer to prevent rusting.

c. Galvanized With Severe Deteriorated Coating or Severe Rusting: Water jet to SSPC-SP WJ-3/NACE WJ-3 degree of cleanliness.

3.3.4 Non-Ferrous Metallic Surfaces

Aluminum and aluminum-alloy, lead, copper, and other nonferrous metal surfaces.

Surface Cleaning: Solvent clean in accordance with SSPC SP 1 and wash with mild non-alkaline detergent to remove dirt and water soluble contaminants.

3.3.5 Terne-Coated Metal Surfaces

Solvent clean surfaces with mineral spirits, ASTM D235. Wipe dry with clean, dry cloths.

3.3.6 Existing Surfaces with a Bituminous or Mastic-Type Coating

Remove chalk, mildew, and other loose material by washing with a solution of 1/2 cup trisodium phosphate, 1/4 cup household detergent, one quart 5 percent sodium hypochlorite solution and 3 quarts of warm water.

3.4 PREPARATION OF CONCRETE AND CEMENTITIOUS SURFACE

3.4.1 Concrete and Masonry

a. Curing: Allow concrete, stucco and masonry surfaces to cure at least 30 days before painting, and concrete slab on grade to cure at least 90 days before painting.

b. Surface Cleaning: Remove the following deleterious substances.

(1) Dirt, Grease, and Oil: Wash new and existing uncoated surfaces with a solution composed of 1/2 cup trisodium phosphate, 1/4 cup household detergent, and 4 quarts of warm water. Then rinse thoroughly with fresh water. Wash existing coated surfaces with a suitable detergent and rinse thoroughly. For large areas, water blasting may be used.

(2) Fungus and Mold: Wash new, existing coated, and existing uncoated surfaces with a solution composed of 1/2 cup trisodium phosphate, 1/4 cup household detergent, one quart 5 percent sodium hypochlorite solution and 3 quarts of warm water. Rinse thoroughly with fresh water.

(3) Paint and Loose Particles: Remove by wire brushing.

(4) Efflorescence: Remove by scraping or wire brushing followed by washing with a 5 to 10 percent by weight aqueous solution of hydrochloric (muriatic) acid. Do not allow acid to remain on the surface for more than five minutes before rinsing with fresh

water. Do not acid clean more than 4 square feet of surface, per workman, at one time.

(5) Removal of Existing Coatings: For surfaces to receive textured coating MPI 42, remove existing coatings including soundly adhered coatings if recommended by textured coating manufacturer.

c. Cosmetic Repair of Minor Defects: Repair or fill mortar joints and minor defects, including but not limited to spalls, in accordance with manufacturer's recommendations and prior to coating application.

d. Allowable Moisture Content: Latex coatings may be applied to damp surfaces, but not to surfaces with droplets of water. Do not apply epoxies to damp vertical surfaces as determined by ASTM D4263 or horizontal surfaces that exceed 3 lbs of moisture per 1000 square feet in 24 hours as determined by ASTM F1869. In all cases follow manufacturer's recommendations. Allow surfaces to cure a minimum of 30 days before painting.

3.4.2 Gypsum Board, Plaster, and Stucco

3.4.2.1 Surface Cleaning

Verify that plaster and stucco surfaces are free from loose matter and that gypsum board is dry. Remove loose dirt and dust by brushing with a soft brush, rubbing with a dry cloth, or vacuum-cleaning prior to application of the first coat material. A damp cloth or sponge may be used if paint is water-based.

3.4.2.2 Repair of Minor Defects

Prior to painting, repair joints, cracks, holes, surface irregularities, and other minor defects with patching plaster or spackling compound and sand smooth.

3.4.2.3 Allowable Moisture Content

Latex coatings may be applied to damp surfaces, but not surfaces with droplets of water. Do not apply epoxies to damp surfaces as determined by ASTM D4263. Verify that new plaster to be coated has a maximum moisture content of 8 percent, when measured in accordance with ASTM D4444, Method A, unless otherwise authorized. In addition to moisture content requirements, allow new plaster to age a minimum of 30 days before preparation for painting.

3.4.3 Existing Asbestos Cement Surfaces

Remove oily stains by solvent cleaning with mineral spirits in accordance with MIL-PRF-680 or ASTM D235. Remove loose dirt, dust, and other deleterious substances by brushing with a soft brush or rubbing with a dry cloth prior to application of the first coat material. Do not wire brush or clean using other abrasive methods. Verify surfaces are dry and clean prior to application of the coating.

3.5 APPLICATION

3.5.1 Coating Application

a. Comply with applicable federal, state and local laws enacted to ensure compliance with Federal Clean Air Standards. Apply coating materials in accordance with SSPC PA 1. SSPC PA 1 methods are applicable to all substrates, except as modified herein.

b. At the time of application, paint must show no signs of deterioration. Maintain uniform suspension of pigments during application.

c. Unless otherwise specified or recommended by the paint manufacturer, paint may be applied by brush, roller, or spray. Use trigger operated spray nozzles for water hoses. Use rollers for applying paints and enamels of a type designed for the coating to be applied and the surface to be coated. Wear protective clothing and respirators when applying oil-based paints or using spray equipment with any paints.

d. Only apply paints, except water-thinned types, to surfaces that are completely free of moisture as determined by sight or touch.

e. Thoroughly work coating materials into joints, crevices, and open spaces. Pay special attention to ensure that all edges, corners, crevices, welds, and rivets receive a film thickness equal to that of adjacent painted surfaces.

f. Apply each coat of paint so that dry film is of uniform thickness and free from runs, drops, ridges, waves, pinholes or other voids, laps, brush marks, and variations in color, texture, and finish. Completely hide all blemishes.

g. Touch up damaged coatings before applying subsequent coats. Broom clean and clear dust from interior areas before and during the application of coating material.

h. Apply paint to new fire extinguishing sprinkler systems including valves, piping, conduit, hangers, supports, miscellaneous metal work, and accessories. Shield sprinkler heads with protective coverings while painting is in progress. Remove sprinkler heads which have been painted and replace with new sprinkler heads. Unfinished spaces include attic spaces, spaces above suspended ceilings, crawl spaces, pipe chases, mechanical equipment room, and space where walls or ceiling are not painted or not constructed of a prefinished material. Upon completion of painting, remove protective covering from sprinkler heads.

i. Piping in Unfinished Areas: Provide primed surfaces with one coat of red alkyd gloss enamel (MPI 9) applied to a minimum dry film thickness of 1.0 mil in attic spaces, spaces above suspended ceilings, crawl spaces, pipe chases, mechanical equipment room, and spaces where walls or ceiling are not painted or not constructed of a prefinished material.

j. Piping in Finished Areas: Provide primed surfaces with two coats of paint to match adjacent surfaces, except provide valves and operating accessories with one coat of red alkyd gloss enamel (MPI 9) applied to a minimum dry film thickness of 1.0 mil or two component gloss polyurethane (MPI 72) in exterior applications.

k. Provide labeling on the surfaces of all feed and cross mains to show the pipe function such as "Sprinkler System", "Fire Department Connection", "Standpipe". For pipe sizes 4-inch and larger provide white painted stenciled letters and arrows, a minimum of 2 in in height and visible from at least two sides when viewed from the floor. For pipe sizes less than 4-inch, provide white painted stenciled letters and arrows, a minimum of 0.75 in in height and visible from the floor.

l. All fire suppression system valves must be marked with permanent tags indicating normally open or normally closed.

m. Drying Time: Allow time between coats, as recommended by the coating manufacturer, to permit thorough drying, but not to present topcoat adhesion problems. Provide each coat in specified condition to receive next coat.

n. Primers, and Intermediate Coats: Do not allow primers or intermediate coats to dry more than 30 days, or longer than recommended by manufacturer, before applying subsequent coats. Follow manufacturer's recommendations for surface preparation if primers or intermediate coats are allowed to dry longer than recommended by manufacturers of subsequent coatings. Cover each preceding coat or surface completely by ensuring visually perceptible difference in shades of successive coats.

o. Finished Surfaces: Provide finished surfaces free from runs, drops, ridges, waves, laps, brush marks, and variations in colors.

p. Thermosetting Paints: Apply topcoats over thermosetting paints (epoxies and urethanes) within the overcoat window recommended by the manufacturer.

q. Floors:

3.5.2 Mixing and Thinning of Paints

Reduce paints to proper consistency by adding fresh paint, except when thinning is mandatory to suit surface, temperature, weather conditions, application methods, or for the type of paint being used. Obtain written permission from the Contracting Officer to use thinners. Verify that the written permission includes quantities and types of thinners to use.

When thinning is allowed, thin paints immediately prior to application with not more than one pint of suitable thinner per gallon. The use of thinner does not relieve the Contractor from obtaining complete hiding, full film thickness, or required gloss. Thinning cannot cause the paint to exceed limits on volatile organic compounds. Do not mix paints of different manufacturers.

3.5.3 Two-Component Systems

Mix two-component systems in accordance with manufacturer's instructions. Follow recommendation by the manufacturer for any thinning of the first coat to ensure proper penetration and sealing for each type of substrate.

3.5.4 Coating Systems

a. Systems by Substrates: Apply coatings that conform to the respective specifications listed in the following Tables:

Table for Interior Applications	
MPI Division	Substrate Application
MPI Division 3	Interior Concrete Paint Table
MPI Division 4	Interior Concrete Masonry Units Paint Table
MPI Division 5	Interior Metal, Ferrous and Non-Ferrous Paint Table
MPI Division 6	Interior Wood Paint Table
MPI Division 9	Interior Plaster, Gypsum Board, Textured Surfaces Paint Table

b. Minimum Dry Film Thickness (DFT): Apply paints, primers, varnishes, enamels, undercoats, and other coatings to a minimum dry film thickness of 1.5 mil each coat unless specified otherwise in the Tables. Coating thickness, where specified, refers to the minimum dry film thickness.

c. Coatings for Surfaces Not Specified Otherwise: Coat unspecified surfaces the same as surfaces having similar conditions of exposure.

d. Existing Surfaces Damaged During Performance of the Work, Including New Patches In Existing Surfaces: Coat surfaces with the following:

(1) One coat of primer.

(2) One coat of undercoat or intermediate coat.

(3) One topcoat to match adjacent surfaces.

e. Existing Coated Surfaces To Be Painted: Apply coatings conforming to the respective specifications listed in the Tables herein, except that pretreatments, sealers and fillers need not be provided on surfaces where existing coatings are soundly adhered and in good condition. Do not omit undercoats or primers.

3.6 COATING SYSTEMS FOR METAL

Apply coatings of Tables in MPI Division 5 for Exterior and Interior.

a. Apply specified ferrous metal primer to steel surfaces on the same day that surface is cleaned, to surfaces that meet all specified surface preparation requirements at time of application.

b. Inaccessible Surfaces: Prior to erection, use one coat of specified primer on metal surfaces that will be inaccessible after erection.

c. Shop-primed Surfaces: Touch up exposed substrates and damaged coatings to protect from rusting prior to applying field primer.

d. Surface Previously Coated with Epoxy or Urethane: Apply MPI 101, 1.5 mils DFT immediately prior to application of epoxy or urethane coatings.

e. Pipes and Tubing: The semitransparent film applied to some pipes and tubing at the mill is not to be considered a shop coat. Overcoat these items with the specified ferrous-metal primer prior to application of finish coats.

f. Exposed Nails, Screws, Fasteners, and Miscellaneous Ferrous Surfaces. On surfaces to be coated with water thinned coatings, spot prime exposed nails and other ferrous metal with latex primer MPI 107.

3.7 COATING SYSTEMS FOR CONCRETE AND CEMENTITIOUS SUBSTRATES

Apply coatings of Tables in MPI Division 3, 4 and 9 for Exterior and Interior.

3.8 COATING SYSTEMS FOR WOOD AND PLYWOOD

a. Apply coatings of Tables in MPI Division 6 for Exterior and Interior.

b. Prior to erection, apply two coats of specified primer to treat and prime wood surfaces which will be inaccessible after erection.

c. Apply stains in accordance with manufacturer's printed instructions.

3.9 PIPING IDENTIFICATION

Piping Identification, Including Surfaces In Concealed Spaces: Provide in accordance with MIL-STD-101. Place stenciling in clearly visible locations. On piping not covered by MIL-STD-101, stencil approved names or code letters, in letters a minimum of 1/2 inch high for piping and a minimum of 2 inches high elsewhere. Stencil arrow-shaped markings on piping to indicate direction of flow using black stencil paint.

3.10 INSPECTION AND ACCEPTANCE

In addition to meeting previously specified requirements, demonstrate mobility of moving components, including swinging and sliding doors, cabinets, and windows with operable sash, for inspection by the Contracting Officer. Perform this demonstration after appropriate curing and drying times of coatings have elapsed and prior to invoicing for final payment.

3.11 WASTE MANAGEMENT

As specified in the Waste Management Plan and as follows. Do not use kerosene or any such organic solvents to clean up water based paints. Properly dispose of paints or solvents in designated containers. Close and seal partially used containers of paint to maintain quality as necessary for reuse. Store in protected, well-ventilated, fire-safe area at moderate temperature. Place materials defined as hazardous or toxic waste in designated containers. Set aside extra paint for future color matches or reuse by the Government. Where local options exist for leftover paint recycling, collect all waste paint by type and provide for delivery to recycling or collection facility for reuse by local organizations.

3.12 PAINT TABLES

All DFT's are minimum values. Use only materials with a MPI GPS-1-14 green check mark having a minimum MPI "Environmentally Friendly" E1 rating based on VOC (EPA Method 24) content levels. Acceptable products are listed in the MPI Green Approved Products List, available at http://www.specifygreen.com/APL/ProductIdxByMPInum.asp.

3.12.1 Interior Paint Tables

3.12.1.1 MPI Division 3: Interior Concrete Paint Table

A. New and uncoated existing and Existing, previously painted Concrete, vertical surfaces, not specified otherwise

Latex					
New, uncoated Existing	Existing, previously painted	Primer	Intermediate	Topcoat	System DFT
MPI INT 3.1A-G2 (Flat)	MPI RIN 3.1A-G2 (Flat)	MPI 3	MPI 44	MPI 44	4 mils
MPI INT 3.1A-G3 (Eggshell)	MPI RIN 3.1A-G3 (Eggshell)	MPI 3	MPI 52	MPI 52	4 mils
MPI INT 3.1A-G5	MPI RIN 3.1A-G5 (Semigloss)	MPI 3	MPI 54	MPI 54	4 mils
Topcoat: Coating to match adjacent surfaces.					

High Performance Architectural Latex					
New, uncoated Existing	Existing, previously painted	Primer	Intermediate	Topcoat	System DFT
MPI INT 3.1C-G2 (Flat)	MPI RIN 3.1J-G2 (Flat)	MPI 3	MPI 138	MPI 138	4 mils
MPI INT 3.1C-G3 (Eggshell)	MPI RIN 3.1J-G3 (Eggshell)	MPI 3	MPI 139	MPI 139	4 mils
MPI INT 3.1C-G4 (satin)	MPI RIN 3.1J-G4	MPI 3	MPI 140	MPI 140	4 mils
MPI INT 3.1C-G5 (Semigloss)	MPI RIN 3.1J-G5 (Semigloss)	MPI 3	MPI 141	MPI 141	4 mils
Topcoat: Coating to match adjacent surfaces.					

Institutional Low Odor / Low VOC Latex

New, uncoated Existing	Existing, previously painted	Primer	Intermediate	Topcoat	System DFT
MPI INT 3.1M-G2 (Flat)	MPI RIN 3.1L-G2 (Flat)	MPI 149	MPI 144	MPI 144	4 mils
MPI INT 3.1M-G3 (Eggshell)	MPI RIN 3.1L-G3 (Eggshell)	MPI 149	MPI 145	MPI 145	4 mils
MPI INT 3.1M-G4 (satin)	MPI RIN 3.1L-G4	MPI 149	MPI 146	MPI 146	4 mils
MPI INT 3.1M-G5 (Semigloss)	MPI RIN 3.1L-G5 (Semigloss)	MPI 149	MPI 147	MPI 147	4 mils
Topcoat: Coating to match adjacent surfaces.					

B. Concrete Ceilings, Uncoated

Latex Aggregate				
New, uncoated	Primer	Intermediate	Topcoat	System DFT
MPI INT 3.1N-G1 (Flat)	N/A	N/A	MPI 42	Per Manufacturer
Texture - Medium . Surface preparation, number of coats, and primer in accordance with manufacturer's instructions. Topcoat: Coating to match adjacent surfaces.				

C. New and uncoated existing and Existing, previously painted Concrete in areas requiring a high degree of sanitation, not otherwise specified except floors

Waterborne Light Industrial Coating					
New, uncoated Existing	Existing, previously painted	Primer	Intermediate	Topcoat	System DFT
MPI INT 3.1L-G3(Eggshell)	MPI RIN 3.1C-G3(Eggshell)	MPI 3	MPI 151	MPI 151	4.8 mils
MPI INT 3.1L-G5(Semigloss)	MPI RIN 3.1C-G5(Semigloss)	MPI 3	MPI 153	MPI 153	4.8 mils
MPI INT 3.1L-G6(Gloss)	MPI RIN 3.1C-G6(Gloss)	MPI 3	MPI 154	MPI 154	4.8 mils
Topcoat: Coating to match adjacent surfaces.					

Alkyd					
New, uncoated Existing	Existing, previously painted	Primer	Intermediate	Topcoat	System DFT
MPI INT 3.1D-G3 (Eggshell)	MPI RIN 3.1D-G3 (Eggshell)	MPI 3	MPI 51	MPI 51	4.5 mils
MPI INT 3.1D-G5 (Semigloss)	MPI RIN 3.1D-G5 (Semigloss)	MPI 3	MPI 47	MPI 47	4.5 mils
MPI INT 3.1D-G6 (Gloss)	MPI RIN 3.1D-G6 (Gloss)	MPI 3	MPI 48	MPI 48	4.5 mils
Topcoat: Coating to match adjacent surfaces.					

Epoxy					
New, uncoated Existing	Existing, previously painted	Primer	Intermediate	Topcoat	System DFT
MPI INT 3.1F-G6 (Gloss)	MPI RIN 3.1E-G6 (Gloss)	MPI 77	MPI 77	MPI 77	4 mils
Note: Primer may be reduced for penetration per manufacturer's instructions.					

D. New and uncoated existing and Existing, previously painted concrete walls and bottom of swimming pools

Chlorinated Rubber					
New and uncoated existing	Existing, previously painted	Primer	Intermediate	Topcoat	System DFT
Chlorinated Rubber	Per Manufacturer	Per Manufacturer	Per Manufacturer	Per Manufacturer	Per Manufacturer
Note: Primer may be reduced for penetration per manufacturer's instructions.					

Epoxy					
New, uncoated Existing	Existing, previously painted	Primer	Intermediate	Topcoat	System DFT
MPI INT 3.1F	MPI RIN 3.1E	MPI 77	MPI 77	MPI 77	4 mils

Note: Primer may be reduced for penetration per manufacturer's instructions.

E. New and uncoated existing and Existing, previously painted concrete floors in following areas

Latex Floor Paint					
New, uncoated Existing	Existing, previously painted	Primer	Intermediate	Topcoat	System DFT
MPI INT 3.2A-G2 (Flat)	MPI RIN 3.2A-G2 (Flat)	MPI 60	MPI 60	MPI 60	5 mils

Alkyd Floor Paint					
New, uncoated Existing	Existing, previously painted	Primer	Intermediate	Topcoat	System DFT
MPI INT 3.2B-G2 (Flat)	MPI RIN 3.2B-G2 (Flat)	MPI 59	MPI 59	MPI 59	5 mils
Note: Primer may be reduced for penetration per manufacturer's instructions.					

Epoxy					
New, uncoated Existing	Existing, previously painted	Primer	Intermediate	Topcoat	System DFT
MPI INT 3.2C-G6 (Gloss)	MPI RIN 3.2C-G6 (Gloss)	MPI 77	MPI 77	MPI 77	5 mils
Note: Primer may be reduced for penetration per manufacturer's instructions.					

3.12.1.2 MPI Division 4: Interior Concrete Masonry Units Paint Table

A. New and uncoated Existing Concrete Masonry

High Performance Architectural Latex					
New, uncoated Existing	Filler	Primer	Intermediate	Topcoat	System DFT
MPI INT 4.2D-G2 (Flat)	MPI 4	N/A	MPI 139	MPI 138	11 mils
MPI INT 4.2D-G3 (Eggshell)	MPI 4	N/A	MPI 139	MPI 139	11 mils
MPI INT 4.2D-G4 (Satin)	MPI 4	N/A	MPI 140	MPI 140	11 mils

MPI INT 4.2D-G5 (Semigloss)	MPI 4	N/A	MPI 141	MPI 141	11 mils
Fill all holes in masonry surface					

Institutional Low Odor / Low VOC Latex					
New, uncoated Existing	Filler	Primer	Intermediate	Topcoat	System DFT
MPI INT 4.2E-G2 (Flat)	MPI 4	N/A	MPI 144	MPI 144	4 mils
MPI INT 4.2E-G3 (Eggshell)	MPI 4	N/A	MPI 145	MPI 145	4 mils
MPI INT 4.2E-G4 (Satin)	MPI 4	N/A	MPI 146	MPI 146	4 mils
MPI INT 4.2E-G5 (Semigloss)	MPI 4	N/A	MPI 147	MPI 147	4 mils
Fill all holes in masonry surface					

B. Existing, Previously Painted Concrete Masonry

High Performance Architectural Latex					
Existing, previously painted	Filler	Primer	Intermediate	Topcoat	System DFT
MPI RIN 4.2K-G2 (Flat)	N/A	MPI 138	MPI 138	MPI 138	4.5 mils
MPI RIN 4.2K-G3 (Eggshell)	N/A	MPI 139	MPI 139	MPI 139	4.5 mils
MPI RIN 4.2K-G4	N/A	MPI 140	MPI 140	MPI 140	4.5 mils
MPI RIN 4.2K-G5 (Semigloss)	N/A	MPI 141	MPI 141	MPI 141	4.5 mils

Institutional Low Odor / Low VOC Latex					
Existing, previously painted	Filler	Primer	Intermediate	Topcoat	System DFT
MPI RIN 4.2L-G2 (Flat)	N/A	MPI 144	MPI 144	MPI 144	4 mils

MPI RIN 4.2L-G3 (Eggshell)	N/A	MPI 145	MPI 145	MPI 145	4 mils
MPI RIN 4.2L-G4 (Satin)	N/A	MPI 146	MPI 146	MPI 146	4 mils
MPI RIN 4.2L-G5 (Semigloss)	N/A	MPI 147	MPI 147	MPI 147	4 mils

C. New and uncoated Existing Concrete masonry units in areas requiring a high degree of sanitation, unless otherwise specified

Waterborne Light Industrial Coating					
New, uncoated Existing	Filler	Primer	Intermediate	Topcoat	System DFT
MPI INT 4.2K-G3(Eggshell)	MPI 4	N/A	MPI 151	MPI 151	11 mils
MPI INT 4.2K-G5(Semigloss)	MPI 4	N/A	MPI 153	MPI 153	11 mils
MPI INT 4.2K-G6(Gloss)	MPI 4	N/A	MPI 154	MPI 154	11 mils
Fill all holes in masonry surface					

Alkyd					
New, uncoated Existing	Filler	Primer	Intermediate	Topcoat	System DFT
MPI INT 4.2K-G3(Eggshell)	MPI 4	MPI 50	MPI 51	MPI 51	12 mils
MPI INT 4.2K-G5(Semigloss)	MPI 4	MPI 50	MPI 47	MPI 47	12 mils
MPI INT 4.2K-G6(Gloss)	MPI 4	MPI 50	MPI 48	MPI 48	12 mils
Fill all holes in masonry surface					

Epoxy					
New, uncoated Existing	Filler	Primer	Intermediate	Topcoat	System DFT
MPI INT 4.2G-G6 (Gloss)	MPI 116	N/A	MPI 77	MPI 77	10 mils

Fill all holes in masonry surface

D. Existing, previously painted, concrete masonry units in areas requiring a high degree of sanitation, unless otherwise specified

Waterborne Light Industrial Coating					
Existing, previously painted	Filler	Primer	Intermediate	Topcoat	System DFT
MPI RIN 4.2G-G3(Eggshell)	N/A	MPI 151	MPI 151	MPI 151	4.5 mils
MPI RIN 4.2G-G5(Semigloss)	N/A	MPI 153	MPI 153	MPI 153	4.5 mils
MPI RIN 4.2G-G6(Gloss)	N/A	MPI 154	MPI 154	MPI 154	4.5 mils

Alkyd					
Existing, previously painted	Filler	Primer	Intermediate	Topcoat	System DFT
MPI RIN 4.2C-G3 (Eggshell)	N/A	MPI 17	MPI 51	MPI 51	4.5 mils
MPI RIN 4.2C-G5 (Semigloss)	N/A	MPI 17	MPI 47	MPI 47	4.5 mils
MPI RIN 4.2C-G6 (Gloss)	N/A	MPI 17	MPI 48	MPI 48	4.5 mils

Epoxy					
Existing, previously painted	Filler	Primer	Intermediate	Topcoat	System DFT
MPI RIN 4.2D-G6 (Gloss)	N/A	MPI 77	MPI 77	MPI 77	5 mils

3.12.1.3 MPI Division 5: Interior Metal, Ferrous and Non-Ferrous Paint Table

A. Interior Steel / Ferrous Surfaces

(1) Metal, and miscellaneous metal items not otherwise specified except floors, hot metal surfaces, and new prefinished equipment

High Performance Architectural Latex				
New, uncoated Existing	Primer	Intermediate	Topcoat	System DFT

MPI INT 5.1R-G2 (Flat)	MPI 76	MPI 138	MPI 138	5 mils
MPI INT 5.1R-G3 (Eggshell)	MPI 76	MPI 139	MPI 139	5 mils
MPI INT 5.1R-G5 (Semigloss)	MPI 76	MPI 141	MPI 141	5 mils
Topcoat: Coating to match adjacent surfaces.				

Alkyd				
New, uncoated Existing	Primer	Intermediate	Topcoat	System DFT
MPI INT 5.1E-G2 (Flat)	MPI 76	MPI 49	MPI 49	5.25 mils
MPI INT 5.1E-G3 (Eggshell)	MPI 76	MPI 51	MPI 51	5.25 mils
MPI INT 5.1E-G5 (Semigloss)	MPI 76	MPI 47	MPI 47	5.25 mils
MPI INT 5.1E-G6 (Gloss)	MPI 76	MPI 48	MPI 48	5.25 mils
Topcoat: Coating to match adjacent surfaces.				

(2) Metal floors (non-shop-primed surfaces or non-slip deck surfaces) with non-skid additive (NSA), load at manufacturer's recommendations

Alkyd (over q.d. Alkyd Primer)				
New, uncoated Existing	Primer	Intermediate	Topcoat	System DFT
MPI INT 5.1E-G5 (Semi-Gloss)	MPI 76	MPI 47	MPI 47	5.25 mils
Topcoat: Coating to match adjacent surfaces.				

Epoxy				
New, uncoated Existing	Primer	Intermediate	Topcoat	System DFT
MPI INT 5.1L-G6 (Gloss)	MPI 101	MPI 101	MPI 101	5.25 mils
Topcoat: Coating to match adjacent surfaces.				

(3) Metal in areas requiring a high degree of sanitation, not otherwise

specified except floors, hot metal surfaces, and new prefinished equipment

Alkyd				
New, uncoated Existing	Primer	Intermediate	Topcoat	System DFT
MPI INT 5.1E-G3 (Eggshell)	MPI 76	MPI 51	MPI 51	5.25 mils
MPI INT 5.1E-G5 (Semigloss)	MPI 76	MPI 47	MPI 47	5.25 mils
MPI INT 5.1E-G6 (Gloss)	MPI 76	MPI 48	MPI 48	5.25 mils
Topcoat: Coating to match adjacent surfaces.				

Alkyd; For Hand Tool Cleaning				
New, uncoated Existing	Primer	Intermediate	Topcoat	System DFT
MPI INT 5.1T-G3 (Eggshell)	MPI 23	MPI 51	MPI 51	5.25 mils
MPI INT 5.1T-G5 (Semigloss)	MPI 23	MPI 47	MPI 47	5.25 mils
MPI INT 5.1T-G6 (Gloss)	MPI 23	MPI 48	MPI 48	5.25 mils
Topcoat: Coating to match adjacent surfaces.				

(4) Ferrous metal in concealed damp spaces or in exposed areas having unpainted adjacent surfaces as follows:

Aluminum Paint				
New, uncoated Existing	Primer	Intermediate	Topcoat	System DFT
MPI INT 5.1M	MPI 76	MPI 1	MPI 1	4.25 mils
Topcoat: Coating to match adjacent surfaces.				

(5) Miscellaneous non-ferrous metal items not otherwise specified except floors, hot metal surfaces, and new prefinished equipment. Match surrounding finish

High Performance Architectural Latex				
New, uncoated Existing	Primer	Intermediate	Topcoat	System DFT
MPI INT 5.4F-G2 (Flat)	MPI 95	MPI 138	MPI 138	5 mils
MPI INT 5.4F-G3 (Eggshell)	MPI 95	MPI 139	MPI 139	5 mils
MPI INT 5.4F-G4 (Satin)	MPI 95	MPI 140	MPI 140	5 mils
MPI INT 5.4F-G5 (Semigloss)	MPI 95	MPI 141	MPI 141	5 mils
Topcoat: Coating to match adjacent surfaces.				

Alkyd				
New, uncoated Existing	Primer	Intermediate	Topcoat	System DFT
MPI INT 5.4J-G2 (Flat)	MPI 95	MPI 49	MPI 49	5 mils
MPI INT 5.4J-G3 (Eggshell)	MPI 95	MPI 51	MPI 51	5 mils
MPI INT 5.4J-G5 (Semigloss)	MPI 95	MPI 47	MPI 47	5 mils
MPI INT 5.4J-G6 (Gloss)	MPI 95	MPI 48	MPI 48	5 mils
Topcoat: Coating to match adjacent surfaces.				

B. Hot Surfaces

(1) Hot metal surfaces subject to temperatures up to 400 degrees F

Heat Resistant Enamel				
New	N/A	Intermediate	Topcoat	System DFT
MPI INT 5.2A	MPI 21	N/A	N/A	Per Manufacturer
Surface preparation and number of coats per manufacturer's instructions.				

(2) Ferrous metal subject to high temperature, up to 750 degrees F

Inorganic Zinc Rich Coating				
New	N/A	Intermediate	Topcoat	System DFT
MPI INT 5.2C	MPI 19	N/A	N/A	Per Manufacturer
Surface preparation and number of coats per manufacturer's instructions.				

Heat Resistant Aluminum Enamel				
New	N/A	Intermediate	Topcoat	System DFT
MPI INT 5.2B (Aluminum Finish)	MPI 2	N/A	N/A	Per Manufacturer
Surface preparation and number of coats per manufacturer's instructions.				

(3) New and Existing Surfaces made bare subject to temperatures up to 1100 degrees F

(1) New surfaces and Existing surfaces made bare cleaning to SSPC SP 10/NACE No. 2 subject to temperatures up to 1100 degrees F:

Heat Resistant Coating					
New	Existing	N/A	Intermediate	Topcoat	System DFT
MPI INT 5.2D	MPI RIN 5.2D	MPI 22	N/A	N/A	Per Manufacturer
Surface preparation and number of coats per manufacturer's instructions.					

3.12.1.4 MPI Division 6: Interior Wood Paint Table

A. Interior Wood and Plywood

(1) New and Existing, uncoated Wood and plywood not otherwise specified

High Performance Architectural Latex				
New, uncoated Existing	Primer	Intermediate	Topcoat	System DFT
MPI INT 6.4S-G3 (Eggshell)	MPI 39	MPI 139	MPI 139	4.5 mils
MPI INT 6.4S-G4 (Satin)	MPI 39	MPI 140	MPI 140	4.5 mils

MPI INT 6.4S-G5 (Semigloss)	MPI 39	MPI 141	MPI 141	4.5 mils
Topcoat: Coating to match adjacent surfaces.				

Alkyd				
New, uncoated Existing	Primer	Intermediate	Topcoat	System DFT
MPI INT 6.4B-G3 (Eggshell)	MPI 45	MPI 51	MPI 51	4.5 mils
MPI INT 6.4B-G5 (Semigloss)	MPI 45	MPI 47	MPI 47	4.5 mils
MPI INT 6.4B-G6 (Gloss)	MPI 45	MPI 48	MPI 48	4.5 mils
Topcoat: Coating to match adjacent surfaces.				

Institutional Low Odor / Low VOC Latex				
New, uncoated Existing	Primer	Intermediate	Topcoat	System DFT
MPI INT 6.3V-G2 (Flat)	MPI 39	MPI 144	MPI 144	4 mils
MPI INT 6.3V-G3 (Eggshell)	MPI 39	MPI 145	MPI 145	4 mils
MPI INT 6.3V-G4 (Satin)	MPI 39	MPI 146	MPI 146	4 mils
MPI INT 6.3V-G5 (Semigloss)	MPI 39	MPI 147	MPI 147	4 mils

(2) Existing, previously painted Wood and plywood not otherwise specified

High Performance Architectural Latex				
Existing, previously painted	Primer	Intermediate	Topcoat	System DFT
MPI RIN 6.4B-G3 (Eggshell)	MPI 39	MPI 139	MPI 139	4.5 mils
MPI RIN 6.4B-G4 (Satin)	MPI 39	MPI 140	MPI 140	4.5 mils

MPI RIN 6.4B-G5 (Semigloss)	MPI 39	MPI 141	MPI 141	4.5 mils
Topcoat: Coating to match adjacent surfaces.				

Alkyd				
Existing, previously painted	Primer	Intermediate	Topcoat	System DFT
MPI RIN 6.4C-G3 (Eggshell)	MPI 46	MPI 51	MPI 51	4.5 mils
MPI RIN 6.4C-G5 (Semigloss)	MPI 46	MPI 47	MPI 47	4.5 mils
MPI RIN 6.4C-G6 (Gloss)	MPI 46	MPI 48	MPI 48	4.5 mils
Topcoat: Coating to match adjacent surfaces.				

Institutional Low Odor / Low VOC Latex				
Existing, previously painted	Primer	Intermediate	Topcoat	System DFT
MPI RIN 6.4D-G2 (Flat)	MPI 39	MPI 144	MPI 144	4 mils
MPI RIN 6.4D-G3 (Eggshell)	MPI 39	MPI 145	MPI 145	4 mils
MPI RIN 6.4D-G4 (Satin)	MPI 39	MPI 146	MPI 146	4 mils
MPI RIN 6.4D-G5 (Semigloss)	MPI 39	MPI 147	MPI 147	4 mils

B. Interior New and Existing, previously finished or stained Wood and Plywood, except floors; natural finish or stained

Natural finish, oil-modified polyurethane					
New	Existing	Primer	Intermediate	Topcoat	System DFT
MPI INT 6.4J-G4	MPI RIN 6.4L-G4	MPI 57	MPI 57	MPI 57	4 mils
MPI INT 6.4J-G6 (Gloss)	MPI RIN 6.4L-G6 (Gloss)	MPI 56	MPI 56	MPI 56	4 mils

Stained, oil-modified polyurethane						
New	Existing	Stain	Primer	Intermediate	Topcoat	System DFT
MPI INT 6.4E-G4	MPI RIN 6.4G-G4	MPI 90	MPI 57	MPI 57	MPI 57	4 mils
MPI INT 6.4E-G6 (Gloss)	MPI RIN 6.4G-G6 (Gloss)	MPI 90	MPI 56	MPI 56	MPI 56	4 mils

Stained, Moisture Cured Urethane						
New	Existing	Stain	Primer	Intermediate	Topcoat	System DFT
MPI INT 6.4V-G2 (Flat)	MPI RIN 6.4V-G2 (Flat)	MPI 90	MPI 71	MPI 71	MPI 71	4 mils
MPI INT 6.4V-G6 (Gloss)	MPI RIN 6.4V-G6 (Gloss)	MPI 90	MPI 31	MPI 31	MPI 31	4 mils

C. Interior New and Existing, previously finished or stained Wood Floors; Natural finish or stained

Natural finish, oil-modified polyurethane					
New	Existing, previously finished or stained	Primer	Intermediate	Topcoat	System DFT
MPI INT 6.5C-G6 (Gloss)	MPI RIN 6.5C-G6 (Gloss)	MPI 56	MPI 56	MPI 56	4 mils

Natural finish, Moisture Cured Polyurethane					
New	Existing, previously finished or stained	Primer	Intermediate	Topcoat	System DFT
MPI INT 6.5K-G6 (Gloss)	MPI RIN 6.5D-G6 (Gloss)	MPI 31	MPI 31	MPI 31	4 mils

Stained, oil-modified polyurethane

New	Existing, previously finished or stained	Stain	Primer	Intermediate	Topcoat	System DFT
MPI INT 6.5B-G6 (Gloss)	MPI RIN 6.5B-G6 (Gloss)	MPI 90	MPI 56	MPI 56	MPI 56	4 mils

Stained, Moisture Cured Urethane						
New	Existing, previously finished or stained	Stain	Primer	Intermediate	Topcoat	System DFT
MPI INT 6.4V-G6 (Gloss)	MPI RIN 6.4V-G6 (Gloss)	MPI 90	MPI 31	MPI 31	MPI 31	4 mils

D. New and Existing, previously coated Wood floors; pigmented finish

Latex Floor Paint					
New	Existing, previously finished	Primer	Intermediate	Topcoat	System DFT
MPI INT 6.5G-G2 (Flat)	MPI RIN 6.5J-G2 (Flat)	MPI 45	MPI 60	MPI 60	4.5 mils
MPI INT 6.5G-G6 (Gloss)	MPI RIN 6.5J-G6 (Gloss)	MPI 45	MPI 68	MPI 68	4.5 mils
Topcoat: Coating to match adjacent surfaces.					

Alkyd Floor Paint					
New	Existing, previously finished	Primer	Intermediate	Topcoat	System DFT
MPI INT 6.5A-G2 (Flat)	MPI RIN 6.5A-G2 (Flat)	MPI 59	MPI 59	MPI 59	4.5 mils
MPI INT 6.5A-G6 (Gloss)	MPI RIN 6.5A-G6 (Gloss)	MPI 27	MPI 27	MPI 27	4.5 mils

Topcoat: Coating to match adjacent surfaces.

E. Interior New and Existing, uncoated wood surfaces in areas requiring a high degree of sanitation, not otherwise specified

High-Build Glaze Coatings

Waterborne Light Industrial				
New, uncoated Existing	Primer	Intermediate	Topcoat	System DFT
MPI INT 6.3P-G5 (Semigloss)	MPI 45	MPI 153	MPI 153	4.5 mils
MPI INT 6.3P-G6 (Gloss)	MPI 45	MPI 154	MPI 154	4.5 mils
Topcoat: Coating to match adjacent surfaces.				

Alkyd				
New, uncoated Existing	Primer	Intermediate	Topcoat	System DFT
MPI INT 6.3B-G5 (Semigloss)	MPI 45	MPI 47	MPI 47	4.5 mils
MPI INT 6.3B-G6 (Gloss)	MPI 45	MPI 48	MPI 48	4.5 mils
Topcoat: Coating to match adjacent surfaces.				

F. Existing, previously painted wood surfaces in areas requiring a high degree of sanitation, not otherwise specified

High-Build Glaze Coatings

Waterborne Light Industrial				
Existing, previously finished	Primer	Intermediate	Topcoat	System DFT
MPI RIN 6.3P-G5 (Semigloss)	MPI 39	MPI 153	MPI 153	4.5 mils
MPI RIN 6.3P-G6 (Gloss)	MPI 39	MPI 154	MPI 154	4.5 mils

Topcoat: Coating to match adjacent surfaces.

Alkyd				
Existing, previously finished	Primer	Intermediate	Topcoat	System DFT
MPI RIN 6.3B-G5 (Semigloss)	MPI 46	MPI 47	MPI 47	4.5 mils
MPI RIN 6.3B-G6 (Gloss)	MPI 46	MPI 48	MPI 48	4.5 mils
Topcoat: Coating to match adjacent surfaces.				

G. Interior New and Existing, previously finished or stained Wood Doors; Natural Finish or Stained

Natural finish, oil-modified polyurethane					
New	Existing, previously finished or stained	Primer	Intermediate	Topcoat	System DFT
MPI INT 6.3K-G4	MPI RIN 6.3K-G4	MPI 57	MPI 57	MPI 57	4 mils
MPI INT 6.3K-G6 (Gloss)	MPI RIN 6.3K-G6 (Gloss)	MPI 56	MPI 56	MPI 56	4 mils
Note: Sand between all coats per manufacturers recommendations.					

Stained, oil-modified polyurethane						
New	Existing, previously finished or stained	Stain	Primer	Intermediate	Topcoat	System DFT
MPI INT 6.3E-G4	MPI RIN 6.3E-G4	MPI 90	MPI 57	MPI 57	MPI 57	4 mils
MPI INT 6.5B-G6 (Gloss)	MPI RIN 6.5B-G6 (Gloss)	MPI 90	MPI 56	MPI 56	MPI 56	4 mils
Note: Sand between all coats per manufacturers recommendations.						

Stained, Moisture Cured Urethane						
New	Existing, previously finished or stained	Stain	Primer	Intermediate	Topcoat	System DFT

MPI INT 6.4V-G2 (Flat)	MPI RIN 6.4V-G2 (Flat)	MPI 90	MPI 71	MPI 71	MPI 71	4 mils
MPI INT 6.4V-G6 (Gloss)	MPI RIN 6.4V-G6 (Gloss)	MPI 90	MPI 31	MPI 31	MPI 31	4 mils
Note: Sand between all coats per manufacturers recommendations.						

H. New and Existing, uncoated Wood Doors; Pigmented finish

Alkyd				
New, uncoated Existing	Primer	Intermediate	Topcoat	System DFT
MPI INT 6.3B-G5 (Semigloss)	MPI 45	MPI 47	MPI 47	4.5 mils
MPI INT 6.3B-G6 (Gloss)	MPI 45	MPI 48	MPI 48	4.5 mils
Note: Sand between all coats per manufacturers recommendations.				

Pigmented Polyurethane				
New, uncoated Existing	Primer	Intermediate	Topcoat	System DFT
MPI INT 6.1E-G6 (Gloss)	MPI 72	MPI 72	MPI 72	4.5 mils
Note: Sand between all coats per manufacturers recommendations.				

I. Existing, previously painted Wood Doors; Pigmented finish

Alkyd				
Existing, previously finished	Primer	Intermediate	Topcoat	System DFT
MPI RIN 6.3B-G5 (Semigloss)	MPI 46	MPI 47	MPI 47	4.5 mils
MPI RIN 6.3B-G6 (Gloss)	MPI 46	MPI 48	MPI 48	4.5 mils
Note: Sand between all coats per manufacturers recommendations.				

3.12.1.5 MPI Division 9: Interior Plaster, Gypsum Board, Textured Surfaces Paint Table

A. Interior New and Existing, previously painted not otherwise specified

Latex					
New	Existing, previously painted	Primer	Intermediate	Topcoat	System DFT
MPI INT 9.2A-G2 (Flat)	RIN 9.2A-G2 (Flat)	MPI 50	MPI 44	MPI 44	4 mils
MPI INT 9.2A-G3 (Eggshell)	RIN 9.2A-G3 (Eggshell)	MPI 50	MPI 52	MPI 52	4 mils
MPI INT 9.2A-G5 (Semigloss)	RIN 9.2A-G5 (Semigloss)	MPI 50	MPI 54	MPI 54	4 mils
Topcoat: Coating to match adjacent surfaces.					

High Performance Architectural Latex - High Traffic Areas					
New	Existing, previously painted	Primer	Intermediate	Topcoat	System DFT
MPI INT 9.2B-G2 (Flat)	MPI RIN 9.2B-G2 (Flat)	MPI 50	MPI 138	MPI 138	4 mils
MPI INT 9.2B-G3 (Eggshell)	MPI RIN 9.2B-G3 (Eggshell)	MPI 50	MPI 139	MPI 139	4 mils
MPI INT 9.2B-G5 (Semigloss)	MPI RIN 9.2B-G5 (Semigloss)	MPI 50	MPI 141	MPI 141	4 mils
Topcoat: Coating to match adjacent surfaces.					

Institutional Low Odor / Low VOC Latex, New

Institutional Low Odor / Low VOC Latex				
New	Primer	Intermediate	Topcoat	System DFT
MPI INT 9.2M-G2 (Flat)	MPI 149	MPI 144	MPI 144	4 mils
MPI INT 9.2M-G3 (Eggshell)	MPI 149	MPI 145	MPI 145	4 mils
MPI INT 9.2M-G4 (Satin)	MPI 149	MPI 146	MPI 146	4 mils
MPI INT 9.2M-G5 (Semigloss)	MPI 149	MPI 147	MPI 147	4 mils
Topcoat: Coating to match adjacent surfaces.				

Institutional Low Odor / Low VOC Latex, Existing, previously painted

Institutional Low Odor / Low VOC Latex				
Existing, previously painted	Primer	Intermediate	Topcoat	System DFT
MPI RIN 9.2M-G2 (Flat)	MPI 144	MPI 144	MPI 144	4 mils
MPI RIN 9.2M-G3 (Eggshell)	MPI 144	MPI 145	MPI 145	4 mils
MPI RIN 9.2M-G4 (Satin)	MPI 144	MPI 146	MPI 146	4 mils
MPI RIN 9.2M-G5 (Semigloss)	MPI 144	MPI 147	MPI 147	4 mils
Topcoat: Coating to match adjacent surfaces.				

B. Interior New and Existing, previously painted in areas requiring a high degree of sanitation, not otherwise specified

Waterborne Light Industrial Coating					
New, uncoated Existing	Existing, previously painted	Primer	Intermediate	Topcoat	System DFT
MPI INT 9.2L-G5(Semigloss)	MPI RIN 9.2L-G5 (Semigloss)	MPI 50	MPI 153	MPI 153	4 mils
Topcoat: Coating to match adjacent surfaces.					

Alkyd					
New, uncoated Existing	Existing, previously painted	Primer	Intermediate	Topcoat	System DFT
MPI INT 9.2C-G5 (Semigloss)	MPI RIN 9.2C-G5 (Semigloss)	MPI 50	MPI 47	MPI 47	4 mils
Topcoat: Coating to match adjacent surfaces.					

Epoxy, New, uncoated Existing

Epoxy				
New, uncoated Existing	Primer	Intermediate	Topcoat	System DFT
MPI INT 9.2E-G6 (Gloss)	MPI 50	MPI 77	MPI 77	4 mils

Topcoat: Coating to match adjacent surfaces.

Epoxy, Existing, previously painted

Epoxy				
Existing, previously painted	Primer	Intermediate	Topcoat	System DFT
MPI RIN 9.2D-G6 (Gloss)	MPI 17	MPI 77	MPI 77	4 mils
Topcoat: Coating to match adjacent surfaces.				

-- End of Section --

SECTION 31 63 33

INJECTION GROUTED MICROPILES

AMENDMENT 0002

PART 1 GENERAL

1.1 DESCRIPTION

This section includes drilled and pressure grouted micropiles and all associated hardware.

1.2 DEFINITION

Injection Grouted Micropiles: Micropiles drilled using a drill bit fitted on a continuous-thread hollow core steel bar to be permanently grouted into the pile and serve as reinforcing. Piles are installed using rotary percussion equipment and are continuously flushed through the hollow bar and drill bit with drilling grout. Upon reaching tip depth the pile is injection grouted through the hollow bar and drill bit with final grout from the bottom of the excavation. For the purposes of this project augercast micropiles are not considered injection grouted micropiles.

1.3 RELATED WORK

A. Materials testing and inspection during construction: Section 01 45 04.10 06, CONTRACTOR QUALITY CONTROL.

B. Concrete: Section 03 30 00, CAST-IN-PLACE CONCRETE.

1.4 PERFORMANCE REQUIREMENTS

A. Structural Performance: Design and provide piles capable of withstanding the design loads indicated within the limits and under the conditions indicated.

B. Design Standards: Design piles using general acceptable engineering practices and in accordance with the following documents. In case of conflict the more stringent requirement shall be used.

1. IBC 2018

2. FHWA NHI-05-039, "Micropile Design and Construction Guidelines".

C. Geotechnical Capacity: The overall length shall be such that the capacity is developed through skin friction or end bearing as in suitable ground stratum as recommended by the geotechnical report.

1. The allowable geotechnical capacity shall be based on a safety factor of 2.0.

2. Piles shall be capable of resisting the required loading in addition to down drag/negative skin friction forces including those resulting from earthquake induced liquefaction.

3. When determining suitable ground stratum the potential for and

the effects of liquefaction shall be accounted for.

D. Lateral Load and Bending: Where lateral loads are indicated the bending moments from those loads shall be determined using a lateral load analysis program that is capable of modeling the linear elastic properties of the soil column.

1. The modeled soil column shall account for the effects of liquefaction.

2. The allowable load shall be one half of the load that results in a pile head deflection of 1".

3. A pinned head condition shall be assumed unless note otherwise.

E. Structural Capacity:

1. The structural capacity of the pile shall be based on the combined strength of the grout column and the central hollow reinforcing bar.
a. Structural capacity shall account for calculated section loss due to corrosion.
b. The minimum annular grout cover over reinforcing shall be 1.5 inches.
c. The design life for purpose of corrosion loss shall be 75 years.
d. The project soil conditions are mildly aggressive.
e. The minimum effective pile diameter shall be as required by design, but not less than indicated.
f. Drill casing shall extend down to competent bedrock.

2. Where lateral load requirements exist the structural capacity shall account for the interaction of bending and compression, and bending and tension.

3. The pile shall be checked for buckling using a lateral load analysis program that is capable of modeling the linear elastic properties of the soil column including the effects of liquefaction.

4. Load transfer devices and connection hardware shall distribute loads to the foundation as indicated in the drawings and such that concrete stresses do not exceed the limits of ACI 318.

F. Tolerance: Locate piles where indicated. The maximum permissible variation of the center of each pile from the required locations is 6 inches at the ground surface. No pile shall be out of required axial alignment by more than 2 percent. Periodically check the required axial alignment of each pile during the drilling operation and after reaching the required tip elevation with not less than 5 feet of the augerflight extending above the ground surface.

1.5 DESIGN MODIFICATIONS

A. Where piles are installed exceeding specified tolerances for plumb or location, the foundation design will be analyzed by the Structural Engineer of Record and if necessary redesigned by the Structural Engineer of Record. Costs for analysis, redesign, and remediation shall be responsibility of Contractor.

B. Additional piles and pile cap modifications necessitated by redesign shall be furnished and installed, at no additional cost to the Government.

1.6 SUBMITTALS

A. Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for Contractor Quality Control approval. Submit in accordance with Section 01 33 00, SUBMITTAL PROCEDURES.

B. SD-02 Shop Drawings:

1. Delegated Design Submittal; G, CE

2. Pile Foundation Plan; G, CE

3. Installation Sequence Narrative; G, CE

4. Reinforcing for Piles and Anchorage to Foundations; G, CE

5. Pile Schedule; G, CE

C. SD-03 Product Data:

1. Hollow Thread Bar; G, CE

2. Couplers; G, CE

3. Centralizers/Spacers; G, CE

4. Drill Bits

5. Nuts; G, CE

6. Plates; G, CE

D. SD-05 Design Data:

1. Grout Mix Design; G, CE

E. SD-06 Test Reports:

1. Pile Load Testing; G, CE

F. SD-07 Certificates:

1. Installer's Qualifications

2. Testing Equipment Calibration

1.7 SUBMITTAL REQUIREMENTS

A. Shop Drawings and Miscellaneous Submittals:

1. The shop drawing package shall be signed and sealed by the professional engineer responsible for the pile design.

2. Show reinforcing for piles and anchorage to foundations. Include material, grade, and dimensional information for steel elements proposed for use.

3. Include a pile schedule for each type or capacity of pile including the following:
a. Product designations threaded hollow reinforcing bar, couplers, drill bits, and all ancillary products to be supplied for that type or capacity of pile
b. Grout body diameter
c. Final grout strength
d. The allowable pile load as limited by mechanical or geotechnical allowable strength.
e. Required pile flexural, compression, and tension capacities
f. The calculated allowable mechanical strength for the pile assembly including connection hardware and load transfer device. The strength shall account for the designed tolerance
g. The allowable theoretical geotechnical capacity.
h. Embedment length
i. Inclination angle tolerance requirement
j. Location tolerance requirement

B. Installation sequence narrative: A written narrative description of equipment, installation procedures, and installation sequences to be used. The narrative shall include any necessary measures for ensuring the micropile hole is held open and clear until final grout is placed including any casings if deemed necessary.

C. Delegated-Design Submittal:

1. Provide structural design data signed and sealed by the qualified professional engineer responsible for their preparation. Calculations shall include the following:
a. Calculated allowable theoretical geotechnical capacity for each type of pile.
b. Calculated allowable structural capacity for each type of pile.
c. Calculated allowable lateral load for each type of pile.
d. Maximum location and inclination angle tolerance accounted for in the design of the pile, load transfer device and connection hardware.

D. Qualification Data:

1. Engineer:
a. Evidence of registration to practice structural engineering in the jurisdiction where Project is located.
b. A written recommendation from the pile manufacturer, pile distributor, or manufacturer's representative for the unit being specified indicating competence in the design using the specific unit.
c. A list of at least three projects which were designed by the engineer within the previous three years using piles of types similar to those required for this project, using equipment similar to that required for this project, and under similar conditions required for this project.

2. Installer:
a. Certificate from unit manufacturer that the installer has been trained and is certified for the installation of the unit.
b. A list of at least three projects completed within the previous three years wherein the installer installed piles of types similar to those required for this project, using equipment similar to that required for this project, and under similar conditions required for this project.

E. Design Mixtures: For each final grout mixture. Submit alternate design mixtures when characteristics of materials, project conditions, weather, test results, or other circumstances warrant adjustments.

1. Mix design submittals shall include field test results and/or trial batch data the meet or exceed the required average compressive strength.

2. Trial batches shall consist of identical cementitious materials, fine and course aggregates, and admixtures to be used for mix design.

F. Installation Equipment Data:

1. Indicate type, make, and rating of equipment to be used for pile installation.
a. Drilling Equipment
b. Grout Mixer and Pump
c. Drill Bits

G. Testing Equipment Data

1. Indicate type, make, and rating of all equipment to be used for testing.

2. Copies of certified calibration report for all loading and measurement devices to be used in testing piles. The calibrations shall be dated within one year of the proposed testing date.

H. As built pile survey

1.8 QUALITY ASSURANCE

A. The Contracting Officer shall retain the services of a Geotechnical Consultant (Consultant) to provide general observation of all pile operations and to provide technical advice to the Contracting Officer with regard to pile operations and performance.

B. Engineering Responsibility: Preparation of Shop Drawings, design calculations, and other structural data by a qualified professional engineer.

C. Professional Engineer Qualifications:

1. A professional engineer who is legally qualified to practice in the jurisdiction where Project is located and who is experienced

in providing engineering services of the kind indicated.

2. Shall be able to obtain a recommendation from the pile reinforcing and accessory manufacturer, distributor, or manufacturer's representative for the type of reinforcing and accessories being specified indicating competence in the design using the specific products.

3. The engineer shall have completed the design for at least three projects within the previous three years using piles of types similar to those required for this project, using similar equipment for this project, and under similar conditions required for this project.

D. Installer's Qualifications:

1. Manufacturer's authorized representative who is trained and approved for installation of units required for this Project.

2. The installer shall have completed at least three projects within the previous three years wherein the installer installed piles of types similar to those required for this project, using similar equipment required for this project, and under similar conditions required for this project.

3. Ability to obtain field support from the pile manufacturer, the pile distributor or the manufacturer's representative.

E. Testing Agency Qualifications: Qualified according to ASTM E 329 for testing indicated.

F. Pre-installation Conference: Conduct conference at Project site.

1. Review test pile requirements

2. Review test pile inspection requirements

3. Review production pile installation requirements

4. Review production pile installation tolerances

5. Review production pile inspection requirements

1.9 DELIVERY, STORAGE, AND HANDLING

A. Deliver store and handle steel reinforcement to prevent bending or damage to threads.

B. Store reinforcement on dunnage elevated off of the ground.

C. Store all pile accessories in manufacturer's packaging until such time they are installed.

1.10 PROJECT CONDITIONS

A. Protect structures, underground utilities, and other construction from damage caused by pile installation.

B. Site Information: A geotechnical report has been prepared for this Project and is provided as an attachment to these specifications for information only.

PART 2 PRODUCTS

2.1 STEEL REINFORCEMENT

A. Continuously Hollow Threaded Bar

1. ASTM A615, ASTM A722, ASTM A519 or equivalent, Grade as required by structural design.
a. Thread pattern meeting or exceeding the bond characteristics of ASTM A615.

B. Steel Casing

1. ASTM A252, Grade3, with minimum yield as required by structural design.

2. Wall thickness: As required by structural design, but not less than 3/16 inch.

2.2 ACCESSORIES

A. Centralizers/Spacers: Rigid and durable mechanical device which fits over and anchors to the hollow threaded bar and maintains the hollow threaded bar central to the drill hole and grout column.

1. Device must be capable of firmly attaching to the hollow thread bar and be sufficiently robust to withstand the drilling and grouting process.

2. Device must be configured such that it allows final grout to flow freely up the drillhole.

3. Materials:
a. Schedule 40 PVC
b. Steel

B. Couplers: Cast steel element of strength, grade, and configuration capable of developing the strength of two joined hollow threaded bars. Thread pattern shall be consistent with hollow threaded bar.

C. Plates: Structural steel plates of grade and size to resist the required forces.

D. Nuts: Cast steel element of strength, grade, and configuration capable of developing the strength of the hollow thread bar.

2.3 GROUT MATERIALS

A. Cementitious Material: Use the following cementitious materials, of the same type, brand, and source, throughout Project:
1. Portland Cement: ASTM C 150, Type I, Type I/II or Type III, Supplement with the following:

B. Fine Aggregates: ASTM C 33.

1. Provide aggregates from a single source with documented service

record data of at least 10 years' satisfactory service in similar applications and service conditions using similar aggregates and cementitious materials.

2. Free of materials with deleterious reactivity to alkali in cement.

C. Water: Potable, and complying with ASTM C94

D. Admixtures: Provide admixtures certified by manufacturer to be compatible with other admixtures.

1. Air-Entraining Admixture: ASTM C 260.

2. Water-Reducing Admixture: ASTM C 494/C 494M, Type A.

3. Retarding Admixture: ASTM C 494/C 494M, Type B.

4. Water-Reducing and Retarding Admixture: ASTM C 494/C 494M, Type D.

5. Water-Reducing and Accelerating Admixture: ASTM C 494/C 494M, Type E.

6. High-Range, Water-Reducing Admixture: ASTM C 494/C 494M, Type F.

7. High-Range, Water-Reducing and Retarding Admixture: ASTM C 494/C 494M, Type G.

8. Plasticizing and Retarding Admixture: ASTM C 1017/C 1017M, Type II.

2.4 GROUT MIXES

A. Prepare design mixtures for each type and strength of concrete, proportioned on the basis of laboratory trial mixture or field test data, or both, according to ACI 301.

1. Use a qualified independent testing agency for preparing and reporting proposed mixture designs based on laboratory trial mixtures.

B. Grout shall be a neat cement grout or sand-cement grout.

C. Approved admixtures may be used in grout if they were included in the mix design.

D. Drilling and Flushing grout shall have a minimum water-cement ratio of 0.70.

E. Final Grout shall have a minimum water-cement ration of 0.45.

F. Compressive Strength: As required by delegated design, but not less than 4000 psi.

G. The grout mix shall be capable of maintaining the solids in suspension, shall be pumped without difficulty, and shall be capable of penetrating and filling voids in the adjacent soils.

H. Limit water-soluble chloride ion content in hardened concrete to 0.15 percent by weight of cement

2.5 INSTALLATION EQUIPMENT

A. General:
1. All equipment shall be capable of accessing the site given the existing limits of accessibility and any other limits that may exists as construction progresses.
2. All equipment shall be capable of installing the piles within the existing space limits, including headroom and any other space limitation that may existing as construction progresses.
3. The installation contractor shall be responsible for visiting the site to confirm the proposed equipment can access the site and work within the space limits.

B. Drilling Equipment: Hydraulic rotary or rotary/percussion drill accommodating the required reinforcing steel sizes, drill bits, and drilling conditions.

C. Grout Mixer and Pump: A system consisting of a separate holding tank and water and cement dosing system to assure continuous grouting independent from mixing.

1. Minimum pressure capability of 150 psi or as required to install the submitted piles.

2. An integral and automated monitor capable of measuring and recording grout pressure and volume

3. The mixer shall be capable of continuous agitation of the grout.

4. The mixer and pump shall be sized to allow for grout to be pumped in one continuous operation.

D. Drill Bits: A bit of type and size required for the expected soil conditions and provided the required clear grout column.

2.6 TESTING EQUIPMENT

A. Reaction System:

1. The reaction system shall be designed by a registered professional engineer, and shall have sufficient strength and stiffness to distribute test loads to the ground.

2. The reaction system shall be designed to minimize movement under load and prevent applying an eccentric load to the pile head.

3. The direction of the load shall be co-linear with the pile at all times.

4. Anchor piles and/or reaction piles of sufficient quantity, strength and stiffness shall be provided to run the required tests.

B. Loading Device:

1. The loading device shall have a load capacity of the allowable load times the required safety factor, but not less than the ultimate capacity of the proposed reinforcing steel.

2. The loading device shall be capable of increasing or decreasing the applied load incrementally.

3. The loading device shall have incremental control to establish and maintain a desired load for sustained, hold period.

4. The stroke of the loading device shall not be less than the theoretical elastic shortening of the pile length at the maximum test load

5. The loading device shall be capable of applying a load not less than two times the proposed design loads.

6. The loading device measurement gauges shall be graduated in 100 psi increments or less.

7. The loading device and any associated gauges shall have been calibrated within a one year period of the expecting testing date.

C. Deflection Measurement:

1. Dial gauges shall be used to measure pile movement during testing.

2. Dial gauges shall have an accuracy of at least +/-0.001 inch and a minimum travel sufficient to measure all helical pile movements without resetting the gauge.

3. Gauges shall be placed with stems parallel to the pile

4. Gauges shall be supported by a reference apparatus to provide an independent fixed reference point. The reference shall be independent of the reaction system and shall not be affected by any movement of the reaction system.

5. All gauges shall have been calibrated within a one year period of the expecting testing date.

PART 3 EXECUTION

3.1 EXAMINATION

A. Site Conditions:

1. Do not start pile installation operations until earthwork fills have been completed or excavations have reached an elevation of 6 to 12 inches above bottom of footing or pile cap.

2. Located and protect existing structures, underground utilities, and other construction from damage caused by pile installation.

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3.2 STATIC PILE TESTS

A. General: Proposed pile lengths and diameter will be verified by load testing.

1. Axial Compressive Static Load Test: ASTM D 1143.

2. Axial Tension Static Load Test: ASTM D 3689.

3. Lateral Load Test: ASTM D 3966.

B. Installing Test Piles: Installation methods, procedures, equipment, and overall length shall be identical to the proposed production pies to the extent practical.

C. Testing Process:

1. Contracting Officer's inspector shall observe and log the installation of test piles.

2. Notify Contracting Officer's inspector at least 96 hours in advance of performing tests.

3. Conduct testing with Contracting Officer's inspector present to audit and record results of testing program and report to the Contracting Officer.

4. If test piles fail to meet the design requirements and associated performance criteria the contractor shall modify the pile design and/or installation methods and complete additional test piles.
a. The cost of the additional test piles and associated anchor or reaction piles shall be the burden of the contractor.
b. The cost of additional testing and inspection to be completed by the Contracting Officer's inspector shall be the burden of the contractor.

5. On completion of testing, remove testing structure, equipment, and instrumentation.

6. Test piles, anchor piles and reaction piles shall be sacrificial, and shall not be used as production piles.

D. Number of Test Piles:
1. Two tension test piles for each proposed combination of length, diameter and reinforcement.

2. Two compression test piles for each proposed combination of length, diameter and reinforcement.
3. One lateral test piles for each proposed combination of length, diameter and reinforcement.

E. Approval Criteria: Pile shall be tested to 2.0 times the design load and shall be deemed approved if they meet all of the following:

1. The net settlement after rebound does not exceed 0.5 inches.

2. There is not a sharp break in the load settlement curve at any time during the loading sequence.

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3.3 INSTALLING PILES

A. General: Installation technique shall be consistent with the geotechnical, logistical, environ-mental and load carrying conditions of the project.

1. Where test piles where employed production piles shall be installed with the same techniques of that used for test piles.

B. Preparation:

1. Stage the necessary reinforcement, couplers and other accessories for the pile prior to commencing pile installation.

2. Stage materials on dunnage out of contact with soil.

3. Place centralizers/spacers on hollow thread bars such that the resulting spacing of centralizers/spacers will not exceed 8 feet on center vertically. Locate one centralizer spacer not more than 4 feet from top and bottom of micropile.

4. Clean all bars, couplers and other devices to ensure they are free of dirt. Extra care shall be taken to ensure the interior of hollow thread bars are clean and free of dirt.

5. Mix sufficient flushing grout and move to holding tank.

C. Drilling:

1. Flushing grout shall be continuously pumped through the drill bit while drilling.

2. Use top hammer as needed where harder ground, boulders or rocks are encountered.

3. Install flush joint threaded drill casing as required for highly unstable soils to prevent the drilled hole from collapsing. Casing shall not advance into bond zone where skin friction is developed, or shall be pulled back from this zone as final grout is placed.

4. Work hollow reinforcing in and out several times for each ten foot length of pile installed.

5. Excess flushing grout and trimmings that exit the hole during drilling and grouting shall be channeled or direct away from the drill hole as to not alter soil stability around the drill hole or aggravate existing environmental conditions.

D. Grouting:

1. Immediately upon advancing the drill to the desired depth the flushing grout shall be stopped and commence pumping final grout.

2. Continue to rotate the drill and work the reinforcing in and

out five to ten feet as final grout is being pumped.

3. Final grout shall be applied under the pressure required to produce the required pile diameter and pile bond values without causing detrimental side effects.

4. Final grout shall be placed until the theoretical quantity is exceeded and there is visual evidence that all flushing grout and trimmings have evacuated the hole, and the final grout has reached the top of the pile.

5. Grouting shall be completed in a continuous operation within the time frame of the initial set time of the grout mixture.

E. Cleanup:

1. Unless approved otherwise all evacuated flushing grout, trimmings, other debris from drilling operations and excess materials shall be removed from the site.

F. Installation Tolerances: Install piles without exceeding the following tolerances, measured at pile heads:

1. Location: 6 inches from specified location,

2. Inclination Angle: Maintain alignment within 2 percent from vertical.

3. Cutoff elevation: Shall not be higher than, and not more than 2 inches below the specified design elevation

G. If a pile quality control criterion is not met for any reason, proposed remedy is subject to the approval of the Contracting Officer prior to implementing the remedy. It shall be the burden of the contractor to substantiate the remedy to the Contracting Officer.

H. Withdraw damaged or defective piles and piles that exceed tolerances and install new piles within driving tolerances.

1. Fill holes left by withdrawn piles as directed by Contracting Officer.

2. If rejected piles cannot be practically withdrawn, abandon and cut off as directed by the Contracting Officer. Install additional piles in new location as directed by the Contracting Officer.

I. As-Built Survey: Contractor shall provide an as-built survey to the engineer of record. The survey shall show all piles with deviations from theoretical pile location, pile alignment and pile butt elevation. All pile locations, alignments and cutoff elevations that are not within tolerance shall be specifically noted.

1. Foundation work should not commence until the as-built survey has been reviewed and/or approved and/or associated foundation revisions have been made to account for any out of tolerance piles.

J. Additional pile and/or foundation work as the resultant of defective or out of tolerance piles shall be completed at no additional cost or time penalty to the Contracting Officer.

K. The cost of additional analysis and design of piles and/or foundations as the resultant of defective or out of tolerance pile shall be the responsibility of the contractor and reimbursed to the Contracting Officer as a change order to the contract.

3.4 FIELD QUALITY CONTROL

A. Testing: Contracting Officer will engage a qualified independent testing and inspecting agency to perform field tests and inspections and prepare test reports in accordance with the schedule of special inspections.

B. Correct deficiencies in Work that test reports and inspections indicate does not comply with the Contract Documents.

C. Additional testing and inspecting, at Contractor's expense, will be performed to determine compliance of replaced or additional work with specified requirements.

-- End of Section --

APPENDICES

APPENDIX B

ELECTRONIC EQUIPMENT PICTURES

B145 SWGR NAMEPLATE

B145 SWGR DIMENSIONS

B146 EXISTING BREAKER

APPENDIX C

RQ OI 32-10 EAST

BY ORDER OF THE DIRECTOR
AIR FORCE RESEARCH LABORATORY (AFRL)

RQ OPERATING INSTRUCTION 32-10_East

12 December 2023

CIVIL ENGINEERING

ELECTRICAL POWER SYSTEMS AT RQ-EAST (Wright-Patterson AFB)

COMPLIANCE WITH THIS PUBLICATION IS MANDATORY

ACCESSIBILITY: Publications and forms are available in the official AFRL/RQ Publications and Forms Libraries at: https://usaf.dps.mil/sites/20072/8/SitePages/RQ-Publications.aspx

RELEASABILITY: There are no releasability restrictions on this publication.

OPR: AFRL/RQOO

Certified by: AFRL/RQO (Ms. Olivia K. Beal)

Pages: 40

This operating instruction (OI) implements Air Force Policy Directive (AFPD) 32-10, *Installations and Facilities.* This instruction provides Lockout Tagout (LOTO) guidance, references to other Air Force Occupational Safety, Fire, and Health Standards, and establishes procedures and policies for the installation and use of all electrical power systems AC and DC voltages above 50V throughout the Directorate. This Aerospace Systems Directorate (RQ) OI establishes standards that will be followed throughout the Directorate for electrical protection, connected loads, and electrical connectors for all RQ electrical power systems (EPSs). This RQ OI provides safety guidance within RQ utilizing the U.S. Air Force Mishap Prevention Program and reference to other USAF Mishap Prevention Program language. Refer recommended changes and questions about this publication to the Office of Primary Responsibility (OPR) using the Air Force Form 847, *Recommendation for Change of Publication*; route AF Form 847s from the field through the appropriate functional chain of command. Submit variance requests through the chain of command to the appropriate tier approval authority, or alternately, to the Publication OPR for non-tiered compliance items. This OI is mandatory; all end users will ensure that all records created as a result of processes prescribed in this publication are maintained in accordance with (IAW) AFI 33-322, *Records Management and Information Governance Program*, and disposed IAW the *Air Force Records Disposition Schedule* located in the *Air Force Records Information Management System*. The use of the name or mark of any specific manufacturer, commercial product, commodity, or service in this publication does not imply endorsement by the Air Force.

SUMMARY OF CHANGES

This document must be completely reviewed. This new instruction replaces the rescinded RZ OI 32-1 and was created to align with AFPD 32-10, *Installations and Facilities,* DAFMAN 91-203, Chapter 21*, Hazardous Energy Control LOTO* and 29 CFR 1910.147.

1. Overview

RQ's OI 32-10 details MANDATORY COMPLIANCE language, under one title, for AC and DC voltage systems and Safety, that will be applied to electrical designs, installations, modifications, Hazardous Energy Control LOTO, and electrical hazardous area determinations (EHADs) by RQ personnel and contractors working with or on RQ's electrical power distribution systems (EPDS) at WPAFB.

2. Roles and Responsibilities

2.1. RQ Personnel

All personnel assigned to or working within RQ facilities at WPAFB are responsible for compliance with this OI.

2.2. Branch Chiefs/Supervisors

Branch Chiefs/Supervisors, specific to each room, space, or area will ensure that personnel are qualified to perform assigned tasks related to RQ's EPDS, have adequate tools, training, and safety equipment, and conform to the requirements of this OI, as well as other applicable electrical codes, standards, and regulations.

3. Procedures

3.1. Application

3.1.1. Unless specifically stated, this OI pertains to new work and modifications of EPDSs and does not apply to existing conditions that were installed following previous guidelines.

3.2. Electrical Work Approval

RQOO must be notified of equipment or receptacles, installed, or modified (both temporary and permanent), that are attached to RQ EPDS. The exception is single pole loads, 16A or less, plugged into existing 15 or 20A 120VAC nominal convenience receptacles. Electrical work requests can be submitted online by anyone in the AFRL/RQ East community by using FIST. Information furnished will be as described in the Design and Analysis section of this OI.

3.3. Design and Analysis

Electrical work will not be performed on EPDSs without first obtaining approval on a submitted electrical design. An electronic copy of an electrical design showing the information listed in this Design and Analysis section of this OI will be provided to RQOO and approved prior to starting any work on the electrical systems described above. The design can be attached to the FIST ticket or submitted separately. If approved, the documentation will be returned to the initiator and noted as approved. If not approved, the documentation will be returned with comments for edits and resubmission of the edited version. RQOO will notify Base Civil Engineering and the Electrical Utility Privatization (UP) Contractor when required. The user will notify RQOO when the work is being performed. RQOO will provide inspection and approval of the work.

3.3.1. A National Electrical Code (NEC) compliant electrical design will be accomplished for new installations and modifications of existing installations. Documentation of this design will be provided electronically as a sketch or drawing showing the following elements:

3.3.1.1. A one-line diagram and electrical equipment layout with information including voltage and phase, load information (current, horsepower and or wattage), description and location.

3.3.1.2. Include electrical size of wires and cables, conduits, breakers, fuses, contactors, overloads, and disconnect switches.

3.3.2. Additions and modifications that add more than 100A of load on EPDSs between 50V and 1000V, and all loads above 1000V will have the following additional analysis provided. Consult with RQOO prior to performing these studies and before establishing the final budget for the work. RQOO will provide guidelines for each analysis.

3.3.2.1. A load analysis to determine the size of the new load and appropriate sizing of the existing distribution system.

3.3.2.2. A short circuit study to determine device specifications prior to purchase.

3.3.2.3. A protective device coordination study.

3.3.2.4. An Arc Flash study.

3.4 Testing

3.4.1. Tests will conform to the latest edition of the International Electrical Testing Association (NETA) Acceptance and Maintenance Testing Specifications (ATS and MTS). Testing will be accomplished by a NETA accredited company. Testing required by 3.4.3 may be accomplished by the installer. Test results will be provided to RQOO prior to energization.

3.4.2. New equipment will be acceptance tested prior to energization. Optional tests are not required. Any relocated equipment will have had tests performed within 3 years or be retested.

3.4.3. Newly installed conductors operating at 460V or greater, all 480V feeds to panels or power centers, all 208Y/120V feeds to panels or power centers, and all DC conductors operating at 100VDC or greater, will have insulation resistance (Megger) testing prior to energization.

3.4.4. Newly installed conductors operating at greater than 1000V will be submitted to an electrical insulation resistance (Megger) test, including Polarization Index, and a Very Low Frequency dielectric withstand resistance test prior to energization.

3.4.5. Conductors with a 600V or less insulation rating, for which the result of the insulation resistance (Megger) test is less than 2.0 megohm (1 minute reading) or show a polarization index of 2.0 or less will be replaced. RQOO will be notified of these conditions.

3.4.6. Medium voltage conductors having an insulation resistance (Megger) reading of 20 megohms or less or show a polarization index of 2.0 or less will be replaced. RQOO will be notified of these conditions.

3.4.7. 3-phase equipment will be tested for phase rotation in accordance with NEMA Standards when wiring is initially installed or has been modified.

3.4.8. Temporary Protective Grounds will comply with ASTM F855.

3.4.9. Testing of Grounds: Grounds will be tested in accordance with the below table and NETA guidelines. RQOO will be notified of grounds that test above the maximum levels listed. All grounds will be labeled with their resistance and the date tested with a standard embossed metal label.

Table 3.1 Ground Testing Requirements

Ground Type	Testing Frequency	Maximum Resistance
Service grounding electrode	6 years	25 ohms
Instrumentation ground	3 years	0.3 ohms
Lightning protection ground	6 years	5 ohms
Fuel farm static grounds	6 years	5 ohms
Equipotential bonding grounds	3 years	1 ohm

3.5 Electrical Outages

3.5.1 Electrical outages required in an AFRL/RQ building will be coordinated with RQOE, RQOO, and RQOR. For whole-building outages and outages that require coordination with the utility company, a minimum notice of 15 working days is required. Localized outages that only affect one room or panel may be coordinated in less time. See Attachment 2 for an example outage request form. This form may be modified as needed. An editable MS Word or Adobe .pdf document can be provided upon request.

4. Safety

4.1. Equipment or materials will not be located, temporarily or otherwise, within a 3-foot radius nor permanently located within a 4-foot radius of any floor-mounted switchgear, transformer, or motor generator set.

4.2. A clear aisle must be left in front of wall-mounted distribution panels and disconnecting means per NEC defined access and working space areas. This area must be a minimum clearance of 36" for 120V devices and 42" for 460 or 480V devices.

4.3. Equipment will not be stored in any transformer vault, switchgear enclosure, electrical panel, bus tie compartment, or electrical equipment room unless designed for that purpose. In all cases, section 4.1 must be observed.

4.4. Before breaking the seal or other tamper resistant device on any power system relay, timer, or meter, written approval must be obtained from RQOO. For circuit breakers with adjustable trip settings, implement NEC 240.6(C) as a minimum for seals on adjustments where cable has a lower amperage rating than the maximum circuit breaker setting. Seals will be Schneider Electric MICROTUSEAL or approved equal tamper resistant device.

4.5. When working on energized circuits above 50V or in locations where unguarded energized circuits are present, strict adherence to National Fire Protection Association (NFPA) 70E and AFMAN 32-1065 is required (reference IEEE 1584 for arc flash evaluation criteria). Submit Energized Electrical Work Permit to RQOO for approval. Refer to Attachment 3 for permit form. An editable MS Word or Adobe .pdf document can be provided upon request.

4.6. LOTO Procedures are performed in accordance with DAFMAN 91-203 and OSHA standards.

4.6.1 Government personnel, for each RQ branch, will maintain written LOTO procedures in coordination with RQY (RQ Safety Office).

4.6.2 Contractors will maintain written LOTO procedures for their personnel.

4.7. Temporary protective grounds will comply with ASTM F855.

5. Installations

5.1. Equipment

5.1.1 Electrical distribution equipment and connected loads will be installed in accordance with the latest edition of the NEC and other applicable electrical codes, standards, and regulations.

5.1.2 When commercially available, electrical supplies, devices and equipment will have markings to show that the device has been tested for conformance by a third-party testing laboratory. These devices will have a mark from a Nationally Recognized Testing Laboratory (NRTL).

5.2. Medium Voltage (greater than 1000V)

5.2.1 Power cables for 6900 and 12470V circuits will be stranded copper, 15 KV rated, type EPR Insulation with PVC jacket, MV-90 or MV-105, copper tape shield with 25% minimum overlap, and 133% insulation.

5.2.2 Control wiring on 6900V or greater switchgear will be no smaller than #14 AWG type SIS (Thermoset cable or Switchboard or Panelboard**)** wire and terminated with uninsulated locking fork-type compression terminals onto compatible terminal blocks. Wiring from Current Transformers (CTs) will be no smaller than #12 AWG type SIS wire for the entire circuit. CTs will be terminated directly onto compatible shorting-type terminal blocks at the first termination point after leaving the CTs. The sole grounding point of the CT circuit will be located at the shorting type of terminal blocks nearest the CTs.

5.2.3 On 6900V and 12470V systems, cables will be restricted from movement by applying cable ties and tie-downs on open cable runs every 3 feet for EPR cable and every foot for unshielded internal hookup cable.

5.2.4 Phase conductors of multiple 6900V and 12470V circuits routed through raceways will be oriented to reduce circuit impedance and coronal effects.

5.3. Low Voltage Systems (1000V or less)

5.3.1 Wiring used on power circuits will be stranded copper, 600V, with insulation rated 75ºC wet and 90ºC dry or better (i.e., THHN/THWN or XHHW). Power cables #2 AWG or larger will be stranded copper type XHHW or XHHW-2 insulation.

5.3.2 Power conductors must be #12 AWG or larger. Control wiring 50V or greater, AC or DC, must be stranded copper and sized as specified for the control circuit load in accordance with the NEC, but no smaller than #16 AWG.

5.3.3 Circuit conductors must be sized based on the 60°C ampacity for circuits rated 100 amperes or less (#12 AWG through #1 AWG) and the 75° C ampacity for circuits rated over 100 amperes (larger than #1 AWG). Higher temperature ampacity ratings for conductors may only be used for ampacity adjustment or correction as permitted by NFPA 70. (UFC 3-520-01)

5.3.4 Control wiring on 600V or less PDCs and LDCs will be no smaller than #14 AWG type SIS wire and terminated with uninsulated locking fork-type compression terminals onto compatible terminal blocks. Wiring from Current Transformers (CTs) will be no smaller than #12 AWG type SIS wire for the entire circuit. CTs will be terminated directly onto compatible shorting-type terminal blocks immediately after leaving the CTs. The sole grounding point of the CT circuit will be located at the shorting-type terminal blocks nearest the CTs.

5.3.5 Insulated equipment grounding conductors (EGC) will be pulled in the same raceway with primary and secondary distribution circuits, and branch circuits. An EGC will also be used for 120V control circuits protected by overcurrent protection devices 15A or greater. The conductor size will be as specified in Article 250, Table 250.122 of the NEC. This EGC wire will be connected to each grounding bushing or grounding hub and enclosure including metal junction boxes, pull boxes, terminations, etc. EGCs entering an enclosure will be bonded to the same point in the enclosure. An enclosure will not purposely be used as an EGC pathway. EGCs for parallel runs will be properly sized based upon the ampere rating of the overcurrent protecting device.

5.3.6 Power conductors will be terminated in compression type 75ºC minimum, mechanical fasteners (lugs) that totally encircle the cable being terminated, unless listed 75ºC mechanical fasteners are provided by the manufacturer. For cables above 1000V, or below 1000V but above 800A, terminations will be lug-type compression. Cables 1000V or less and #2 AWG or larger, the crimp will be hexagonal and fully formed in one pass. A 15-ton hexagonal crimper will be used with die size used as shown on the lug. For applications over 1000V, a hydraulic 15-ton elliptical crimp will be used. This crimp requires two passes, 90 degrees apart to form a full crimp. A 15-ton ratchet-type or hydraulic-type crimper that only releases when full rated crimp pressure is attained, is required for all crimps.

5.3.7 No aluminum screw lugs will be used on #2 AWG or larger cable terminations. In addition, recessed hex-type clamping screws are preferred.

5.3.8 No split-bolt (i.e., "Kearney") connections may be used.

5.3.9 Drawout-type circuit breakers will be used for the main breaker of LDCs and for all breakers of PDCs with bus ratings of 2000A or above. Drawout-type circuit breakers will include electronic trip units and trip cause indicators. All other circuit breakers in the 480V or less systems will be molded case type. Molded case type circuit breakers used in Lighting Distribution Panels (LDPs), LDCs, Power Distribution Panels (PDPs), and PDCs will have their line side directly bolted to the switchgear bus and, for LDCs and PDCs, have provisions for 75ºC bolt-on copper lugs for load side connections.

5.3.10 To satisfy NEC 240.87 for Arc Energy Reduction for new equipment, ZSI (zone-selective interlocking) is the preferred method of protection.

5.3.11 Circuit breakers 1000A and larger will have LSIG (long time, short time, instantaneous, ground) protection on solidly grounded systems to satisfy NEC 230.13

5.3.12 Bolt-on molded case circuit breakers will be the only type used in Lighting Panels (LPs) and Power Panels (PPs). These circuit breakers will have 75ºC load terminals for breakers larger than 100 A.

5.3.13 All circuit breakers rated greater than 225A will be equipped with electronic trip units.

5.3.14 Only one-time fuses of the same manufacturer, size, and class will be used when replacing fuses. Replaceable element fuses will not be used and when found, immediately replaced. RK1 fuses for standard loads and RK5 fuses for motor loads will be used if available in the required size and form factor. No glass fuses may be used except as manufacturer's replacements in equipment control circuits.

5.3.15 Power distribution (PDP's, PDC's, buses, etc.) will have meters capable of displaying the peak demand power and current.

5.3.16 New panelboards will be Y connected unless a variance is approved. Y-connected panelboards will be supplied with an isolated neutral bus and separate ground bus. Bonding of the system neutral to ground will occur only at the transformer feeding the panel.

5.3.17 New panelboards will have a hinged "door-in-door" cover with the following:

5.3.17.1 Lockable interior hinged door with hand-operated latch or latches as required to provide access to circuit breaker operating handles only; not to energized parts.

5.3.17.2 Outer hinged door to provide access to the entire enclosure, including deadfront and all wiring gutters.

5.3.17.3 Outer door must be securely mounted to the panelboard box with factory bolts, screws, clips, or other fasteners requiring a tool for entry; hand-operated latches are not acceptable.

5.3.18 Panel schedules for existing panels will be updated after adding and/or removing loads.

5.3.19 208Y/120V or 120/240V Branch Circuit Panelboards shall have 200% rated neutral bus per WPAFB IFS Appendix G10, July 2021.

5.3.20 Adjustable speed drives on rotating equipment must be within the harmonic distortion levels listed in section 6 Restrictions, or have a reactor installed at its line input terminals and mounted either internal to the drive or outside the drive in a separate enclosure. If required, the reactor impedance will be 5% or greater of the drive load rating. See Attachment 4 for a list of recommended manufacturers.

5.3.21 Conductors entering a motor starter, contactor, disconnect switch or relay enclosure will terminate on a terminal strip or on the enclosed device. No splices will be made inside the enclosure. No through-conductors will be permitted in any device enclosure.

5.3.22 Receptacles wired to an emergency generator will be red and/or have red coverplates.

5.3.23 Receptacles wired to an UPS will be orange and/or have orange coverplates.

5.3.24 Phase sequence will be verified prior to and after the energization of new equipment and documented with an adhesive label on the equipment including the date as either A-B-C or A-C-B.

5.3.25 In installations where extreme power distribution flexibility is desired, such as in an open Lab area, a track busway with plug-in units may be provided. See Attachment 4 for recommended manufacturers. This can only be used for 208Y/120V service. No single circuit tap can be greater than 60A. 480Y/277V plug bus is allowed when designed and installed in accordance with this OI.

5.3.26 Industrial control panels will follow UL 508A Standard for Industrial Control Panels.

5.3.26.1 Lamps for indication will have incandescent bases (BA9s, for example) with LED lamps.

5.3.26.2 Control panels will have main power disconnecting means.

5.3.26.3 Control panels will be marked with SCCR (short-circuit current rating).

5.3.27 Cable tray is allowed where the installation is compliant with the NEC. See NEC article 392 for maximum fill area tables and other cable tray-specific requirements. A label stating maximum allowable fill area will be attached to the cable tray.

5.4. Raceways

5.4.1 Rigid conduit runs will be terminated in grounding type bushings or grounding type hubs.

5.4.2 Electrical distribution circuits and control circuits that are run outdoors in rigid conduit will use NEMA 4X enclosures. Rigid conduit installations indoors will use NEMA 12 or better metallic enclosures. All other indoor enclosures will be NEMA 1.

5.4.3 **Conduit and Enclosures:** The selection of raceways and enclosures is based upon the level of protection needed for the environment and the level of hazard posed by the circuits in the raceway. The following Environmental Conditions describe the areas where the raceway and enclosure are to be installed and the minimum required conduit and enclosure rating that is to be used in that area. For areas classified as hazardous (i.e., requiring explosion-proof devices), only the specific raceways and enclosures referenced in the NEC are to be used.

5.4.3.1 **Environmental Condition 1:** This is where raceways and enclosures are guarded from physical or environmental damage OR are to be installed in office areas.

5.4.3.1.1 Only ¾ inch or larger conduit is to be installed except for office lighting where ½ inch is allowed for flexible connections to fixtures.

5.4.3.1.2 Electrical Metallic Tubing is used.

5.4.3.1.3 Knock-out boxes of sufficient volume per NEC are to be used.

5.4.3.1.4 Flexible Metallic Conduit may be used for light fixtures but is limited to lengths of 6 feet or less.

5.4.3.2 **Environment Condition 2:** This is where conduits and enclosures may endure physical damage but otherwise are in a dry location without any corrosive substances. Example: Indoor high bays or indoor mechanical spaces without corrosives (water or chemicals).

5.4.3.2.1 Only ¾ inch or larger conduit is to be installed.

5.4.3.2.2 Rigid Galvanized Steel (RGS) Conduit is used.

5.4.3.2.3 Knock-out boxes may be used in runs where the conduit size is 1 inch or less. For runs using conduit larger than 1-inch, solid enclosures are to be used.

5.4.3.2.4 Grounding bushings or grounding hubs are used at every enclosure.

5.4.3.2.5 Liquid tight flexible metallic conduit may be used for areas requiring flexibility but should be minimized and not exceed 6 feet.

5.4.3.3 **Environmental Condition 3:** This is where raceways and enclosures may endure physical damage or damage from water, oils, dusts, or any corrosive substance. Example: Outdoors and indoor areas subject to corrosive (water or chemicals) damage.

5.4.3.3.1 Only ¾ inch or larger conduit is to be installed.

5.4.3.3.2 RGS Conduit is used.

5.4.3.3.3 NEMA 12 enclosures are to be used indoors and NEMA 4X enclosures are to be used outdoors.

5.4.3.3.4 Grounding and waterproof conduit bushings or grounding and waterproof conduit hubs are to be used at every enclosure.

5.4.3.3.5 Liquid-tight flexible metallic conduit may be used for areas requiring flexibility but limited to lengths of 3 feet or less.

5.4.3.4 **Environmental Condition 4:** This is where raceway is under the slab or buried underground.

5.4.3.4.1 Underground conduit must be PVC-coated RGS conduit; with appropriate fittings and installed per UL listing. Note that transition fittings may have a reduced opening size.

5.4.3.4.2 Duct banks may be Schedule 80 PVC or specialized duct raceway.

5.4.3.5 Regardless of Environmental Condition, the following circuits will use Environmental Condition 2 raceways and enclosures when located indoors, and Environmental Condition 3 raceways and enclosures if any portion of the circuit is located outdoors: 460V or greater circuits; 240V panel and power center feeds and subfeeds; and 208V panel and power center feeds and subfeeds.

5.5. Motors

5.5.1 No motor larger than 1/3hp may be connected to any 208Y/120V or 240V system unless these distribution systems are specifically designed for these types of loads.

5.5.2 New motors 10hp or more, 3 phase, will satisfy the following requirements:

5.5.2.1 External thermal overload protection that is manually resettable.

5.5.2.2 Motors above 500hp and all motors connected to medium voltage must have internal winding embedded RTDs, 4-wire, three per phase, 100ohm Pt or 10ohm Cu.

5.5.2.3 Motors connected to low voltage; 3-phase systems will have no less than 0.80 lagging power factor at rated load. If below 0.80 lagging, the power factor must be raised to 0.95 lagging using correction capacitors, connected per NEC Section 430. This does not apply to Adjustable Speed Drive (ASD) supplied motors.

5.5.3 New motors 10hp or more, 3 phase, and designated for 60 minutes or more of continuous use will satisfy the following additional requirements:

5.5.3.1 TEFC construction.

5.5.3.2 Locked Rotor Code H or less.

5.5.3.3 Class F insulation (155ºC) or better.

5.5.3.4 NEMA Premium Efficiency Motors are required.

5.5.4 If the motor is supplied by an ASD, shielded multi-conductor Variable Frequency Drive (VFD) cable must be used on AC circuits between the ASD and the motor. This VFD cable must have the shield connected only at one end.

5.5.5 For motors supplied by an ASD and having a local disconnecting means near the motor, an auxiliary contact will be provided in the disconnect and connected to the ASD where the ASD is immediately shut down upon this contact opening.

5.5.6 Motors 2hp and above that are fed by ASDs will have shaft grounding brushes.

5.5.7 Motors, regardless of horsepower, that are driven by an ASD will be rated for invertor duty.

5.5.8 Motors being fed by ASDs will not be operated at overrated speed conditions.

5.5.9 Motors 20hp and above shall be provided with a reduced voltage starter. No open transition Wye-Delta starters are allowed.

5.5.10 For 600V or less systems, running time meters will be installed on all rotating equipment rated over 75 hp except for air handling equipment, doors, and cranes.

5.5.11 Motor starters and magnetic contactors operating at voltages of 220V or greater, but less than 600V will be equipped with control circuit transformers with primary and secondary fuse blocks to provide control power. Control circuits will be 120 VAC, 24 VAC, or 24 VDC. Medium voltage motors follow existing building standard for control.

5.6. Labeling and Identification

5.6.1 Refer to Attachment 5 for WPAFB AFRL RQO Naming and Labeling Instructions for panelboards, LDCs or PDCs, switchgear, Motor Control Centers (MCCs), disconnects, bus plugs, and other equipment.

5.6.2 Receptacles and light switches will be identified with the panel and circuit number that the device is fed from. Tape labels or machine printed tags with adhesive back are acceptable. Handwritten tags are not acceptable. Minimum text size is 3/16 inches.

5.6.3 Format of identification tags will comply with Attachment 5 or Attachment 7 as applicable.

5.6.4 LPs, PPs, LDPs, PDPs, LDCs, PDCs and switchgear will have a manufacturer's nameplate that includes, in addition to standard data such as volts, amps, model number, etc., the fault current withstand rating of internal bussing and interrupting rating in amps RMS Symmetrical for the unit and for each overcurrent device.

5.6.5 Conductor Identification.

5.6.5.1 Wire markers with conductor identification numbers will be provided on all control conductors within each enclosure where a tap, splice, or termination is made. Hand lettering or marking is not acceptable. Machine-printed self-laminating vinyl wire markers, plastic clip-on "C" type markers, or machine printed slip-on markers are the only approved identification means.

5.6.5.2 Phase conductor color-coding will be as follows: EPDSs less than or equal to 240V will have phases coordinated A phase - Black, B phase - red, C phase - Blue, and the grounded (neutral) conductor - white and distribution systems greater than 240V will have phases coordinated A phase - Brown, B phase - orange, C phase - yellow, and the grounded (neutral) conductor will be grey.

5.6.6 A preprinted label or other permanent marker will be placed inside any device using fuses. The label will clearly show the manufacturer, type, brand, and rating of the fuses to be used.

5.6.7 A printed panel schedule will be placed in and maintained in Lighting Panels and Power Panels that identifies all loads in the panels. Handwritten panel schedules are not acceptable.

5.6.8 In EPDSs with complete arc flash studies, provide Arc flash labels in accordance with UFC 3-560-01 change 5. Figure 1-11 where incident energy above 40 cal/cm2 is a red danger label, and Figure 1-7 where incident energy is 40 cal/cm2 or below is a yellow warning label. Labels will be 6” wide by 4” high.

5.7. Grounding

5.7.1 Grounding of the distribution system, installation of ground rods, and testing of grounds will be coordinated through RQOO.

5.7.2 All grounding electrode conductors are bonded to the grounding electrode or ground grid by means of an exothermic weld or by an irreversible compression connection using a 15-ton hexagonal crimp (C-tap).

5.7.3 Grounding electrode conductors will be protected from damage by routing through PVC raceway where they are routed independent of phase wiring.

5.7.4 Ground rods will be separated, when connected to the same ground point, by at least twice the depth to which they are driven, as long as that depth is at least 6 feet.

5.8. Wiring Devices

5.8.1 The standardized connectors or equal (mechanically and electrically) listed in Attachment 6, will be the only ones used on the described power system.

5.8.2 In non-hazardous areas, three wire grounded type polarized, paralleled slots, Specification Grade, 20A receptacles will be used for 120V service. These receptacles, NEMA 5-20R, when mounted vertically will be oriented with the ground pin up. For horizontal mounted orientation the grounded conductor larger slot, neutral parallel slot, will be up.

5.8.3 Loads requiring current in excess of those listed for receptacles will be directly connected to power systems through fused disconnects, circuit breakers, or combination magnetic starters.

5.8.4 Contact RQOO for coordination of connectors that differ from these standards.

6. Restrictions

6.1. Loads connected to 208Y/120V and 240V panelboards by means of a wall receptacle will be limited to 16A and nominal 120V. No motor larger than 1/3hp or heaters in excess of 1 kilowatt (kW) may be connected to the 208Y/120V and 240V panelboards. Exceptions:

6.1.1. An existing receptacle of the correct rating already exists.

6.1.2. A design has been reviewed and approved by RQOO. Designs could include, but are not limited to, a dedicated circuit and receptacle or a direct connection with a disconnect switch.

6.2. For loads in excess of 16A, the 460V or 480Y/277V system will be used wherever available in preference to other lower voltage systems.

6.3. 208V equipment should be specified if available rather than 240V.

6.4. Single-phase loads will be powered from single-phase connections to the 3-phase system.

6.6. Motor starting or load establishment will be limited to no more than 10% RMS voltage drop as seen from the primary distribution point of the load based upon this point's nominal voltage. Reduced voltage starting devices are used to satisfy this requirement.

6.7. Power distribution connected devices capable of generating harmonic signals back onto the power distribution system, e.g., UPSs, adjustable speed drives (ASDs), electric heater controllers and power supplies will be limited. The total harmonic voltage will not exceed 5% of the nominal system voltage. The total harmonic current will not exceed 8% of the full load amps of the load. These two conditions must be met throughout the operating range of the equipment.

6.8. Power controllers will satisfy the guidelines and restrictions for harmonic generating loads detailed elsewhere in this OI. This can be accomplished with isolation transformers, zero-crossing controllers, minimum 12 pulse rectifiers, or other mitigation techniques.

6.9. Use of combustible materials will comply with the guidelines of an EHAD. EHADs will be made by RQOO in consultation with AFRL/RQY. Contact RQOO for templates and further information.

7. Special Site Conditions

7.1 Building 65, 68 and surrounding related areas:

7.1.1 Responsibilities

7.1.1.1 AFRL/RQVV Facility Team: reviews designs, drawings and sketches submitted for approval. The RQVV Facility Team is the review board for Building 65 complex electrical systems. The Facility Team will maintain the master one-line diagram/Easy Power and hard copy files for electrical systems. Copies will be sent to RQOO for record-keeping.

7.1.1.2 AFRL/RQVV Facility Functional Area Manager (FAM): The Facility FAM will be the lead of the RQVV Facility Team. The Facility FAM will authorize electrical work being performed in the building 65 complex including power outages. The Facility FAM will coordinate with RQOO for electrical installations 100A or greater, all distribution power systems, and all installations 1000V or greater.

7.1.1.3 Electrical design information will be furnished to the Facility Team for approval prior to the purchase of any equipment or start of any work on electrical systems. This documentation will be signed by the Facility FAM and returned to the initiator for posting on the job site or be returned to the originator with comments.

7.1.2 Wiring used on lamp banks will be stranded copper, 600V, UL 5107 compliant high-temperature wire with insulation rated minimum 450°C.

7.1.3 Naming and labeling conventions for equipment in these areas will conform to the guidelines laid out in Attachment 7 RQVV Labeling and Identification Standards.

7.1.4 Wiring Devices:

7.1.4.1 Orange Receptacles will be used for Instrument Power outlets. Instrument power will not be used for general purpose power including any motor load, tools, lights, etc.

7.1.4.2 Three phase plugs and receptacles in excess of 20A will be DS series Meltric Decontactors and DSN series Meltric Decontactors for all 208V instrument power, 480V house power, and 240V.

7.1.4.3 Indoor installations in building 65 will meet Environmental Condition 1 at a minimum. Circuits rated 400A and above will meet Environmental Condition 2.

7.2. Buildings 24A-C, 25A-D, 26, 27, 455 and 456 (Wind Tunnel Complex):

7.2.1 Responsibilities

7.2.1.1 AFRL/RQVX Facility Working Group (FWG): reviews designs, drawings and sketches submitted for approval. The RQVX FWG is the review board for the Wind Tunnel Complex electrical systems. The RQVX FWG will maintain the master one-line diagram and hard copy files for electrical systems. Copies will be sent to RQOO for record-keeping.

7.2.1.2 AFRL/RQVX FWG Lead or RQVX Branch Chief will authorize electrical work being performed in the Wind Tunnel Complex including power outages. The FWG Lead or Branch Chief will coordinate with RQOO for electrical installations.

7.2.1.3 Electrical design information will be furnished to the FWG for approval prior to the purchase of any equipment or start of any work on electrical systems.

7.2.1.4 Motor supplied with adjustable speed drives in excess 75hp will require a disconnect. Coordinate with the RQVX FWG.

7.2.1.5 RQVX will be responsible for sealing of equipment for relays, timers, meters, and OCPD.

7.2.1.6 RQVX FWG will be responsible for relay testing and coordination with substation relays. Test results will be maintained by the FWG. Copies will be sent to RQOO.

7.2.1.7 RQVX FWG will perform coordination studies, short circuit studies, and arc flash studies prior to specifying new equipment. Utility contributions will be coordinated with AES. Copies of studies will be sent to RQOO.

7.2.1.8 Motors will only be required to have hour meters where their integration is practical.

7.2.1.9 Wiring used on power circuits will be stranded copper, 600V, THNN, THWN, or XHHW. MTW will be allowed only in accordance with NEC table 310.4(1).

8. Approval/ Exceptions

8.1 Designs, drawings, and sketches will be submitted for review and approval through RQOO.

8.2 Requests for exceptions to the requirements and standards of this OI will be referred to RQOO for evaluation by submitting Attachment 8 RQ OI 32-10 Variance Request.

MICHAEL R. GREGG, PhD, SES
Director
Aerospace Systems Directorate

BEGIN ATTACHMENT

ATTACHMENT 1

GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION

References

29 CFR 1910.147, OSHA*, The Control of Hazardous Energy, (Lockout/Tagout) - Inspection Procedures and Interpretive Guidance*, 11 Sep 1990

AFI 33-322, *Records Management and Information Governance Program,* 23 Mar 2020

AFI 91-202AFMCSUP, *Mishap Prevention Program,* 31 Mar 2022

AFI 91-202 AFMCSUP_AFRLSUP, *Mishap Prevention Program,* 11 Oct 2023

AFMD - 1, *Headquarters Air Force (HAF),* 5 Aug 2016

AFPD 32-10, *Installations and Facilities,* 20 Jul 2020

ASTM F855, Temporary Protective Grounds to Be Used on De-energized Electric Power Lines and Equipment, 15 Sep 2023

DAFI 91-202, *Mishap Prevention Program* 12 Mar 2020

DAFMAN 91-203, Air Force Occupational Safety Fire and Health Standards, 25 Mar 2022

DAFPD 90-1, *Policy, Publications, and DoD Issuance Management*, 24 Mar 2023

DoDI 5025.01, *DoD Issuances Program*, 1 Aug 2016, Change 4 – 7 Jun 2023

IEEE 1584, Arc Flash Calculations, 2018

NFPA 70, *National Electric Code (NEC),* 2023

NFPA 70E, *Standard for Electrical Safety in the Workplace,* 2021

NFPA 79, *Electrical Standard for Industrial Machinery,* 2021

Wright-Patterson AFB IFS, 2021

Prescribed Forms

None

Adopted Forms

None

Abbreviations and Acronyms

A - Amps - Amperes

AC - Alternating Current

AF - Air Force

AFI - Air Force Instruction

AFRIMS - Air Force Records Information Management System

AFMD - Air Force Mission Directive

AFPD - Air Force Policy Directive

AFRL - Air Force Research Laboratory

AHJ - Authority Having Jurisdiction

AO - Action Officer

ASD - Adjustable Speed Drive

ASTM - American Society for Testing Materials

AWG - American Wire Gage

BCE - Base Civil Engineering

CFR - Code of Federal Regulations

COR - Contracting Officer Representative

CT - Current Transformer

Cu - Copper

DAF - Department of the Air Force

DAFI - Department of the Air Force Instruction

DAFPD - Department of the Air Force Policy Directive

DAFPMan - Department of the Air Force Manual

DC - Direct Current

DoD - Department of Defense

DoDI - Department of Defense Instruction

EGC - Equipment Grounding Conductor

EHAD - Electrical Hazardous Area Determination

EPS - Electrical Power System(s)

EPD - Electrical Power Distribution

EPDS - Electrical Power Distribution System

EPR - Ethylene Propylene Rubber

FAM - Functional Area Manager

FWG - Facility Working Group

FIST - Facility Infrastructure Support Tool

GEC - Grounding Electrode Conductor

HHQ - Higher Headquarters

IAW - In Accordance With

IEEE - Institute of Electrical and Electronics Engineers

IFS - Installation Facilities Standards

LDC - Lighting Distribution Center or Low Voltage Distribution Center

LP - Lighting Panel or Low Voltage Panel

LOTO - Lock Out Tag Out

LSIG - Long Time, Short Time, Instantaneous, and Ground Fault

MCC - Motor Control Center

MTW - Machine Tool Wire

NEC - National Electrical Code

NEMA - National Electrical Manufacturers Association

NETA - International Electrical Testing Association

NFPA - National Fire Protection Association

NRTL - Nationally Recognized Testing Laboratory

OI - Operating Instruction

OPR - Office of Primary Responsibility

PD - Policy Directive or Power Distribution

PDC - Power Distribution Center

POC - Point of Contact

PP - Power Panel

Pt - Platinum

PVC - Polyvinyl Chloride

RDS - Records Disposition System

RGS - Rigid Galvanized Steel

RMC - Rigid Metal Conduit

RMS - Root Mean Square

RQ - Aerospace Systems Directorate

RQO - Aerospace Systems Directorate, Integration & Operations Division

RQOE - Aerospace Systems Directorate, Integration & Operations Division, Engineering & Design Branch

RQOO - Aerospace Systems Directorate, Integration & Operations Division, Research Operations Support Branch

RQOR - Aerospace Systems Directorate, Integration & Operations Division, Facilities Support Branch

RQV - Aerospace Systems Directorate, Aerospace Vehicles Division

RQVV - Aerospace Systems Directorate, Aerospace Vehicles Division, Structural Validation Branch

RQVX - Aerospace Systems Directorate Aerospace Vehicles Division, Aero Validation Branch

RQY - Aerospace Systems Directorate, Safety Office

RTD - Resistance Temperature Detector

RZ - Propulsion Directorate

SIS - Structural Insulated Sheathing

SCCR - Short Circuit Current Rating

THHN - Thermoplastic High Heat Nylon

THWN - Thermoplastic Heat Water Nylon

XHHW - Cross-Linked Polyethylene High Heat Water

UL - Underwriter Laboratories

UP - Utility Privatization

UPS - Uninterruptible Power Supply

V - Volts

VFD - Variable Frequency Drive

WO - Work Order

ZSI - Zone-Selective Interlocking

Terms

Approving Official - Approves the release of publications for compliance and enforcement and is solely responsible for ensuring the publications and forms are necessary, information is current, and in conformance with existing laws, policy, guidance, and Department of the Air Force Mission.

Certifying Official - The official who certifies the need for the publication within the numbered publication subject series. This official also certifies the publication's consistency with departmental policy and with the assigned responsibilities in antecedent Policy Directive and/or implemented Higher Headquarters publication.

Department of the Air Force Publications - Officially produced, published, and distributed publications issued for compliance, implementation, and/or information.

FIST – Online database for gathering new facilities requirements, tracking work orders, and managing projects, intended for use by the entire RQ East community. The link can be found by searching for FIST in RQoogle: https://rqintranet.wpafb.af.mil/rqoogle/index.cfm or here: https://rqtools.wpafb.af.mil/cfdocs2/rqfpt/main_rqfpt.cfm?new_menu=1&site=rqfpt

Headquarters Air Force - The senior headquarters of the Department of the Air Force, comprised of three major entities: the Secretariat (including the SecAF and the Secretary's principal staff), the Air Staff, headed by the Chief of Staff, and the Space Staff, headed by the Chief of Space Operations.

WPAFB IFS – This document is part of the Air Force Corporate Facilities Standards (AFCFS) program and provides facility design standards for Wright-Patterson Air Force Base. It can be accessed at the following link: https://www.wbdg.org/ffc/af-afcec/installation-facilities-standards-ifs/wright-patterson-afb-ifs

END ATTACHMENT

BEGIN ATTACHMENT
Attachment 2

OUTAGE REQUEST FORM

AFRL/RQ Wright Patterson Air Force Base, Ohio

Project Title: **WO# or FIST#:**

Contractor Submitting Request:	
POC Name:	**POC Phone Number:**
Date Submitted:	**Outage Requested Date(s):**
Building #:	**Time Requested:**
Risk Assessment Code:	**Estimated Duration:**
Interim Life Safety Measures Required (i.e.: Fire Sprinkler Motor Pump, Fire Alarm System, Egress Lighting, etc. affected)**:** Y or N	**Floor/Wing:**
Reason for Outage:	
All Equipment & Rooms Effected:	
System(s) Shut Down: List all systems to be shut down.	1. 2. 3. 4.
Description of Work: Briefly describe what is being shut off and what is anticipated to be affected. ***Provide a detailed step by step work plan on following page or attach plan.***	
Owner/User Required Involvement: i.e. – Shut down computers, Notify their group of outage	
Attachments: Attach all relevant sketches showing location of work and areas that will be affected	1. i.e.: One-Line Diagram(s) Highlighting the Affected System(s) 2. i.e.: Electrical Print(s) Highlighting the Affected Area/Room(s) 3. i.e.: Specific Panel Schedules

Acknowledgment of System Outage Request

(contractor) Foreman:		**Date:**
(contractor) Electrical Engineer:		**Date:**
(contractor) Project Manager:		**Date:**
AFRL – RQOE or RQOO COR:		**Date:**
Building Owner/User Affected:		**Date:**

Approved Date: ______________ Approved Start Time: ______________

Detailed Work Plan (state if detailed work plan is attached)

Security: (937) 257-6517
Fire: (937) 257-3033

Fire and security will be called at the start of shift and end of shift to communicate the outage start (start of shift) and the outage end (end of shift) each day. The message to security and fire shall include the building (###) that will be shut down in the morning to warn of possible fire & security alarm messages that they may receive. At the end of the day, security and fire will be made aware that the power to building (###) has been restored and to confirm fire and security alarms have been cleared.

Alternate/Back Up Plan **(Think…What do you do if equipment does not re-energize or fails)**

Any other miscellaneous info not listed above that is pertinent to this scheduled electrical outage:

********Contractor to Provide digital copy to RQOE or RQOO COR For Review********

END ATTACHMENT

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Attachment 3

RQ ELECTRICAL ENERGIZED WORK PERMIT

RQ ENERGIZED ELECTRICAL WORK PERMIT

PART I. TO BE COMPLETED BY THE REQUESTER

ORGANIZATION/UNIT/COMPANY	JOB/W.O.#

1. DESCRIPTION OF CIRCUIT/EQUIPMENT/JOB LOCATION:

2. DESCRIPTION OF WORK TO BE DONE:

3a. CAN THE WORK BE DEFERRED UNTIL THE NEXT SCHEDULED OUTAGE? *YES* *NO* *If the work can wait until a future planned outage do not continue form*	3b. DATE OF FUTURE PLANNED OUTAGE

3c. JUSTIFICATION OF WHY THE CIRCUIT/EQUIPMENT CANNOT BE DE-ENERGIZED OR THE WORK DEFERRED UNTIL A PLANNED OUTAGE:

REQUESTOR NAME / TITLE	SIGNATURE	DATE

PART II: COMPLETED BY THE ELECTRICALLY QUALIFIED PERSON ACCOMPLISHING THE WORK

4. DETAILED JOB DESCRIPTION PROCEDURE TO BE USED IN PERFORMING THE ABOVE DETAILED WORK (WORK PLAN MAY BE ATTACHED):

5. DESCRIPTION OF THE SAFE WORK PRACTICES BEING EMPLOYED:

6. RESULTS OF THE SHOCK RISK ASSESSMENT:

6a. VOLTAGE TO WHICH PERSONNEL WILL BE EXPOSED:

6b. LIMITED APPROACH BOUNDARY:	6c. RESTRICTED APPROACH BOUNDARY:

6d. NECESSARY SHOCK, PERSONAL, AND OTHER PROTECTIVE EQUIPMENT TO SAFELY PERFORM ASSIGNED TASK:

7. RESULTS OF THE ARC FLASH RISK ASSESSMENT:

7a. AVAILABLE INCIDENT ENERGY AT WORKING DISTANCE OR ARC FLASH PPE CATEGORY:	
7b. ARC FLASH BOUNDARY	

7c. ARC FLASH PERSONAL AND OTHER PROTECTIVE EQUIPMENT NEEDED TO SAFELY PERFORM ASSIGNED TASK:

8. PLAN TO RESTRICT ACCESS OF UNQUALIFIED PERSONS FROM THE WORK AREA:

9. JOB SAFETY BRIEFING TO INCLUDE DISCUSSION OF ANY JOB-RELATED HAZARDS:

10. EMERGENCY EGRESS PLAN (DETAILED PLAN ATTACHED IF NEEDED):

RQ ENERGIZED ELECTRICAL WORK PERMIT

11. CAN THE ABOVE-DESCRIBED WORK BE DONE SAFELY? *(If no, return to requestor.)*	YES	NO

QUALIFIED ELECTRICAL PERSONNEL	SIGNATURE	DATE
QUALIFIED ELCTRICAL PERSONNEL SUPERVISOR	SIGNATURE	DATE
	SIGNATUE	DATE

PART III. APPROVAL TO PERFORM WORK WHILE ELECTRICALLY ENERGIZED

Work on energized electrical equipment is prohibited except in circumstances justified and approved by the RQ AHJ in accordance with AFI 32-1064, Electrical Safe Practices. The electrical hazards listed above are to be mitigated by implementing the controls described in the work plan and safety plan. To comply with the 5-step Air Force Risk Management Process, a qualified electrical supervisor must evaluate the activity and if additional hazards are identified while the work is underway and stop work if the risk cannot be mitigated by the controls as implemented.

RQ AHJ (NAME/TITLE)	SIGNATURE	DATE

PART IV. IMPLEMENTATION, SUPERVISION AND EVALUATION

DESCRIPTION	FOREMAN'S INITIALS
The required personnel to perform the work are assigned to the job and will be on site until the work on energized circuits or equipment is complete.	
The hazards listed and identified above are accurate and the required PPE and equipment listed in the safety plan are on hand. Do not perform the work if the risk calculations are incorrect.	
The job safety briefing effectively covered the hazards, plan to keep unqualified personnel isolated from the job site and the emergency evacuation plan.	

EVALUATION OF WORK, LESSONS LEARNED AND SAFETY CONCERNS FOR FUTURE WORK (USE ADDITIONAL FORMS AS NEEDED IF MORE SPACE IS REQUIRED)

JOB FOREMAN (NAME/TITLE)	SIGNATURE	DATE

END ATTACHMENT

BEGIN ATTACHMENT

Attachment 4

AFRL/RQ RECOMMENDED MANUFACTURERS

A4.1 Purpose
The intent of this list is not to require certain manufacturers. It does not provide sole source justification, if required that will need to be obtained separately. The equipment and manufacturers listed below have proven to have quality products that meet some of the unique needs of ARL/RQ's research objectives. Purchasing of standard equipment also eases the burden of maintenance due to having interchangeable and spare parts on hand.

A4.2 Recommended Products and Manufacturers
A4.2.1 Bus in open Lab spaces: StarLine Track Bus is recommended in open Lab spaces where flexibility is desired. The most commonly used bus size is 100A. No single circuit tap should be greater than 60A. Contact RQOO for specific building standards.

A4.2.2 Harmonic Filters and Reactors: MTE Corp.

A4.2.3 Panelboards: Square D

A4.2.4 Medium voltage protection relays: SEL 751 (Schweitzer Engineering Laboratories)

A4.2.5 UPS: Liebert supplied by Vertiv

END ATTACHMENT

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Attachment 5

LABELING AND IDENTIFICATION STANDARD FOR ELECTRICAL EQUIPMENT

A5.1 Naming Conventions
When adding or removing new equipment, a feed to a panel, or other type of gear such as disconnects or transformers, the following naming conventions will be utilized for the new equipment:

A5.1.1 Electrical Gear Names: FFFFF_RRRR_XXX_DD

FFFFF – Facility ID number

RRRR – Location Designator:

- Room number
- Location such as mezzanine (MEZZ) or basement (BSMT)

XXX – Identification of Type of Gear
- SUB (Substation)
- PDP, LDP or other (Power Distribution Panel) *see Note 1 below
- PP, LP or other (Panelboard) *see Note 1 below
- MCC (Motor Control Center)
- DS (Disconnect Switch)
- FS (Fused Switch)
- BP (Bus Plug)

DD – Alphanumeric Designator for Areas with more than one of that type of gear (Engineering decision by AHJ will be made at time of install if this is necessary)

*Note 1: In an existing building follow the established building standard. In new buildings, buildings that are completely renovated, or those with no clear standard use the following:
PDP or PP for panels at 460VAC or greater
LDP or LP for panels at less than 460VAC
YDP or YP for panels at 100VDC or greater
CP for panels at 100VDC or less (control panels)

Examples:
- 20490_105_PP_A: 480V Power panel in building 490, Rm 105, Occurrence #1
- 21624_MCC-1: MCC in building 21624 (71D), Occurrence #1

A.5.1.2 Transformers: FFFFF_RRRR_“T”SSS_D

FFFFF – Facility ID number

RRRR – Location Designator:
- Room number
- Location such as mezzanine (MEZZ) or basement (BSMT)

SSS – Size of Transformer in KVA
- 10 (10 KVA)
- 75 (75 KVA)
- 112 (112.5 KVA)
- 150 (150 KVA)

D – Numerical Designator for Areas with more than one of that type of gear (Engineering decision by AHJ will be made at time of install if this is necessary)

Examples:
- 21603_T112_BSMT_B: 112.5 KVA Transformer, 18E basement, Occurrence #2
- 20045_T150_104_1: 150 KVA Transformer, Bldg. 45, RM 104, Occurrence #1

A.5.1.3 Other Equipment: FFFFF_RRRR _XXX_D_YYYY

FFFFF – Facility ID number

RRRR – Location Designator:
- Room number
- Location such as mezzanine (MEZZ) or basement (BSMT)

XXX – Identification of Type of Equipment
- ACP (Air Compressor)
- AHU (Air Handling Unit)
- CND (Condensate Pump)
- CWP (Chilled Water Pump)
- DS (Disconnect Switch)
- HTR (Electric Heater)
- HVAC (Air Conditioning System)
- HYD (Hydraulic Pump)
- OHC (Overhead Crane)
- SCR (Silicon Controlled Rectifier)
- SMP (Sump Pump)
- UPS (Uninterruptible Power Supply)

D – Numerical Designator for Areas with more than one of that type of gear (Optional - Engineering decision by AHJ will be made at time of install if this is necessary)

YYYY – Size of Load of Equipment
- 1/3HP (1/3 Horsepower)
- 5HP (5 Horsepower)
- 75HP (75 Horsepower)
- 400A (400A FLA)
- 36KB (36,000 BTU)
- 150T (150 Ton Crane; if multiple hooks, named for largest)

Examples:
- 20023_105_CWP_1_5HP: Chilled water pump 1, Bldg. 23, 5HP, room 105
- 210601_BSMT_UPS_80KW: 80KW UPS in 18C basement

A5.1.4 AFRL/RQ Building number cross reference chart

AFRL/RQ Directorate Building Cross Reference
WPAFB, OH

Facility ID	Building	Facility ID	Building	Facility ID	Building
20005	5	20145	145	21402	24C
20018	18	20146	146	21500	25A
20019	19	20196	196	21501	25B
20020	20	20252	252	21502	25C
20021	21	20253	253	21503	25D
20023	23	20255	255	21600	18A
20026	26	20352	352	21601	18C
20027	27	20435	435	21603	18E
20045	45	20455	455	21604	18B
20056	56	20456	456	21605	20A
20062	62	20458	458	21609	18H
20064	64	20461	461	21610	18D
20065	65	20490	490	21611	18G
20068	68	20493	493	21621	71A
20071	71	20496	496	21622	71B
20092	92	21400	24A	21624	71D
		21401	24B		

A5.2 Labelling Practices for Electrical Equipment

All tags will be two layer plastic, with white text on a black textured surface finish. Additionally, they shall be a minimum of 1/16” thick and have a 45° beveled edge. RQOO has the capability to make small quantities of labels upon request.

A5.2.1 Panelboards

Provide minimum 2” high by 3” wide tag, with 1/2” top and bottom margins and 1/4” side margins.

Line 1 will be the name tag as identified in A5.1 Naming Conventions. The name will be in an underlined, Arial font not less than 0.3 inches tall.

The remaining lines will be an Arial font not less than 0.2 inches tall.

Line 2 will be the operating voltage.

Line 3 will be the source, “FROM FFFFF_XXX_RRRR_D” for example.

Line 4 will be the work order number associated with the work. For example, “FFFFF-RQ-YYYY-####” if the work order was generated from the FIST tool. If the work is managed through base CE or USACE this will be the project number beginning with “ZHTV”

Line 5 will be the installation date.

A5.2.2 Power Distribution Panels (PDP) and Unit Substations

Provide minimum 2” high by 3” wide tag, with 1/2” top and bottom margins and 1/4” side margins.

Line 1 will be the name tag as identified in A5.1 Naming Conventions. The name will be in an underlined, Arial font not less than 0.3 inches tall.

The remaining lines will be an Arial font not less than 0.2 inches tall.

Line 2 will be the operating voltage.

Line 3 will be the source, “FROM FFFFF_XXX_RRRR_D” for example

Line 4 will be the work order number associated with the work. For example, “FFFFF-RQ-YYYY-####” if the work order was generated from the FIST tool. If the work is managed through base CE or USACE this will be the project number beginning with “ZHTV”

Line 5 will be the installation date.

Primary/Secondary Main Breakers:

Provide minimum 2” high by 3” wide tag, with 1/2” top and bottom margins and 1/4” side margins.

Line 1 will state “Main Breaker” in underlined, Arial font not less than 0.3 inches tall.

Line 2 will state the trip setting amperage in underlined, Arial font not less than 0.2 inches tall

Secondary Side Load Breakers

Provide 1” high by 3” wide tag, with 1/2” top and bottom margins and 1/4” side margins.

Line 1 will state load name in underlined, Arial font not less than 0.2 inches tall.

A5.2.3 Transformers

Provide minimum 2” high by 3” wide tag, with 1/2” top and bottom margins and 1/4” side margins.

Line 1 will be the name tag as identified in A5.1 Naming Conventions. The name will be in an underlined, Arial font not less than 0.3 inches tall.

The remaining lines will be an Arial font not less than 0.2 inches tall.

Line 2 will be the primary and secondary voltage for transformers.

Line 3 will be the source, “FROM FFFFF_XXX_RRRR_D” for example.

Line 4 will be the work order number associated with the work. For example, “FFFFF-RQ-YYYY-####” if the work order was generated from the FIST tool. If the work is managed through base CE or USACE this will be the project number beginning with “ZHTV”

Line 5 will be the installation date.

A5.2.4 Disconnects and other equipment.

Provide minimum 2” high by 3” wide tag, with 1/2” top and bottom margins and 1/4” side margins.

Line 1 will be the name tag as identified in A5.1 Naming Conventions. The name will be in an underlined, Arial font not less than 0.3 inches tall.

The remaining lines will be an Arial font not less than 0.2 inches tall.

Line 2 will be the operating voltage, or primary and secondary voltage for transformers.

Line 3 will be the source, “FROM FFFFF_XXX_RRRR_D” for example.

Line 4 will be the work order number associated with the work. For example, “FFFFF-RQ-YYYY-####” if the work order was generated from the FIST tool. If the work is managed through base CE or USACE this will be the project number beginning with “ZHTV”

Line 5 will be the installation date.

END ATTACHMENT

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Attachment 6 Begins Next Page

Table 6.1 UNCLASSIFIED AREAS RECEPTACLES AND PLUGS

NONHAZARDOUS LOCATIONS: NEC UNCLASSIFIED AREAS

SYSTEM VOLTAGE	OUTLET RATING	FREQUENCY	POLES (P) / WIRES (W)	Crouse-Hinds		Arrow-Hart, Pass & Seymour Hubbell or Bryant		Meltric (For Motor Loads)	
(VOLTS)	(AMPS)	Hz (CPS)		Receptacle	Plugs	Receptacle	Plugs	Receptacle	Plugs
480	60	60	3P4W	ARE 6424	APJ 6485	26410*	26419*		
480	30	60	3P4W			20443*	20445B*		
277/480	60	60	4P5W			560R7W*	560P7W*	33-64047*	33-68047*
277/480	30	60	4P5W			530R7W*	530P7W*	33-34047*	33-38047*
240	60	60	3P4W	ARE 6424-S4	APJ 6485-S4				
240	30	60	3P4W	ARE 3422	APJ 3485	20403*	21415B*		
120	20	60	2P3W		NEMA # 5-20 Specification Grade*				
28	30	DC	2P2W	ARE 3212	[illegible]				
120	30	400	3P4W	ARE 3422-S4	APJ 3485-S4				
120/208	30	60/400	4P5W			[illegible]		25415B*	
120/208	60	60	4P5W			[illegible]		26519*	

* Preferred for New Construction

Table 6.2 CLASS I GROUPS C & D RECEPTACLES AND PLUGS

HAZARDOUS LOCATIONS: NEC CLASS I GROUPS C&D (DIVISION I OR II)

SYSTEM VOLTAGE (VOLTS)	OUTLET RATING (AMPS)	FREQUENCY Hz (CPS)	POLES / WIRES	Crouse-Hinds Receptacle	Crouse-Hinds Plugs
480	30	60	3P4W	DRE 6424	NPJ 6484 NPJ 6485
240	60	60	3P4W	CES 4234	CPH 7734 CPH 7934
240	30	60	3P4W	CES 4234-S4	CPH 7734-S4 CPH 7934-S4
120	20	60	2P3W	CES 2214	CPH 7714 CPH 7914
28	30	DC		CPS 152-201	CPP 316 CPP 516
120	30	400	3P4W	CES 2213	CPH 7713 CPH 7913
120/208	30	400	3P5W	CES 2214-S4	CPH 7714-S4 CPH 7914-S4

Table 6.3 CLASS I GROUP B RECEPTACLES AND PLUGS

HAZARDOUS LOCATIONS: NEC CLASS I GROUP B
(DIVISION I OR II)

SYSTEM VOLTAGE (VOLTS)	OUTLET RATING (AMPS)	FREQUENCY Hz (CPS)	POLES / WIRES	Crouse-Hinds Receptacle	Crouse-Hinds Plugs	Appleton Electric Co Receptacle	Appleton Electric Co Plugs
480	30	60	3P4W	BHR3482D* BHR3483D	BHP3483D BHP3485D		
240	60	60	3P4W	BHR6484D* BHR6485D	BHP6483D BHP6485D		
240	30	60	3P4W	BHR3482NX* BHR3483NX	BHP3483NX BHP3485NX		
120	20	60	2P3W	BHR3382N BHR3383N	BHP3383N BHP3385N	EFSB175-2023M*	ECP-2023
28	30	DC					
120	30	400	3P4W				
120/208	30	400	3P5W	BHR3583NW* BHR3584NW	BHP3583NW BHP3585NW		

* Preferred for New Construction

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Attachment 7

RQVV LABELING AND IDENTIFICATION STANDARD FOR ELECTRICAL EQUIPMENT, BUILDING 20065 AND 20068 ONLY

A7.1 Naming Conventions

When adding or removing a feed to a panel or other type of gear such as disconnects or transformers, the following naming conventions will be utilized:

A7.1.1 Electrical Gear Names: XXX_VVV_RRR_D

XXX – Identification of Type of Gear

- PDP (Power Distribution Panel) *All distribution gear 400A and greater main bus
- PNL (Panel) *All distribution gear less than 400A main bus
- MCC (Motor Control Center)
- DSC(Disconnect)

VVV – Secondary Voltage:

- 600 – 600 Volts 3 Phase
- 480 – 480 Volts 3 Phase
- 277 – 480Y/277 Volts 3 Phase Wye
- 208 – 208Y/120 Volts 3 Phase Wye
- 240 – 240 Volts 3 Phase
- 240S – 240/120 Single Phase
- IP – 208Y/120 Volts 3 Phase Wye Instrument Power

RRR – Location Designator:

- Room number if applicable (202N, 301E, 102TF etc.…)
- Building Areas
  - 1EA – 1st Floor East Alcove
  - 1EW – 1st Floor East Warehouse
  - 1ET – 1st Floor Extended Temperature (…2ET, 3ET, 4ET, 5ET)
  - 2EM – 2nd Floor East Mezzanine
  - ETC1 – Exterior of ETC1 on Test Floor (…ETC2, ETC3, ETC4)
  - TFW/TFE/TFN – West/East/North walls of the Test Floor
  - TOC – Exterior of TOC on Test Floor
  - HTF – High Bay of Test Floor
  - REW – Roof of East Warehouse (…RET, RW, RE, RTF, RN)
  - B68 – Building 68

D – Numerical Designator for Areas with more than one of that type of gear (Engineering decision by AHJ will be made at time of install if this is necessary)

A.7.1.2 Transformers: "T" SSS_VVV_RRR_D

SSS – Size of Transformer in KVA

- 10 (10 KVA)
- 75 (75 KVA)
- 112 (112.5 KVA)
- 150 (150 KVA)
- 2.5K (2500 KVA)

VVV – Secondary Voltage:

- 600 (600 Volts 3 Phase)
- 480 (480 Volts 3 Phase)
- 277 (480Y/277 Volts 3 Phase Wye)
- 208 (208Y/120 Volts 3 Phase Wye)
- 240 (240 Volts 3 Phase)
- 240S (240/120 Single Phase)

RRR – Location Designator:

- Room number if applicable (202N, 301E, 10[illegible]TF etc.…)
- Building Areas
  - 1EA – 1st Floor East Alcove
  - 1EW – 1st Floor East Warehouse
  - 1ET – 1st Floor Extended Temperature (…2ET, 3ET, 4ET, 5ET)
  - 2EM – 2nd Floor East Mezzanine
  - ETC1 – Exterior of ETC1 on Test Floor (…ETC2, ETC3, ETC4)
  - TFW/TFE/TFN – West/East/North walls of the Test Floor
  - TOC – Exterior of TOC on Test Floor
  - HTF – High Bay of Test Floor
  - REW – Roof of East Warehouse (…RET, RW, RE, RTF, RN)
  - B68 – Building 68

D – Numerical Designator for Areas with more than one of that type of gear (Engineering decision by AHJ will be made at time of install if this is necessary)

Examples:

- T112_240_1ET_2: 112.5 KVA Transformer 240 3-Phase in 1ET Occurrence #2
- T150_208_4ET: 150 KVA Transformer 208 3-Phase Wye in 4ET
- T75_277_115E: 75 KVA Transformer 480/277 3 Phase Wye in 115E

A.7.1.3 Other Equipment: XXX_YYYY _RRR_D

XXX – Identification of Type of Equipment

- ACP (Air Compressor)

- AHU (Air Handling Unit)
- CND (Condensate Pump)
- EXB (Exhaust Blower)
- HTR (Electric Heater)
- HVAC (Air Conditioning System)
- HYD (Hydraulic Pump)
- OHC (Overhead Crane)
- SCR (Silicon Controlled Rectifier)
- SMP (Sump Pump)
- UPS (Uninterruptible Power Supply)

YYYY – Size of Load of Equipment:

- 1/3HP (1/3 Horsepower)
- 5HP (5 Horsepower)
- 75HP (75 Horsepower)
- 400A (400A FLA)
- 36KB (36,000 BTU)
- 150T (150 Ton Crane; if multiple hooks, named for largest)

RRR – Location Designator:

- Room number if applicable (202N, 301E, 102TF etc.…)
- Building Areas
    - 1EA – 1st Floor East Alcove
    - 1EW – 1st Floor East Warehouse
    - 1ET – 1st Floor Extended Temperature (…2ET, 3ET, 4ET, 5ET)
    - 2EM – 2nd Floor East Mezzanine
    - ETC1 – Exterior of ETC1 on Test Floor (…ETC2, ETC3, ETC4)
    - TFW/TFE/TFN – West/East/North walls of the Test Floor
    - TOC – Exterior of TOC on Test Floor
    - HTF – High Bay of Test Floor
    - REW – Roof of East Warehouse (…RET, RW, RE, RTF, RN)
    - B68 – Building 68

D – Numerical Designator for Areas with more than one of that type of gear (Engineering decision by AHJ will be made at time of install if this is necessary)

Examples:

- OHC_150T_HTF: 150T Overhead Crane on the High Bay of the Test Floor
- CND_5HP_1EA: 5 Horsepower Condensate Pump in the East Alcove
- SCR_3KA_3ET_1: 3000A Silicon Controlled Rectifier on 3ET 1st Occurrence
- HVAC_36KB_102TF: 36,000 BTU Air Conditioner w/ Condenser in 102TF

A7.2 Labelling Practices for Electrical Gear

All tags will be two-layer plastic, with white text on a black textured surface finish. Additionally, they shall be a minimum of 1/16” thick and have a 45° beveled edge. All names will be in all capital letters.

A7.2.1 Unit Substations:

A7.2.1.1 Unit Sub Name

Will be identified on the face of the cabinet with a 2” high by 3” wide tag, with 1/2” top and bottom margins and 1/4” side margins.

Tags will have two lines of text not less than 3/8” in height:

Line 1 will state “UNIT SUB #” in a bold, underlined, Arial font,

Line 2 will state “FED BY: SUB X-XXX” in an Arial font, “SUB”, Substation Letter(X), and Breaker number (XXX) will be underlined.

A7.2.1.2 Primary/Secondary Main Breakers

Will be identified with a 2” high by 3” wide tag, with 1/2” top and bottom margins and 1/4” side margins.

Tags will have two lines of text not less than 3/8” in height:

Line 1 will state “Main Breaker” in a bold, underlined, Arial font.

Line 2 will state the trip setting amperage and breaker voltage in an Arial font.

A7.2.1.3 Secondary Side Load Breakers

Will be identified with a 1” high by 3” wide tag, with 1/4” top, bottom and side margins.

Tags will have one line of text not less than 3/8” in height.

Will state load name in a bold, underlined, Arial font.

A7.2.2 Power Distribution Panels (PDP)

A7.2.2.1 PDP Name

Will be identified on the face of the cabinet with a 2” high by 3” wide tag, with 1/2” top and bottom margins and 1/4” side margins.

Tags will have two lines of text not less than 3/8” in height:

Line 1 will state the PDP Name in a bold, underlined, Arial font,

Line 2 will state “FED BY:” in an Arial font, followed by feeder gear name in an underlined, Arial font.

A7.2.2.2 Main Breaker

Will be identified with a 2” high by 3” wide tag, with 1/2” top and bottom margins and 1/4” side margins.

Will have two lines of text not less than 3/8” in height:

Line 1 will state “Main Breaker” in a bold, underlined, Arial font.

Line 2 will state the trip setting amperage and primary voltage in an Arial font.

A7.2.2.3 Secondary Side Load Breakers

Will be identified with a 1” high by 3” wide tag, with 1/4” top, bottom and side margins.

Will have one line of text not less than 3/8” in height:

Will state load name in a bold, underlined, Arial font.

A7.2.3 Panel Boards (PNL)

A7.2.3.1 PDP Name

Will be identified on the face of the cabinet with a 2” high by 3” wide tag, with 1/2” top and bottom margins and 1/4” side margins.

Tags will have two lines of text not less than 3/8” in height.

Line 1 will state the panel name in a bold, underlined, Arial font.

Line 2 will state “FED BY:” in an Arial font, followed by feeder gear name in an underlined, Arial font.

A7.2.3.2 Main Breaker

Will be identified with a 2” high by 3” wide tag, with 1/2” top and bottom margins and 1/4” side margins.

Will have two lines of text not less than 3/8” in height:

Line 1 will state “Main Breaker” in a bold, underlined, Arial font.

Line 2 will state the trip setting amperage, “-“, and primary voltage in an Arial font.

A7.2.3.3 Secondary Side Load Breakers

Will be identified with a panel schedule clearly labeling all loads with breaker numbers leading out of the panel.

Schedules will be updated when any new circuit is run or an old one removed.

A7.2.4 Examples

A7.2.4.1 Unit Substation

	------------ 3 Inches in Width -------- ----	
--- 2 Inches in Height ---	**<u>UNIT SUB 11</u>** FED BY: <u>SUB E-609</u>	Top Margin 1/2"
		Row 1 Bold Text Underlined Row 2 Thin Text: Name Underlined
		Bottom Margin 1/2"
	Side Margins 1/4"	

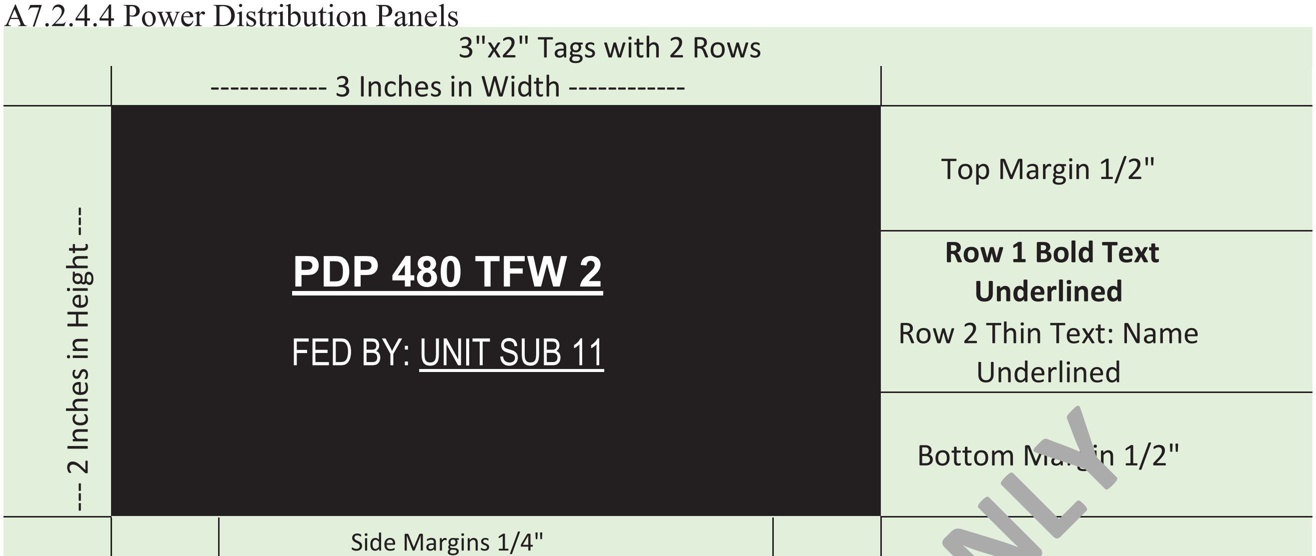
A7.2.4.2 Main Breaker

	------------ 3 Inches in Width --------- ---	
--- 2 Inches in Height ---	**<u>MAIN BREAKER</u>** 3200A – 480V	Top Margin 1/2"
		Row 1 Bold Text Underlined Row 2 Thin Text: Amperage and Voltage
		Bottom Margin 1/2"
	Side Margins 1/4"	

A7.2.4.3 Secondary Breakers

	3"x1" Tags with 1 Row ------------ 3 Inches in Width ------------	
--- 1 Inch ---	**<u>PDP 480 4ET</u>**	Top Margin 1/4"
		Row 1 Bold Text Underlined
		Bottom Margin 1/4"
	Side Margins 1/4"	

A7.2.4.4 Power Distribution Panels



END ATTACHMENT

BEGIN ATTACHMENT
Attachment 8

AFRL/RQ ELECTRICAL OI 32-10 VARIANCE REQUEST SAMPLE

Section that Variance is being requested from:

Physical Location of Variance:

Description of Circumstances:

Authorized by:

Name:______________________________

Date: ______________________________

Position: ___________________________

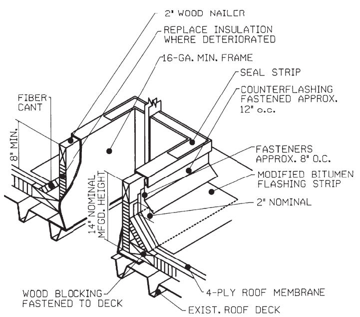
Org: ______________________________

Phone: ______________________________

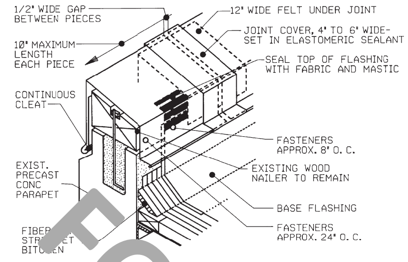
END ATTACHMENT

APPENDIX D

F20145 ROOF INSULATION

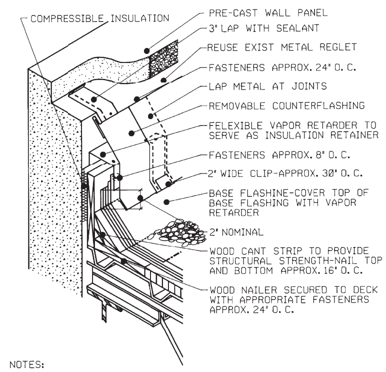


ROOF CURB DETAIL (A)



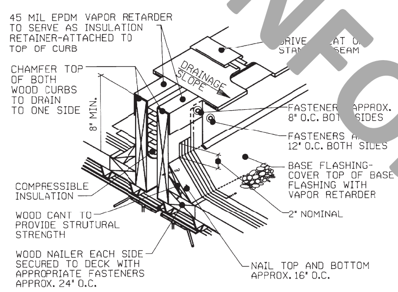
NOTE:
THIS DETAIL ... USED ... WHEN THE DECK IS SUPPORTED BY THE WALL ... SIMILAR TO THE EXPANSION JOINT DETAIL ... SUPPORTED DECK.

LIGHT-METAL PARAPET CAP (B)

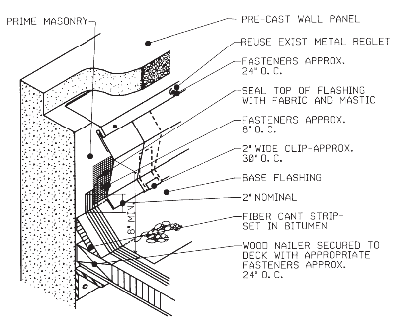


NOTES:
THIS DETAIL IS TO BE USED WHERE THERE IS A POSSIBILITY THAT DIFFERENTIAL MOVEMENT WILL OCCUR BETWEEN THE DECK AND A VERTICAL SURFACE, SUCH AS AT A PENTHOUSE WALL. THE VERTICAL WOOD MEMBER IS TO BE FASTENED TO THE DECK ONLY.

BASE FLASHING FOR NON-WALL-SUPPORTED DECK (C)

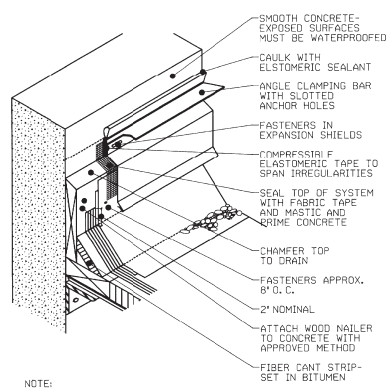


EXPANSION JOINT (D)



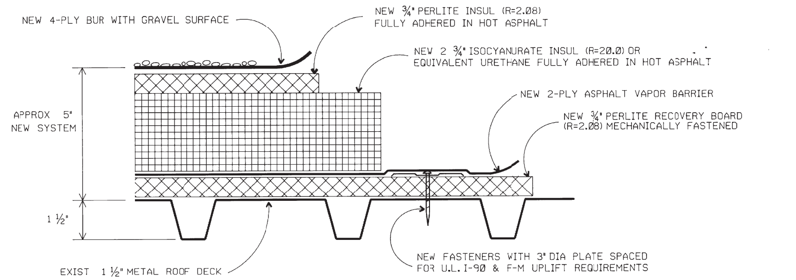
NOTES:
THIS DETAIL IS TO BE USED ONLY WHERE THE DECK IS SUPPORTED BY THE WALL.

BASE FLASHING FOR WALL-SUPPORTED DECK (E)

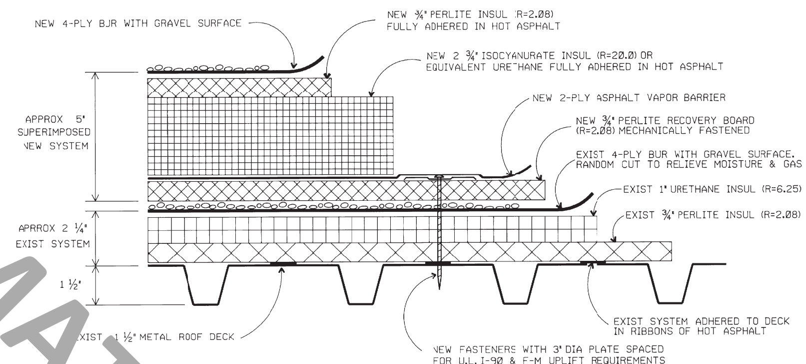


NOTE:
WHERE DECK IS SUPPORTED BY AND FASTENED TO THE CONCRETE WALL, VERTICAL WOOD NAILER IS TO BE SECURED TO THE WALL WITH SUITABLE FASTENERS.
THIS DETAIL TO BE USED ONLY WHERE AN EXIST REGLET CAN NOT BE REUSED OR DOES NOT EXIST.

COUNTERFLASHING FOR CONCRETE WALLS OR PARAPETS (F)



NEW ROOF SYSTEM - TEAR OFF TO EXISTING DECK
SCALE : HALF SIZE



ALTERNATE ROOF SYSTEM - SUPERIMPOSED OVER EXISTING
SCALE : HALF SIZE

CHIEF-ENG. CONST.				CHIEF-ENG. SECTION			COORDINATOR
DATE	NO.	REVISION	BY	DATE	NO.	REVISION	BY

WPAFB
WRIGHT-PATTERSON AIR FORCE BASE

USAF CIVIL ENGINEERING

REPLACE ROOF - FACILITY 20145
SABER W.O.# 78675
ROOF DETAILS

DESIGNED	DRAWN	CHECKED	DATE
JDK	JDK	JDK	14 AUG 91

A.2
2 OF 2

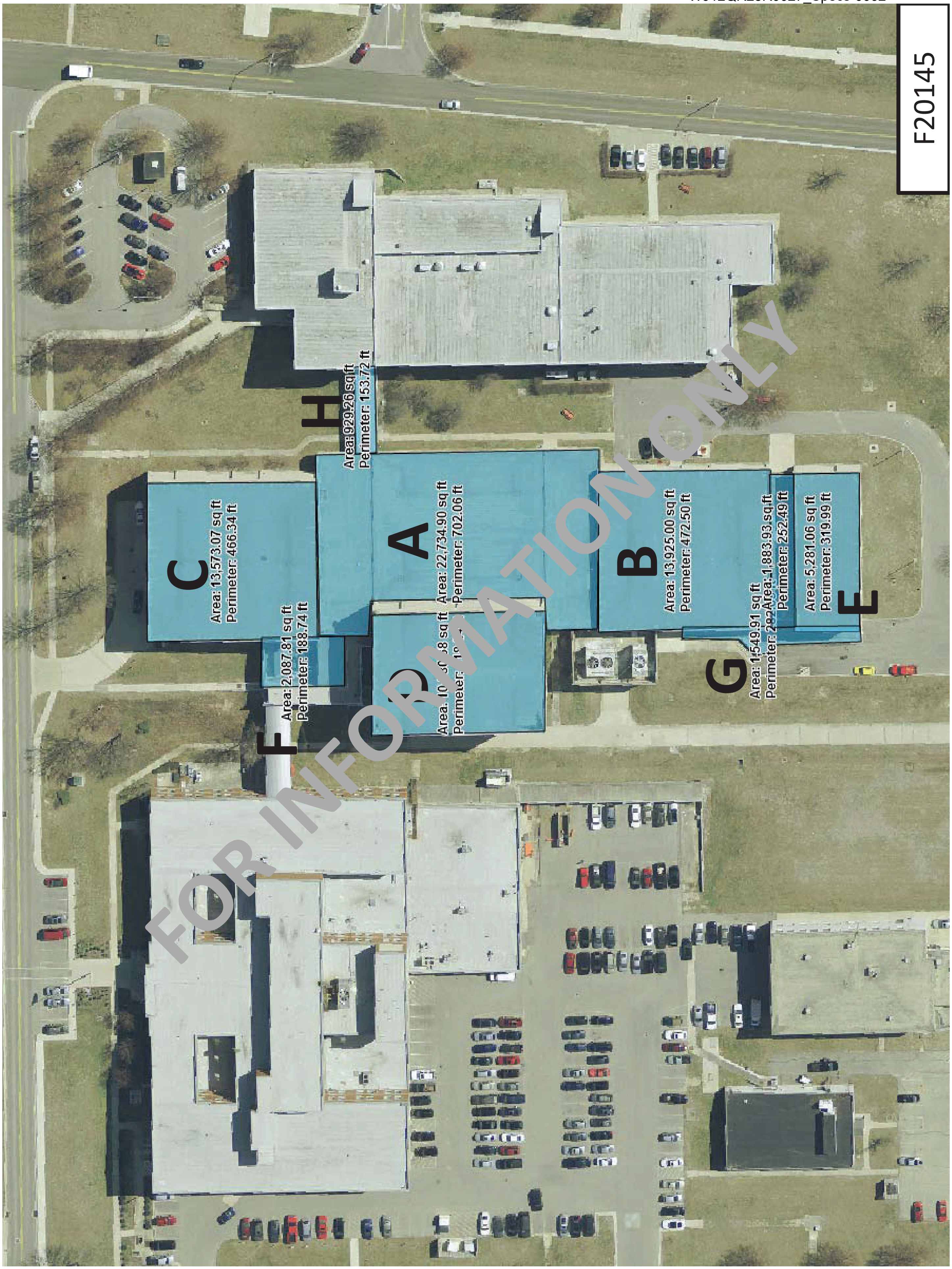
APPENDIX E

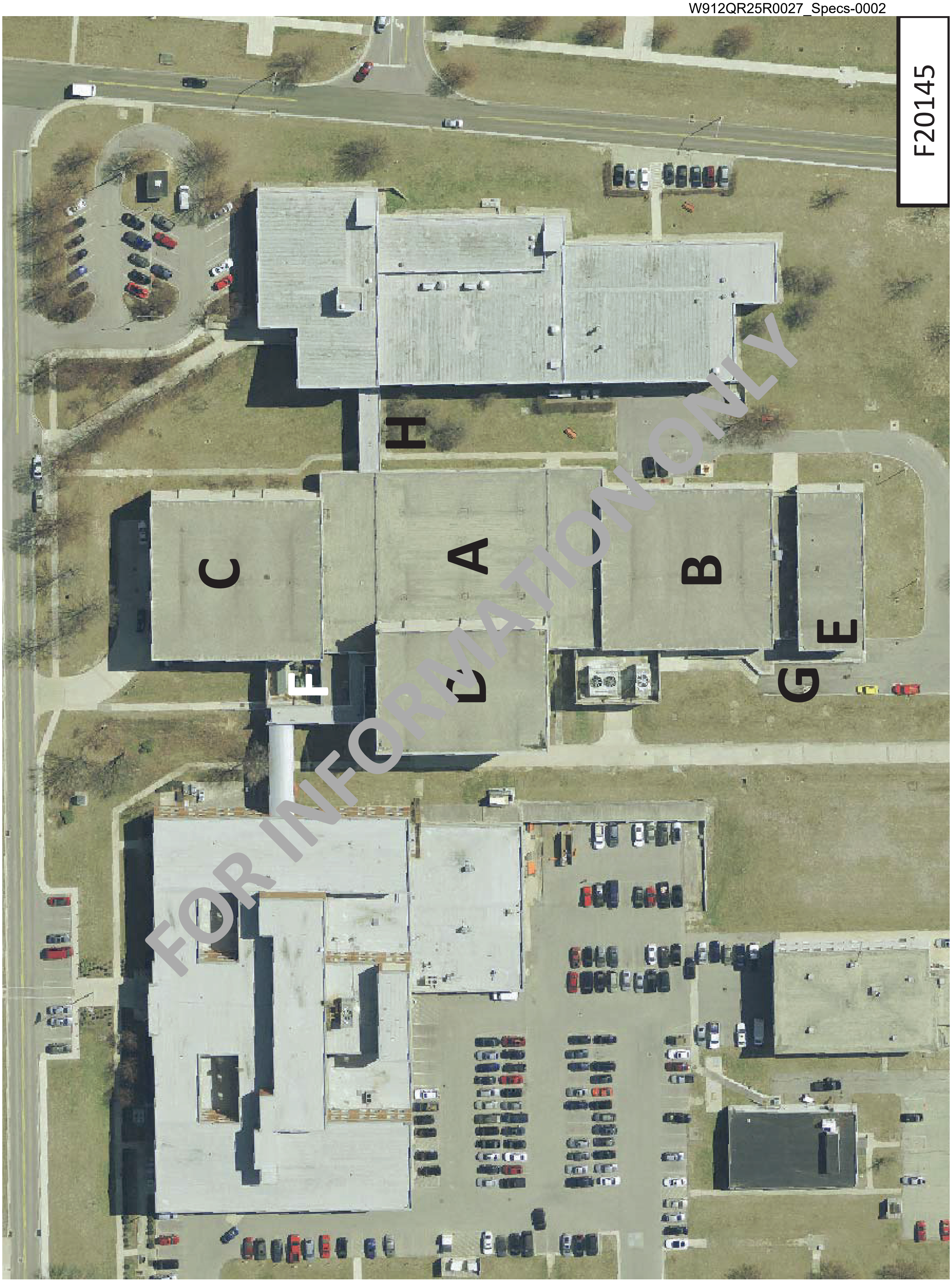
F20145 CHILLER ROOF PICTURE

F20145 Chiller Roof

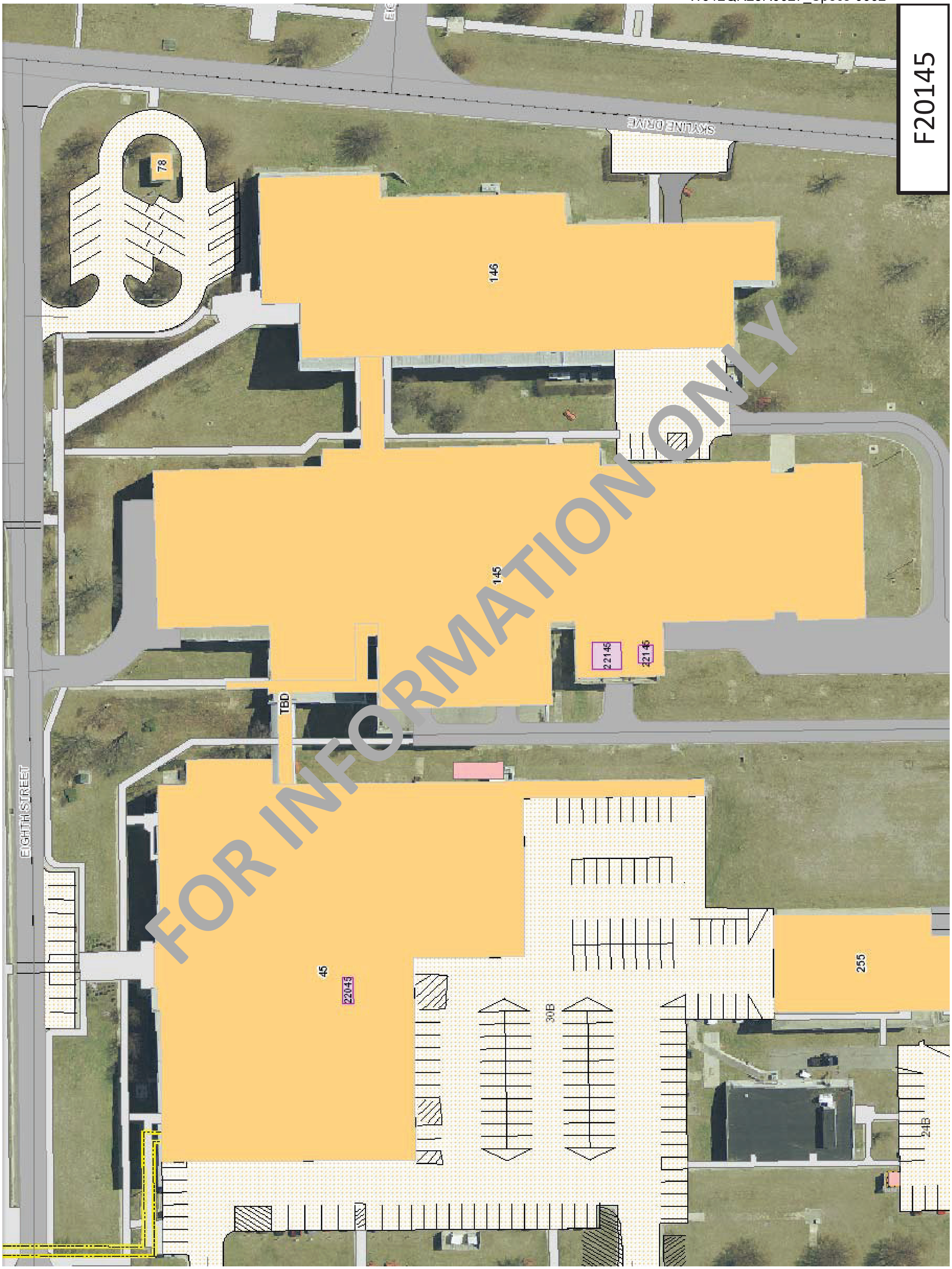
APPENDIX F

F20145 ROOF PLAN


F20145
H
Area: 929.26 sq ft
Perimeter: 153.72 ft
C
Area: 13,573.07 sq ft
Perimeter: 466.34 ft
A
Area: 22,734.90 sq ft
Perimeter: 702.06 ft
B
Area: 13,925.00 sq ft
Perimeter: 472.50 ft
Area: 1,883.93 sq ft
Perimeter: 252.49 ft
E
Area: 5,281.06 sq ft
Perimeter: 319.99 ft
G
Area: 1,549.91 sq ft
F
Area: 2,087.81 sq ft
Perimeter: 188.74 ft
FOR INFORMATION ONLY


F20145
H
C
A
B
E
G
F
D
FOR INFORMATION ONLY


F20145
FOR INFORMATION ONLY


F20145
SKYLINE DRIVE
EIGHTH STREET
78
146
145
22145
22145
TBD
45
22045
30B
255
24B
FOR INFORMATION ONLY